



Allwinner DE2.0 Specification

Revision 1.0

Jan.23, 2018

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Revision History

Revision	Date	Description
1.0	Jan. 23,2018	Initial Release Version

Glossary

The glossary is intended to cover the acronyms used in the document.

Term	Definition
R	Read only/non-Write
R/W	Read/Write
RC	Read-Clear/non-Write
RS	Read-Set/non-Write
RC/W	Read-Clear/Write
RS/W	Read-Set/Write
R/WAC	Read/Write-Automatic-Clear, Clear the bit automatically when the operation of complete.Writing 0 has no effect.
R/WC	Read/Write-Clear
R/WS	Read/Write-Set
RC/WS	Read-Clear/Write-Set
RS/WC	Read-Set/Write-Clear
R/W1C	Read/Write 1 to Clear, Write 0 has non-effect
R/W1S	Read/Write 1 to Set, Write 0 has non-effect
R/W1T	Read/Write 1 to Flip, Write 0 has non-effect
R/W0C	Read/Write 0 to Clear, Write 1 has non-effect
R/W0S	Read/Write 0 to Set, Write 1 has non-effect
R/W0T	Read/Write 0 to Flip, Write 1 has non-effect
RC/W1S	Read-Clear/Write 1 to Set, Write 0 has non-effect
RS/W1C	Read-Set/Write 1 to Clear, Write 0 has non-effect
RC/W0S	Read-Clear/Write 0 to Set, Write 1 has non-effect
RS/W0C	Read-Set/Write 0 to Clear, Write 1 has non-effect
W	Write only/non-Read
WC	Write-Clear/non-Read
WS	Write-Set/non-Read
W1	After reset, Write at the first time, non-Write after the first time/Read
W01	After reset, Write at the first time, non-Write after the first time/non-Read

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1 .H8 Display_Engine_Top

1.1 Overview

The Display Engine(DE) is a hardware composer to transfer image layers from a local bus or a video buffer to the *LCD* interface, The DE support four overlay windows to blending, and support image post-processing in the video channel.

Feature:

- Support output size up to 4096x4096
- Support four alpha blending channels for main display, two channels for aux display.
- Support four overlay layers in each channel, and has a independent scaler.
- Support potter-duff compatible blending operation.
- Support input format YUV422/YUV420/YUV411/ARGB8888/XRGB8888/RGB888/ARGB4444/ ARGB1555 and RGB565.
- Support Frame Packing/Top-and-Bottom/Side-by-Side Full/Side-by-Side Half 3D format data.
- Support SmartColor2.0 for excellent display experience.
 - Adaptive edge sharpening
 - Adaptive color enhancement
 - Adaptive contrast enhancement and fresh tone rectify.
- Support write back and rotation for high efficient dual display and miracast.

1.2 Block Diagram

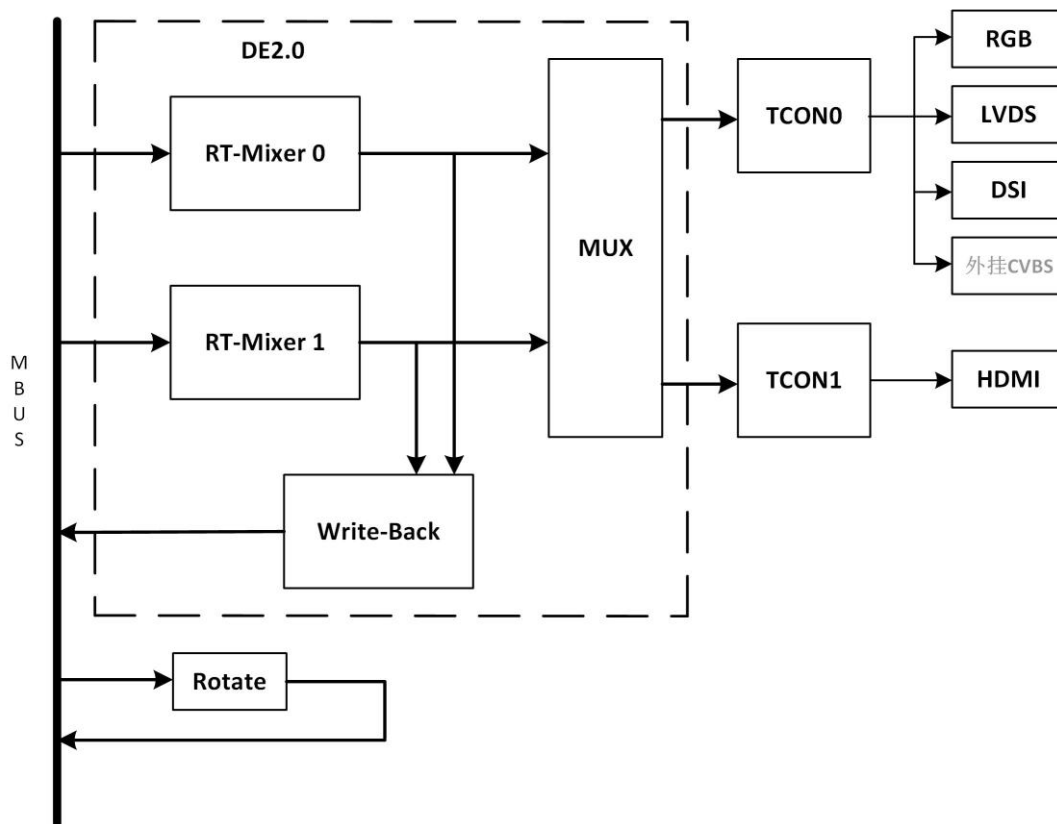


Figure 1-1. Display Top Level Block Diagram

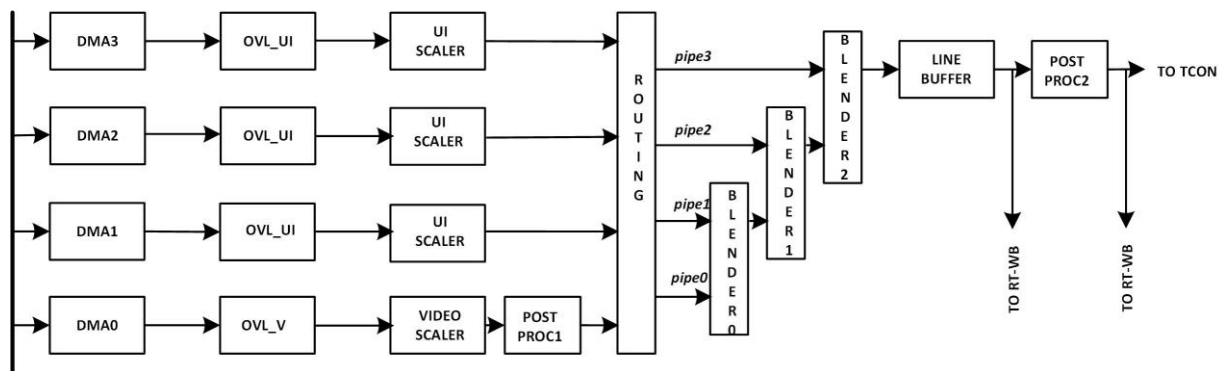


Figure 1-2. RT-MIXER0 Block Diagram

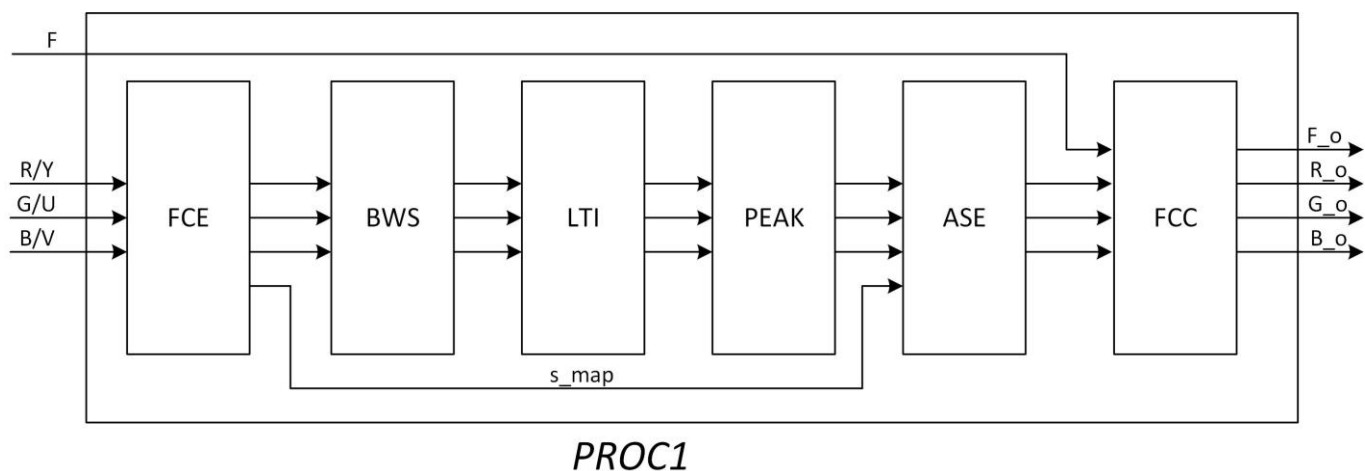


Figure 1-3. POST_PROC1 Block Diagram

DE2.0 post processing, include:

FCE	Fresh and Contrast enhancement
BWS	Black and white stretch
LTI	Luminance transient improvement
PEAK	Luma Peaking
ASE	Adaptive Saturation Enhancement
FCC	Fancy color curvature change

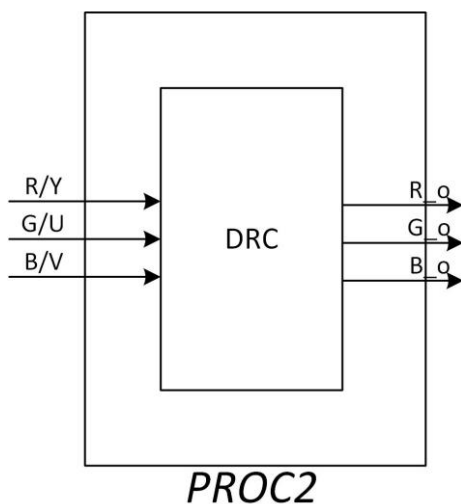


Figure 1-4. POST_PROC2 Block Diagram

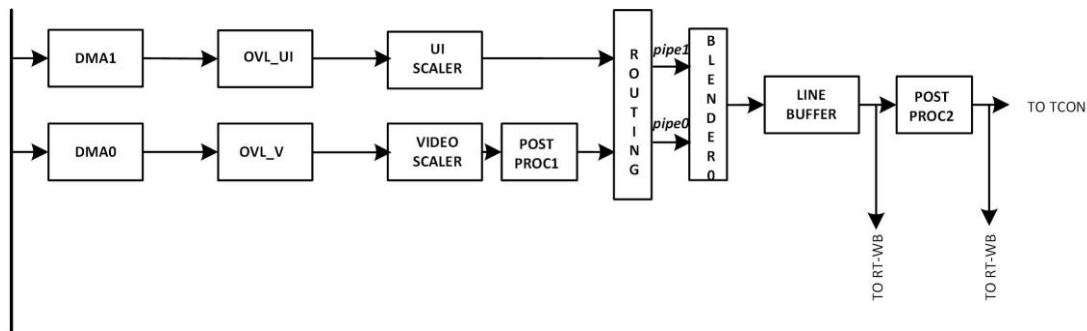


Figure 1-5. RT-MIXER1 Block Diagram

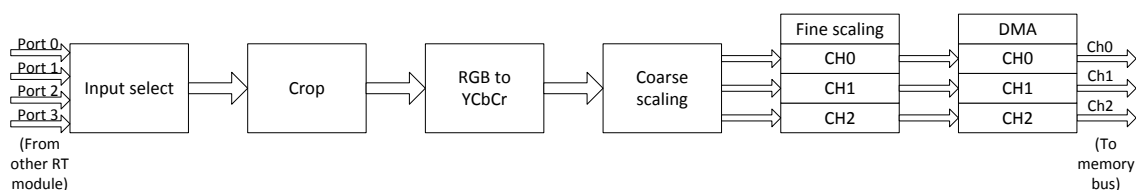


Figure 1-6. RT-WB Block Diagram

- Support RGB888 and YUV444 input data format.
- Support input size from 8x4 to 2048x2048.
- Support output size from 8x4 to 2048x2048.
- Support fine down scaling ratio from 1x to 1/2x, and the anti-aliasing filter is 16-phase 4-tap in horizontal, 16-phase 2-tap/4-tap filter in vertical.
- Support coarse down scaling.

1.3 Display system configurations and requirements

1.3.1 Memory Space Arrangement:

Module name	Base address	
DE	0x01000000	0x013FFFFFFF

Module name	Offset Address	Memory Range
Display system	0x0000000	64K
RT-WB	0x0100000	64K
Rotate	0x0200000	64K
DI	0x0300000	128K
RT-Mixer0	0x1000000	1M
RT-Mixer1	0x2000000	1M

1.4 Register list

Display system top register list:

Register name	Offset	Description
---------------	--------	-------------

SCLK_GATE	0x000	DE SCLK Gating Register
HCLK_GATE	0x004	DE HCLK Gating Register
AHB_RESET	0x008	DE AHB Reset register
SCLK_DIV	0x00C	DE SCLK Division register
DE2TCON_MUX	0x010	DE2TCON MUX register
CMD_CTL	0x014	CMD Control register
DI_CTL	0x01C	DI Control register

1.5 Register Description

1.5.1 SCLK_GATE

Offset: 0x000			Register Name: SCLK_GATE
Bit	Read/Write	Default/Hex	Description
31:4	/	/	Reserved
3	R/W	0x0	ROT_SCLK_GATE 0: clock gate 1: clock pass
2	R/W	0x0	RT_WB_SCLK_GATE 0: clock gate 1: clock pass
1	R/W	0x0	CORE1_SCLK_GATE 0: clock gate 1: clock pass
0	R/W	0x0	CORE0_SCLK_GATE 0: clock gate 1: clock pass

1.5.2 HCLK_GATE

Offset: 0x004			Register Name: HCLK_GATE
Bit	Read/Write	Default/Hex	Description
29:4	/	/	Reserved
3	R/W	0x0	ROT_HCLK_GATE 0: clock gate 1: clock pass
2	R/W	0x0	RT_WB_HCLK_GATE 0: clock gate 1: clock pass
1	R/W	0x0	CORE1_HCLK_GATE 0: clock gate 1: clock pass
0	R/W	0x0	CORE0_HCLK_GATE

			0: clock gate 1: clock pass
--	--	--	--------------------------------

1.5.3 AHB_RESET

Offset: 0x008			Register Name: AHB_RESET
Bit	Read/Write	Default/Hex	Description
29:4	/	/	Reserved
3	R/W	0x0	ROT_SCLK_RESET 0: reset on 1: reset off
2	R/W	0x0	RT_WB_SCLK_RESET 0: reset on 1: reset off
1	R/W	0x0	CORE1_SCLK_RESET 0: reset on 1: reset off
0	R/W	0x0	CORE0_HCLK_RESET 0: reset on 1: reset off

1.5.4 SCLK_DIV

Offset: 0x00C			Register Name: SCLK_DIV
Bit	Read/Write	Default/Hex	Description
31:16	/	/	Reserved
15:12	R/W	0x0	ROT_SCLK_DIV
11:8	R/W	0x0	RT_WB_SCLK_DIV
7:4	R/W	0x0	CORE1_SCLK_DIV
3:0	R/W	0x0	CORE0_SCLK_DIV

1.5.5 DE2TCON_MUX

Offset: 0x010			Register Name: DE2TCON_MUX
Bit	Read/Write	Default/Hex	Description
31:1	/	/	Reserved
0	R/W	0x0	DE2TCON_MUX 0: MIXER0-〉 TCON0; MIXER1-〉 TCON1 1: MIXER0-〉 TCON1; MIXER1-〉 TCON0

1.5.6 CMD_CTL

Offset: 0x014			Register Name: CMD_CTL
Bit	Read/Write	Default/Hex	Description
31:20	/	/	Reserved
19:16	R/W	0x0	RT_WB_CMD_CTL
15:8	/	/	/
7:4	R/W	0x0	CORE1_CMD_CTL
3:0	R/W	0x0	CORE0_CMD_CTL

Note: The number of sending command to DRAM are about N+2.

1.5.7 DI_CTL

Offset: 0x01C			Register Name: DI_CTL
Bit	Read/Write	Default/Hex	Description
31:5	/	/	/
4	R/W	0x0	CORE1_DI_CTL 0: DI disable 1: DI enable Note: if DI enable, VSU use DI data, otherwise use overlay data.
3:0	/	/	/
0	R/W	0x0	CORE0_DI_CTL 0: DI disable 1: DI enable Note: if DI enable, VSU use DI data, otherwise use overlay data.

2 .H3 Display_Engine_Top

2.1 Overview

- Support input layer size up to 2048x2048, and output size up to 2048x2048.
- Support four alpha blending channel for main display, two channel for aux display.
- Support 4 overlay layer in each channel, and has a independent scaler.
- Support potter-duff compatible blending operation.
- Support display enhancement 2.0 for excellent display experience.
- Support write back for high efficient dual display.

2.2 Block Diagram

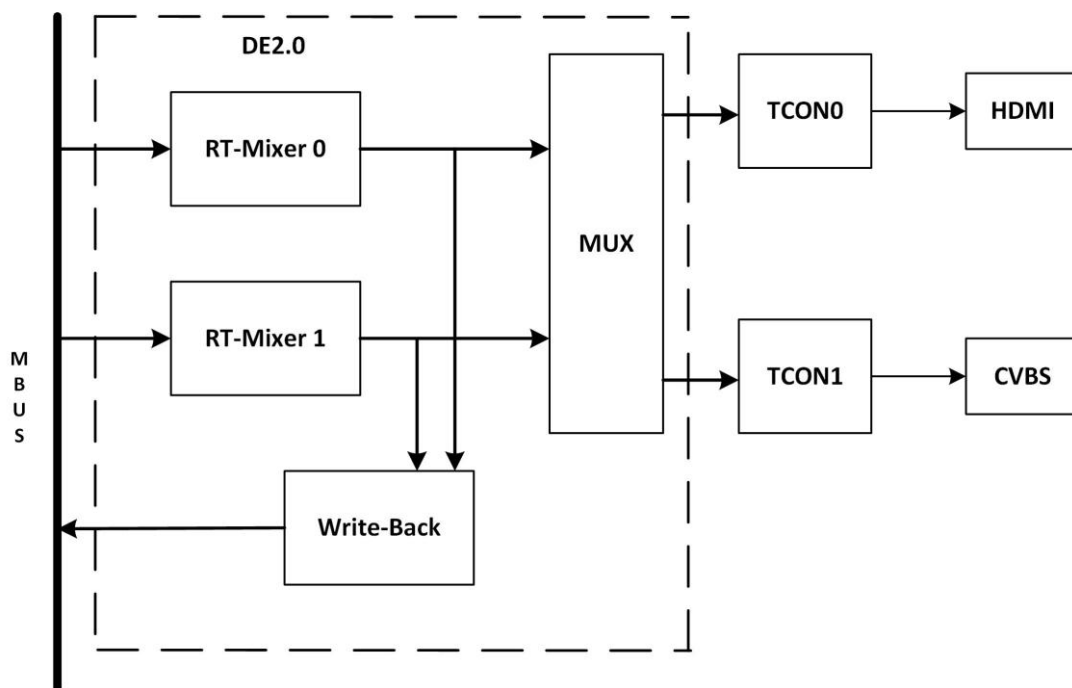


Figure 2-1. Display Top Level Diagram

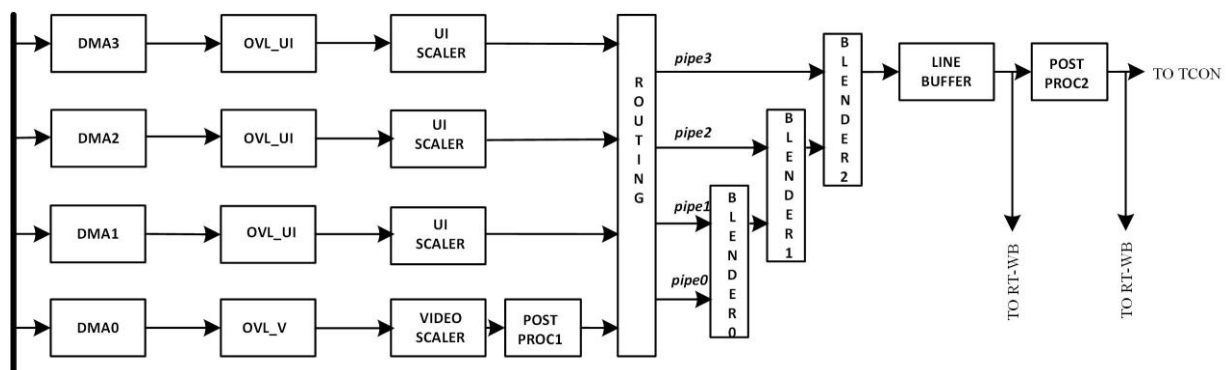


Figure 2-2. RT-MIXER0 Block Diagram

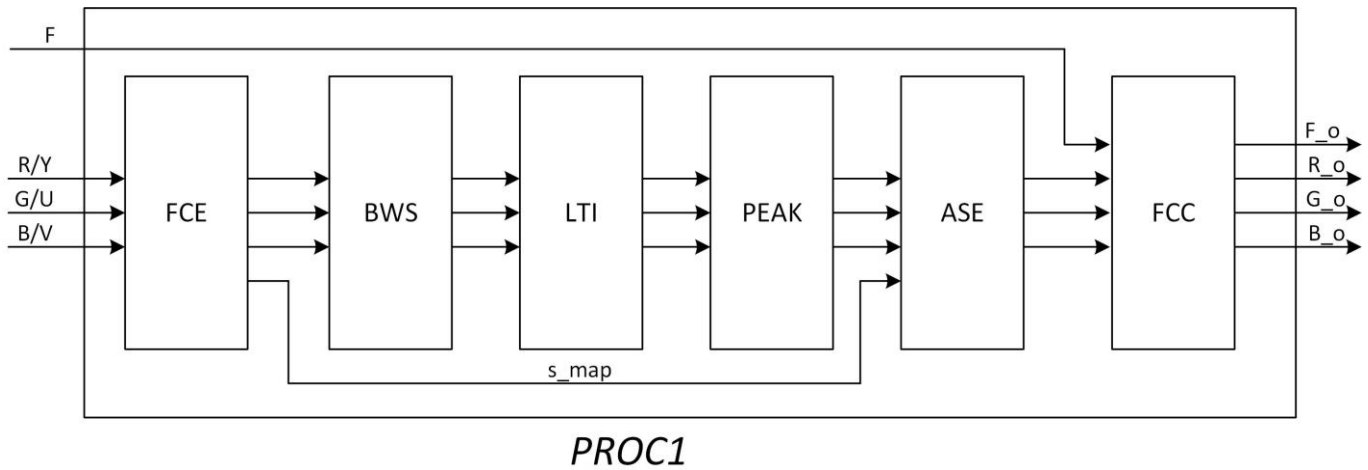


Figure 2-3. POST_PROC1 Block Diagram

DE2.0 post processing, include:

FCE : Fresh and Contrast enhancement

BWS : Black and white stretch

LTI : Luminance transient improvement

PEAK: Luma Peaking

ASE : Adaptive Saturation Enhancement

FCC : Fancy color curvature change

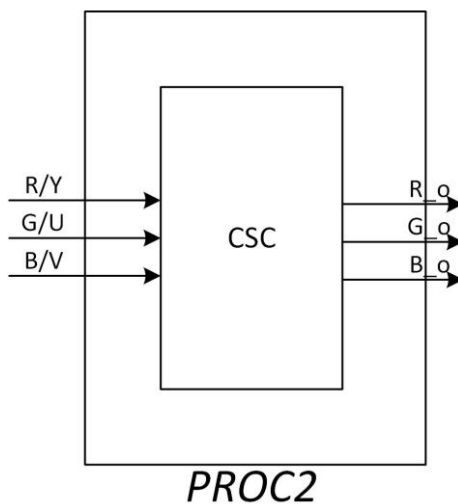


Figure 2-4. POST_PROC2 Block Diagram

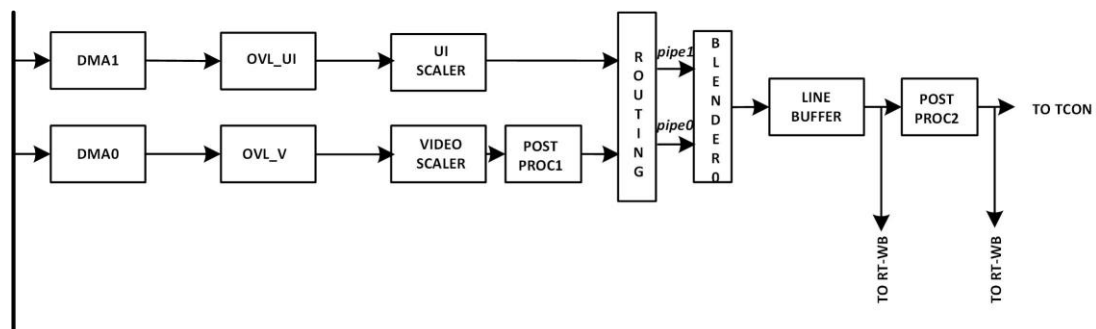


Figure 2-5. RT-MIXER1 Block Diagram

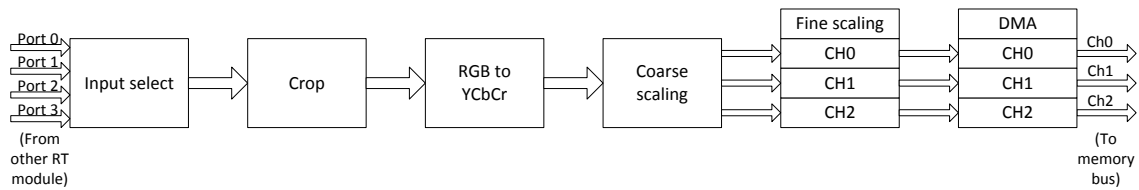


Figure 2-6. RT-WB Block Diagram

- Support RGB888 and YUV444 input data format.
- Support input size from 8x4 to 2048x2048.
- Support output size from 8x4 to 2048x2048.
- Support fine down scaling ratio from 1x to 1/2x, and the anti-aliasing filter is 16-phase 4-tap in horizontal, 16-phase 2-tap/4-tap filter in vertical.
- Support coarse down scaling.

2.3 Display System Configurations and requirements

2.3.1 Speed requirements:

Module Name	Speed(Mhz)	Description
RT-MIXER0	250	
RT-MIXER1	250	
RT-WB	250	

2.3.2 Memory Space Arrangement:

Module name	Base address	
DE	0x01000000	0x012FFFFFFF

Module name	Memory Range	Offset Address
Display system	64K	0x000000
RT-WB	64K	0x010000
RT-Mixer0	1M	0x100000
RT-Mixer1	1M	0x200000

2.4 Register list

Display system top register list:

Register name	Offset	Description
SCLK_GATE	0x000	
HCLK_GATE	0x004	
AHB_RESET	0x008	

SCLK_DIV	0x00C	
DE2TCON_MUX	0X010	

2.5 Register Description

2.5.1 SCLK_GATE

Offset: 0x000			Register Name: SCLK_GATE
Bit	Read/Write	Default/Hex	Description
31:3	/	/	Reserved
2	R/W	0x0	RT_WB_SCLK_GATE 0: clock gate 1: clock pass
1	R/W	0x0	CORE1_SCLK_GATE 0: clock gate 1: clock pass
0	R/W	0x0	CORE0_SCLK_GATE 0: clock gate 1: clock pass

2.5.2 HCLK_GATE

Offset: 0x004			Register Name: HCLK_GATE
Bit	Read/Write	Default/Hex	Description
29:3	/	/	Reserved
2	R/W	0x0	RT_WB_HCLK_GATE 0: clock gate 1: clock pass
1	R/W	0x0	CORE1_HCLK_GATE 0: clock gate 1: clock pass
0	R/W	0x0	CORE0_HCLK_GATE 0: clock gate 1: clock pass

2.5.3 AHB_RESET

Offset: 0x008			Register Name: AHB_RESET
Bit	Read/Write	Default/Hex	Description
29:4	/	/	Reserved
2	R/W	0x0	RT_WB_SCLK_RESET

			0: reset on 1: reset off
1	R/W	0x0	CORE1_SCLK_RESET 0: reset on 1: reset off
0	R/W	0x0	CORE0_HCLK_RESET 0: reset on 1: reset off

2.5.4 SCLK_DIV

Offset: 0x00C			Register Name: SCLK_DIV
Bit	Read/Write	Default/Hex	Description
31:12	/	/	Reserved
11:8	R/W	0x0	RT_WB_SCLK_DIV
7:4	R/W	0x0	CORE1_SCLK_DIV
3:0	R/W	0x0	CORE0_SCLK_DIV

2.5.5 DE2TCON_MUX

Offset: 0x010			Register Name: DE2TCON_MUX
Bit	Read/Write	Default/Hex	Description
31:1	/	/	Reserved
0	R/W	0x0	DE2TCON_MUX 0: MIXER0-〉 TCON0; MIXER1-〉 TCON1 1: MIXER0-〉 TCON1; MIXER1-〉 TCON0

2.5.6 CMD_CTL

Offset: 0x014			Register Name: CMD_CTL
Bit	Read/Write	Default/Hex	Description
31:20	/	/	Reserved
19:16	R/W	0x0	RT_WB_CMD_CTL
15:8	/	/	/
7:4	R/W	0x0	CORE1_CMD_CTL
3:0	R/W	0x0	CORE0_CMD_CTL

Note: The number of sending command to dram are about N+2

3 .H5 Display_Engine_Top

3.1 Overview

The Display Engine(DE) is a hardware composer to transfer image layers from a local bus or a video buffer to the *LCD* interface, The DE support four overlay windows to blending, and support image post-processing in the video channel. The display system block diagram show as the Figure1-1.

Feature:

- Support output size up to 4096x4096
- Support four alpha blending channels for main display, two channels for aux display.
- Support four overlay layers in each channel, and has a independent scaler.
- Support potter-duff compatible blending operation.
- Support input format YUV422/YUV420/YUV411/ARGB8888/XRGB8888/RGB888/ARGB4444/ ARGB1555 and RGB565.
- Support Frame Packing/Top-and-Bottom/Side-by-Side Full/Side-by-Side Half 3D format data.
- Support SmartColor2.0 for excellent display experience.
 - Adaptive edge sharpening
 - Adaptive color enhancement
 - Adaptive contrast enhancement and fresh tone rectify.
- Support write back for high efficient dual display and miracast.

3.2 Block Diagram

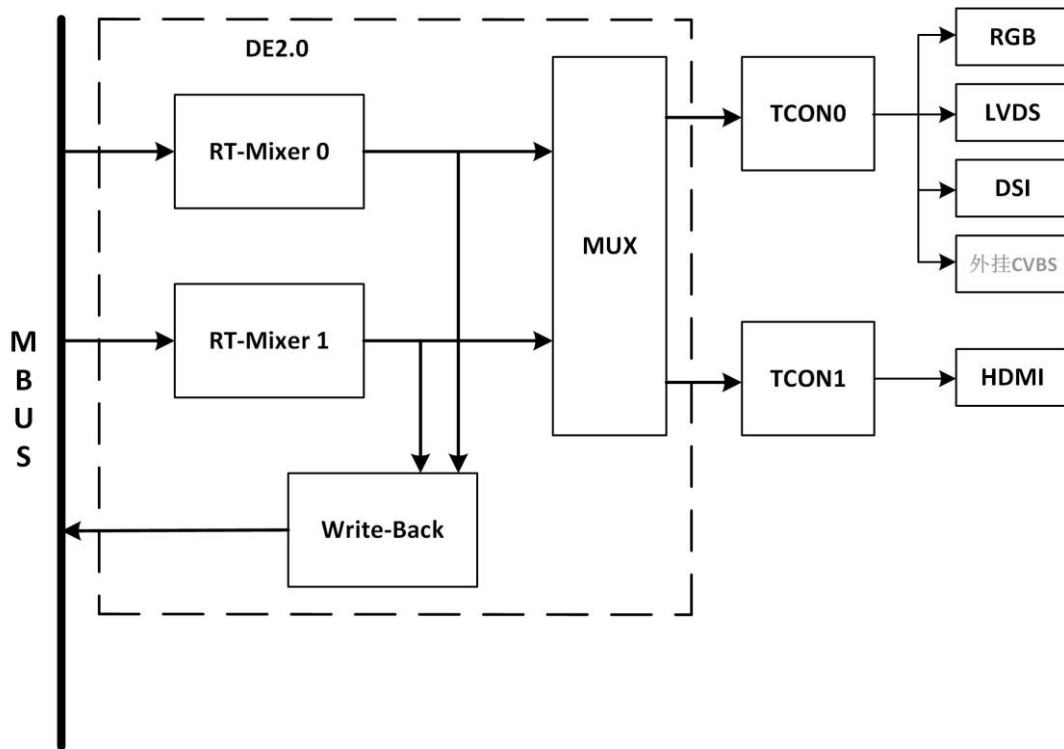


Figure 3-1. DE System Block Diagram

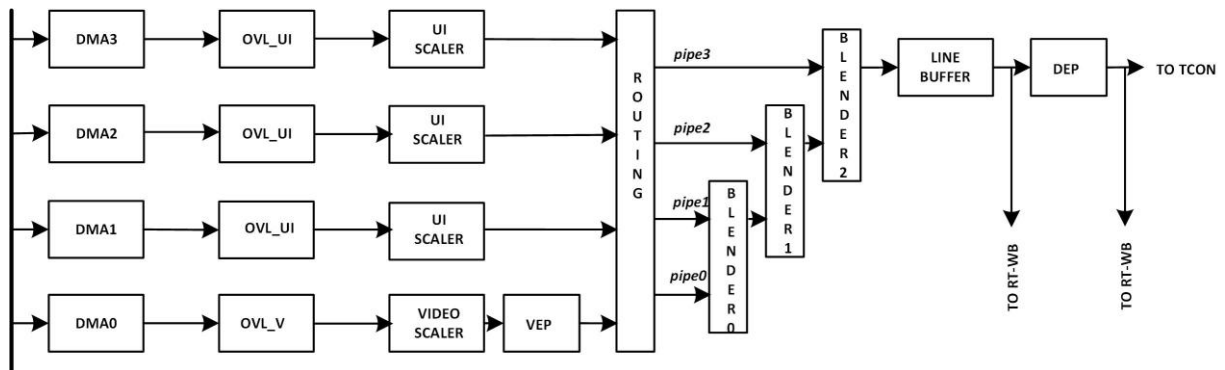


Figure 3-2. RT-Mixer0 Block Diagram

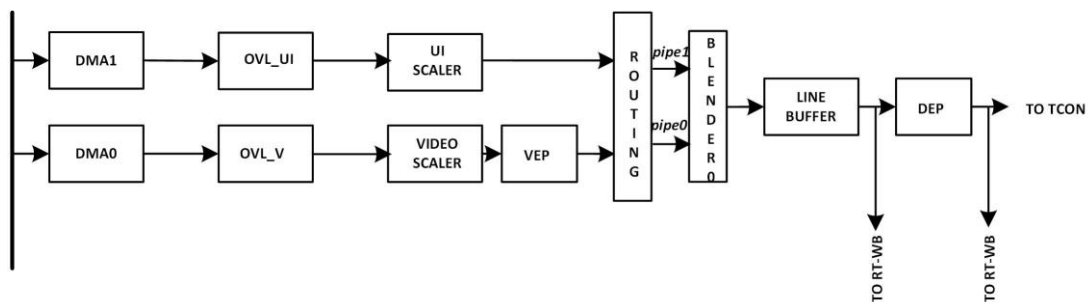


Figure 3-3. RT-Mixer1 Block Diagram

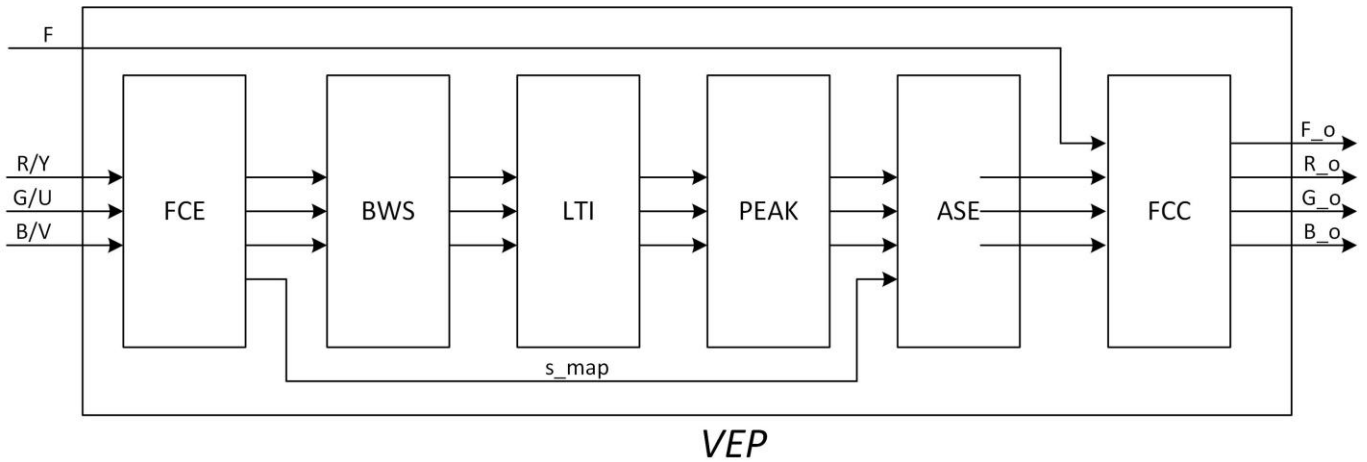


Figure 3-4. VEP Block Diagram

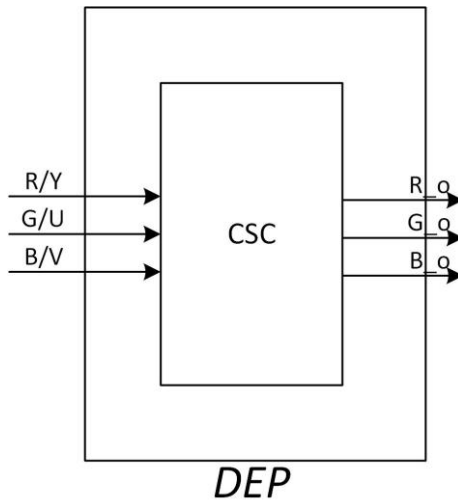


Figure 3-5. DEP Block Diagram

3.3 Operations and Functional Descriptions

3.3.1 Configuration and Requirements

The following table describes the configuration and speed of **DE System**.

Table 3-1. RT-Mixer Configuration

Module	Pipe	Overlay	FBD	DI	DNR	Scaler Type/ Line Buffer	VEP	Blend	DEP
RT-Mixer0	CH0	4 layers	N	N	N	VScaler/4096	Y	Y	CSC
	CH1	4 layers	N	N	N	GScaler/2048	N		
	CH2	4 layers	N	N	N	GScaler/2048	N		
	CH3	4 layers	N	N	N	GScaler/2048	N		
RT-Mixer1	CH0	4 layers	N	N	N	VScaler/2048	CSC	Y	CSC
	CH1	4 layers	N	N	N	GScaler/2048	N		

Table 3-2. RT-WB Configuration

Module	Input	Scaler Type/ Line Buffer
RT-WB	RT-Mixer0 Blending output RT-Mixer0 Post-Proc2 output RT-Mixer1 Blending output RT-Mixer1 Post-Proc2 output	GScaler/2048

Table 3-3. DE Memory and Speed requirements

Module Name	Base Address	Memory Range	Speed(MHz)
Display System	0x01000000	32K	432M
RT-WB	0x01010000	32K	432M
RT-Mixer0	0x01100000	1M	432M
RT-Mixer1	0x01200000	1M	432M

3.3.2 Clock Sources

Display Engine controller get two different clocks,Users can select one of them to make DE Clock Source.The following table describes the clock sources for DE.Users can see **Clock Controller Unit(CCU)** for clock setting, configuration and gating information.

Table 3-4. DE Clock Sources

Clock Sources	Description
PLL_DE	PLL_DE,default value is 250MHz for DE.
PLL_PERIPH0(2x)	Peripheral Clock0 source, divide to about 250MHz for DE.

3.4 DE Register List

Module Name	Base Address
DE	0x01000000

Register Name	Offset	Description
DE_SCLK_GATE	0x00	DE SCLK Gating Register
DE_HCLK_GATE	0x04	DE HCLK Gating Register
DE_AHB_RESET	0x08	DE AHB Reset register
DE_SCLK_DIV	0x0c	DE SCLK Division register
DE2TCON_MUX	0x10	DE2TCON MUX register

3.5 DE Register Description

3.5.1 DE SCLK Gating Register(Default Value: 0x0000_0000)

Offset: 0x00			Register Name: DE_SCLK_GATE
Bit	Read/Write	Default/Hex	Description
31:3	/	/	Reserved
2	R/W	0x0	RT_WB_SCLK_GATE 0: clock gate 1: clock pass
1	R/W	0x0	CORE1_SCLK_GATE 0: clock gate 1: clock pass
0	R/W	0x0	CORE0_SCLK_GATE 0: clock gate 1: clock pass

3.5.2 DE HCLK Gating Register(Default Value: 0x0000_0000)

Offset: 0x04			Register Name: DE_HCLK_GATE
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/
2	R/W	0x0	WB_HCLK_GATE 0: clock gate 1: clock pass
1	R/W	0x0	CORE1_HCLK_GATE 0: clock gate 1: clock pass
0	R/W	0x0	CORE0_HCLK_GATE 0: clock gate 1: clock pass

3.5.3 DE AHB Reset Register(Default Value: 0x0000_0000)

Offset: 0x08			Register Name: DE_AHB_RESET
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/
2	R/W	0x0	RT_WB_SCLK_RESET 0: reset on 1: reset off

1	R/W	0x0	CORE1_SCLK_RESET 0: reset on 1: reset off
0	R/W	0x0	CORE0_HCLK_RESET 0: reset on 1: reset off

3.5.4 DE SCLK Division Register(Default Value: 0x0000_0000)

Offset: 0x0C			Register Name: DE_SCLK_DIV
Bit	Read/Write	Default/Hex	Description
31:12	/	/	/
11:8	R/W	0x0	WB_SCLK_DIV
7:4	R/W	0x0	CORE1_SCLK_DIV
3:0	R/W	0x0	CORE0_SCLK_DIV

3.5.5 DE2TCON MUX Register(Default Value: 0x0000_0000)

Offset: 0x10			Register Name: DE2TCON_MUX
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	DE2TCON_MUX 0: MIXER0-→ TCON0; MIXER1-→ TCON1 1: MIXER0-→ TCON1; MIXER1-→ TCON0

4 .A83 Display_Engine_Top

4.1 Overview

- Support output size up to 2048x2048
- Support four alpha blending channels for main display, two channels for aux display.
- Support four overlay layers in each channel, and has a independent scaler.
- Support potter-duff compatible blending operation.
- Support input format YUV422/YUV420/YUV411/ARGB8888/XRGB8888/RGB888/ARGB4444/ ARGB1555 and RGB565.
- Support Frame Packing/Top-and-Bottom/Side-by-Side Full/Side-by-Side Half 3D format data.
- Support SmartColor2.0 for excellent display experience.
 - Adaptive edge sharpening
 - Adaptive color enhancement
 - Adaptive contrast enhancement and fresh tone rectify.
- Support write back and rotation for high efficient dual display and miracast.

4.2 Block Diagram

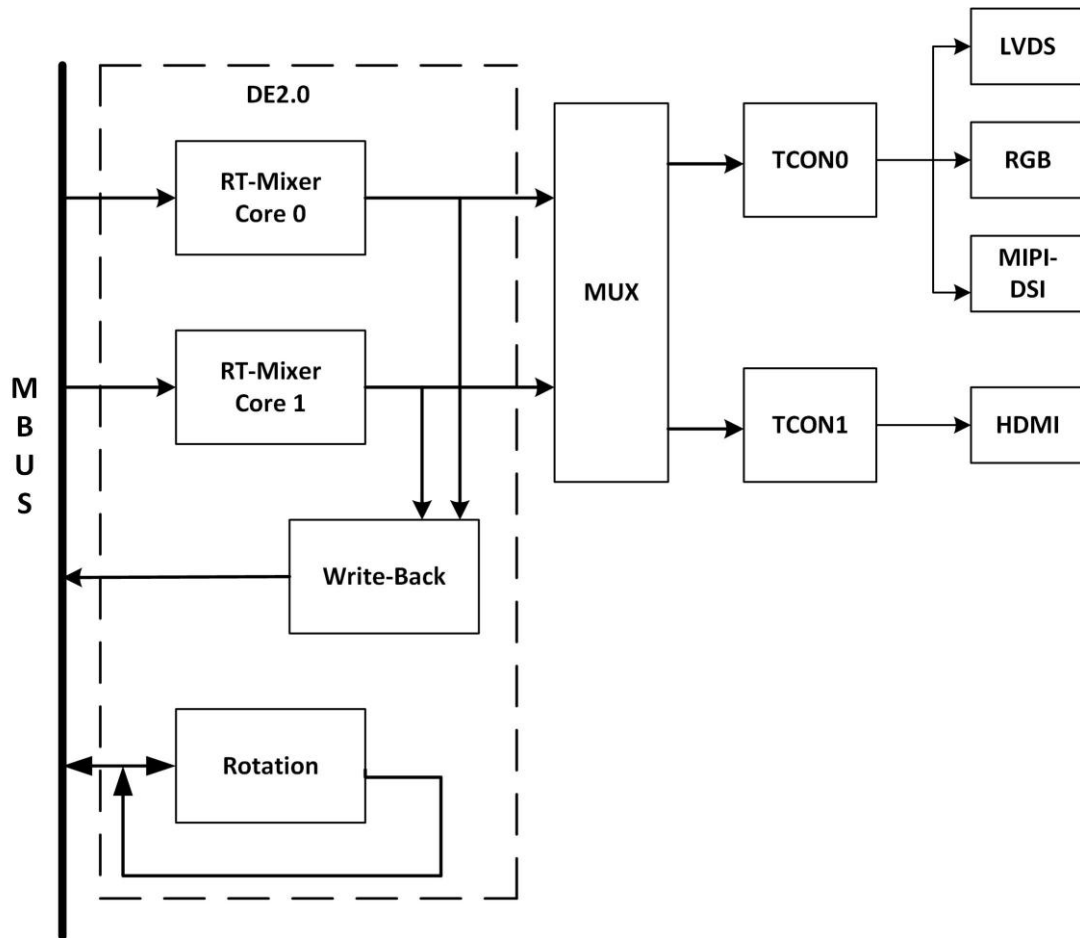


Figure 4-1. DE System Block Diagram

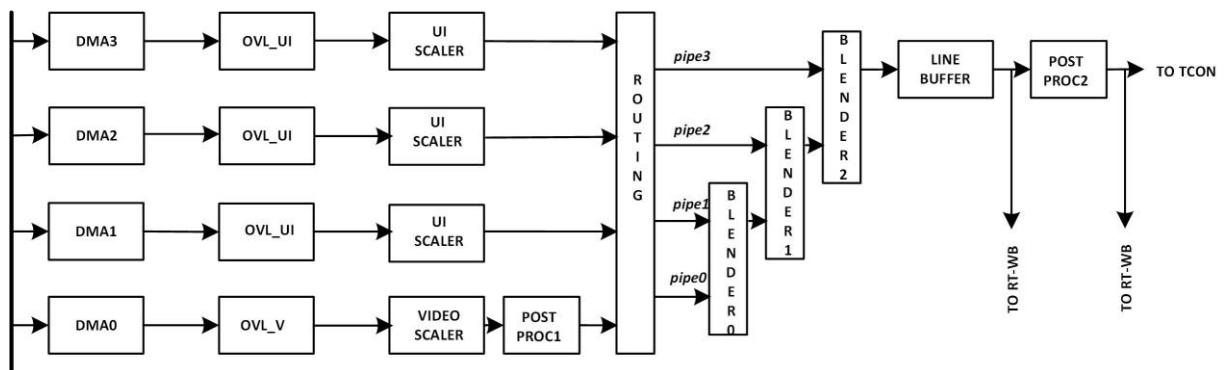


Figure 4-2. RT-MIXER Core0 Block Diagram

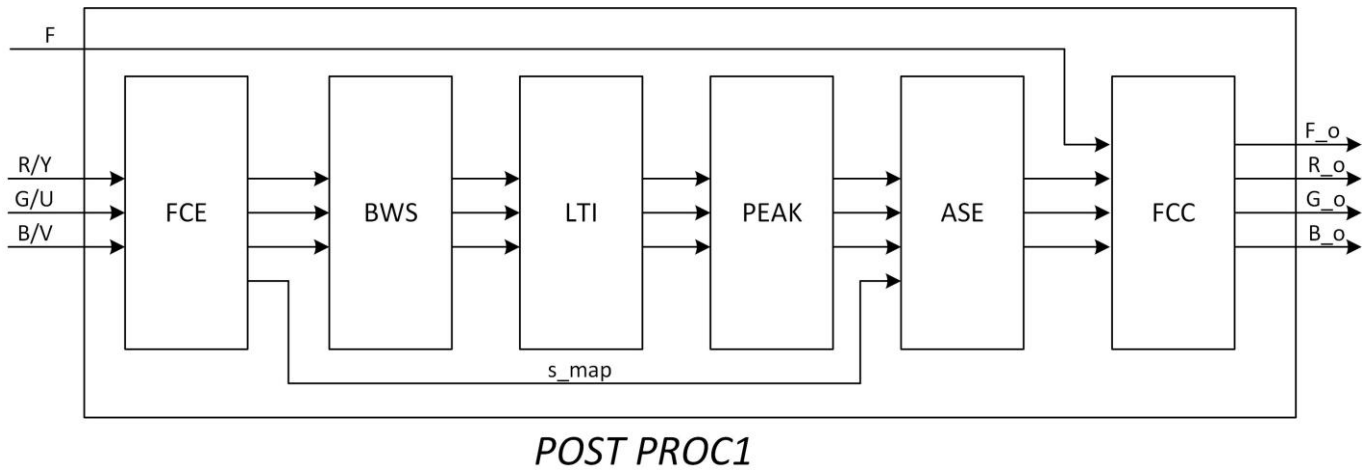


Figure 4-3. POST_PROC1 Block Diagram

DE2.0 post processing, include:

- FCE : Fresh and Contrast enhancement
- BWS: Black and white stretch
- LTI : Luminance transient improvement
- PEAK: Luma Peaking
- ASE : Adaptive Saturation Enhancement
- FCC : Fancy color curvature change

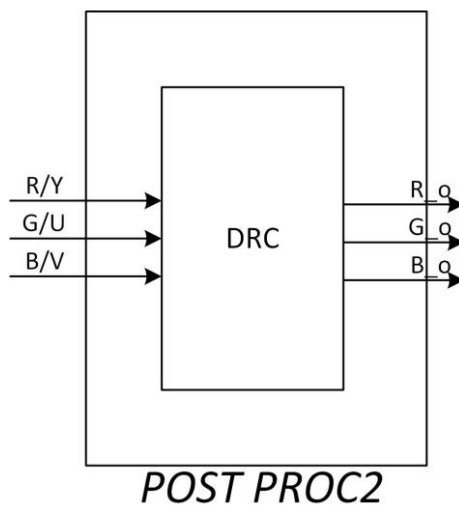


Figure 4-4. POST_PROC2 Block Diagram

NOTE: DRC: Dynamic Range Controller

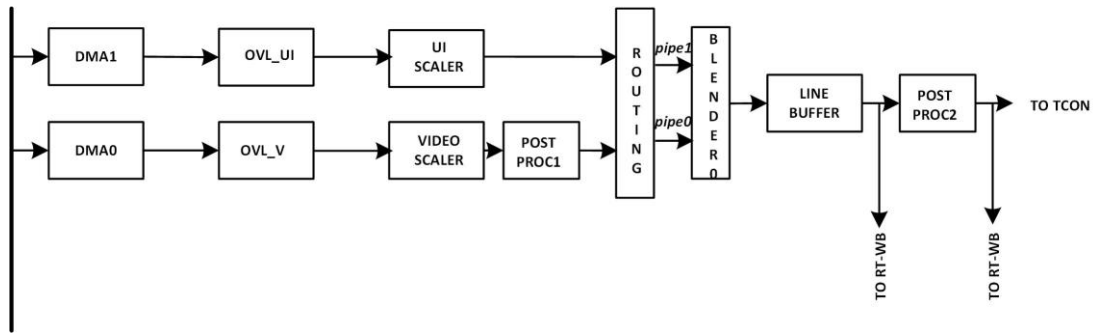


Figure 4-5. RT-MIXER Core1 Block Diagram

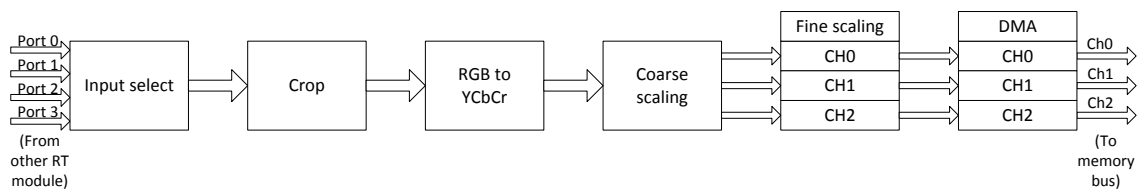


Figure 4-6. RT-WB Block Diagram

- Support RGB888 and YUV444 input data format
- Support input size from 8×4 to 4096×4096
- Support output size from 8×4 to 1920×1200
- Support fine down scaling ratio from 1× to 1/2×, and the anti-aliasing filter is 16-phase 4-tap in horizontal , 16-phase 2-tap/4-tap filter in vertical
- Support coarse down scaling.

Memory to Memory rotation

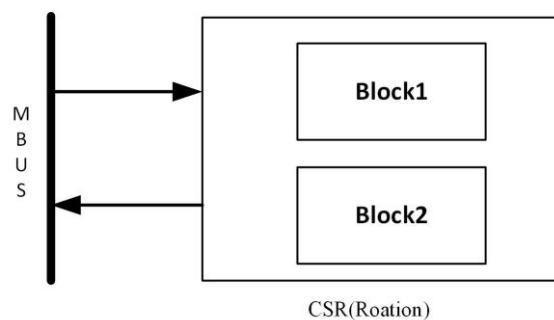


Figure 4-7. CSR(Rotation) Block Diagram

Input Format :

- 1.16/32bit of RGB and ARGB type ,
- YUV422 interleave

- YUV420 combine and planar
- 4.24bit RGB

Output Format :

- If the input is RGB type, the output format is same as the input format
- If the input format is YUV type, the output format is YUV planar(YUV is stored at 3 address, respectively)

4.3 Display system configurations and requirements

4.3.1 Memory Space Arrangement:

Module name	Base address	
DE	0x01000000	0x013F0000

Module name	Memory Range	Offset Address
Display system	64K	0x000000
RT-WB	64K	0x010000
CSR	64K	0x020000
RT-Mixer0	1M	0x100000
RT-Mixer1	1M	0x200000

4.4 Register list

Display system top register list:

Register name	Offset	Description
SCLK_GATE	0x000	
HCLK_GATE	0x004	
AHB_RESET	0x008	
SCLK_DIV	0x00C	
DE2TCON_MUX	0x010	

4.5 Register Description

4.5.1 SCLK_GATE

Offset: 0x000	Register Name: SCLK_GATE
---------------	--------------------------

Bit	Read/Write	Default/Hex	Description
31:4	/	/	Reserved
3	R/W	0x0	CSR_SCLK_GATE 0: clock gate 1: clock pass
2	R/W	0x0	RT_WB_SCLK_GATE 0: clock gate 1: clock pass
1	R/W	0x0	CORE1_SCLK_GATE 0: clock gate 1: clock pass
0	R/W	0x0	CORE0_SCLK_GATE 0: clock gate 1: clock pass

4.5.2 HCLK_GATE

Offset: 0x004			Register Name: HCLK_GATE
Bit	Read/Write	Default/Hex	Description
29:4	/	/	Reserved
3	R/W	0x0	CSR_HCLK_GATE 0: clock gate 1: clock pass
2	R/W	0x0	RT_WB_HCLK_GATE 0: clock gate 1: clock pass
1	R/W	0x0	CORE1_HCLK_GATE 0: clock gate 1: clock pass
0	R/W	0x0	CORE0_HCLK_GATE 0: clock gate 1: clock pass

4.5.3 AHB_RESET

Offset: 0x008			Register Name: AHB_RESET
Bit	Read/Write	Default/Hex	Description
29:4	/	/	Reserved
3	R/W	0x0	CSR_SCLK_RESET 0: reset on 1: reset off
2	R/W	0x0	RT_WB_SCLK_RESET 0: reset on 1: reset off

1	R/W	0x0	CORE1_SCLK_RESET 0: reset on 1: reset off
0	R/W	0x0	CORE0_HCLK_RESET 0: reset on 1: reset off

4.5.4 SCLK_DIV

Offset: 0x00C			Register Name: SCLK_DIV
Bit	Read/Write	Default/Hex	Description
31:16	/	/	Reserved
15:8	R/W	0x0	CSR_SCLK_DIV
11:8	R/W	0x0	RT_WB_SCLK_DIV
7:4	R/W	0x0	CORE1_SCLK_DIV
3:0	R/W	0x0	CORE0_SCLK_DIV

4.5.5 DE2TCON_MUX

Offset: 0x010			Register Name: DE2TCON_MUX
Bit	Read/Write	Default/Hex	Description
31:1	/	/	Reserved
0	R/W	0x0	DE2TCON_MUX 0: MIXER0-〉 TCON0; MIXER1-〉 TCON1 1: MIXER0-〉 TCON1; MIXER1-〉 TCON0

5 .Sub_Module_Specification

5.1 DE ASE Specification

5.1.1 Register List

Module name	Memory Range	Offset Address
ASE	8K	0xA8000

Register name	Offset	Description
ASE_CTRL_REG	0x000	Global control register
ASE_SIZE_REG	0x004	ASE size register
ASE_WIN0_REG	0x008	Windows top position
ASE_WIN1_REG	0x00c	Windows bottom position
ASE_WEIGHT	0x010	ASE gain register

5.1.2 Register Description

5.1.2.1 Global Control Register

Offset: 0x000			Register Name: ASE_CTL_REG
Bit	Read/Write	Default/Hex	Description
31:2	/	/	Reserved
1	R/W	0x0	WINDOW_EN LCE window function enable 0:Disable 1:Enable Note: When window function enable, only the area inside the window will be processed.
0	R/W	0x0	ASE_EN 0: disable 1: enable

5.1.2.2 ASE size register

Offset: 0x004			Register Name: ASE_SIZE_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	Reserved
27:16	R/W	0x0	HEIGHT The real height = The value of these bits add 1

15:12	/	/	/
11:0	R/W	0x0	WIDTH The real width = The value of these bits add 1

5.1.2.3 ASE windows0 register

Offset: 0x008			Register Name: ASE_WIN0_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0	WIN_TOP Window Top position Top position is the left-top y coordinate of display window in pixels
15:12	/	/	/
11:0	R/W	0	WIN_LEFT Window Left position Left position is left-top x coordinate of display window in pixels

5.1.2.4 ASE windows1 register

Offset: 0x00C			Register Name: ASE_WIN1_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0	WIN_BOT Window Bottom position Bottom position is right-bottom y coordinate of display window in pixels
15:12	/	/	/
11:0	R/W	0	WIN_RIGHT Window Right position Right position is right-bottom x coordinate of display window in pixels

5.1.2.5 ASE gain register

Offset: 0x010			Register Name: ASE_WEIGHT
Bit	Read/Write	Default/Hex	Description
31:6	/	/	Reserved
5:0	R/W	0x0	ASE_WEIGHT

5.2 DE BWS Specification

5.2.1 Overview

The BWS(Black and White Stetch) features:

- Support data format with 8-bit pre channel.
- Support Maximum input frame size 4096×2160
- Interface: Stream-to-stream pixel interface

5.2.2 Register List

Module name	Memory Range	Offset Address
BWS	8K	0xA2000

Register name	Offset	Description
BWS_GCTRL_REG	0x000	Control register
BWS_SIZE_REG	0x004	Size setting register
BWS_WINO_REG	0x008	Window setting register0
BWS_WIN1_REG	0x00C	Window setting register1
BWS_THRO_REG	0x020	BWS threshold setting register0
BWS_THR1_REG	0x024	BWS threshold setting register1
BWS_SLP0_REG	0x028	BWS slope setting register0
BWS_SLP1_REG	0x02C	BWS slope setting register1

5.2.3 Register Description

5.2.3.1 BWS_GCTRL_REG

Offset: 0x000			Register Name: GCTRL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	WINDOW_EN BWS window function enable 0:Disable 1:Enable Note: When window function enable, only the area inside the window will be processed.
30:1	/	/	/
0	R/W	0x0	EN BWS module enable 0: Disable 1: Enable Note: When module disable, the clock of the calculation circuit will be gated

automatically.

Note: All bits in **BWS_CTRL_REG** is double-buffered.

5.2.3.2 BWS_SIZE_REG

Offset: 0x004			Register Name: BWS_SIZE_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0x0	HEIGHT Input height The actual height is register value + 1
15:12	/	/	/
11:0	R/W	0x0	WIDTH Input width The actual width is register value + 1

5.2.3.3 BWS_WIN0_REG

Offset: 0x008			Register Name: BWS_WIN0_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0	WIN_TOP Window Top position Top position is the left-top y coordinate of display window in pixels
15:12	/	/	/
11:0	R/W	0	WIN_LEFT Window Left position Left position is left-top x coordinate of display window in pixels

5.2.3.4 BWS_WIN1_REG

Offset: 0x00C			Register Name: BWS_WIN1_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0	WIN_BOT Window Bottom position Bottom position is right-bottom y coordinate of display window in pixels
15:12	/	/	/
11:0	R/W	0	WIN_RIGHT Window Right position Right position is right-bottom x coordinate of display window in pixels

5.2.3.5 BWS_THRO_REG

Offset: 0x020			Register Name: BWS_THRO_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/

23:16	R/W	0x0	BLACK Black level stretch threshold 1
15:8	/	/	/
7:0	R/W	0x0	MIN Black level stretch threshold 0

5.2.3.6 BWS_THR1_REG

Offset: 0x024			Register Name: BWS_THR1_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	MAX White level stretch threshold 1
15:8	/	/	/
7:0	R/W	0x0	WHITE White level stretch threshold 0

5.2.3.7 BWS_SLP0_REG

Offset: 0x028			Register Name: BWS_SLP0_REG
Bit	Read/Write	Default/Hex	Description
31:26	/	/	/
25:16	R/W	0x0	SLOPE1 Black level stretch slope 1
15:10	/	/	/
9:0	R/W	0x0	SLOPE0 Black level stretch slope 0

5.2.3.8 BWS_SLP1_REG

Offset: 0x02C			Register Name: BWS_SLP1_REG
Bit	Read/Write	Default/Hex	Description
31:26	/	/	/
25:16	R/W	0x0	SLOPE3 White level stretch slope 1
15:10	/	/	/
9:0	R/W	0x0	SLOPE2 White level stretch slope 0

5.3 DE CSC Specification

5.3.1 Overview

The CSC features:

- Support 33-bit data format with 8-bit pre channel.
- Interface: Stream-to-stream pixel interface

5.3.2 Register List

Module	Offset Address	Memory Range
CSC	0x011F0000	8K

Register name	Offset	Description
CSC_BYPASS_REG	0x000	CSC bypass setting register
CSC_COEFF0_REG	0x010, 0x014, 0x018	CSC CH0 coefficients registers
CSC_CONST0_REG	0x01C	CSC CH0 constant register
CSC_COEFF1_REG	0x020, 0x024, 0x028	CSC CH1 coefficients registers
CSC_CONST1_REG	0x02C	CSC CH1 constant register
CSC_COEFF2_REG	0x030, 0x034, 0x038	CSC CH2 coefficients registers
CSC_CONST2_REG	0x03C	CSC CH2 constant register
GLB_ALPHA_REG	0x040	Direct output alpha value register

5.3.3 Register Description

5.3.3.1 CSC_BYPASS_REG

Offset: 0x00			Register Name: CSC_BYPASS_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	CSC_BYPASS CSC bypass 0: Bypass 1: Not bypass

5.3.3.2 CSC_COEFF0_REG

Offset:	Register Name: CSC_COEFF0_REG
C00 component: 0x10 C01 component: 0x14	

C02 component: 0x18			
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0	COEFF The value equals to coefficient*2 ¹⁰

5.3.3.3 CSC_CONST0_REG

Offset: 0x1C			Register Name: CSC_CONST0_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	CONST The value equals to coefficient*2 ¹⁰

5.3.3.4 CSC_COEFF1_REG

Offset: C10 component: 0x20 C11 component: 0x24 C12 component: 0x28			Register Name: CSC_COEFF1_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0	COEFF The value equals to coefficient*2 ¹⁰

5.3.3.5 CSC_CONST1_REG

Offset: 0x2C			Register Name: CSC_CONST1_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	CONST The value equals to coefficient*2 ¹⁰

5.3.3.6 CSC_COEFF2_REG

Offset: C20 component: 0x30 C21 component: 0x34 C22 component: 0x38			Register Name: CSC_COEFF2_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0	COEFF The value equals to coefficient*2 ¹⁰

5.3.3.7 CSC_CONST2_REG

Offset: 0x3C			Register Name: CSC_CONST2_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	CONST The value equals to coefficient*2 ¹⁰

5.3.3.8 GLB_ALPHA_REG

Offset: 0x40			Register Name: GLB_ALPHA_REG
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0xFF	GLOBAL ALPHA The direct output global alpha value
23:0	/	/	/

5.4 DE DRC Specification

5.4.1 Overview

The dynamic range controller (DRC) is a post-processing module which adjusts the image mapping curve according to the histogram frame by frame. The control function can be defined by the software driver according to the application. A typical application is Content-based backlight control.

Feature:

- Support data format with 8-bit per channel
- Support 2048*2048 input/output
- Support HISTOGRAM and DRC in HSV color space

5.4.2 Register List

Module	Offset Address	Memory Range
DRC	0x011b0000	64K

Register Name	Offset	Description
GNECTL_REG	0x0000	Module general control register
DRC_SIZE_REG	0x0004	DRC size setting register
DRC_CTL_REG	0x0010	DRC control register
DRC_SET_REG	0x0018	DRC setting register
DRC_WP0_REG	0x001C	DRC window position register0
DRC_WP1_REG	0x0020	DRC window position register1
LH_CTL_REG	0x0030	Luminance histogram control register
LH_THR0_REG	0x0034	Luminance histogram threshold setting register0
LH_THR1_REG	0x0038	Luminance histogram threshold setting register1
LH_SLUMN_REG	0x0040 + N*4 (N=0 ~ 7)	Luminance histogram statistics luminance recording register
LH_SCNTN_REG	0x0060 + N*4 (N=0 ~ 7)	Luminance histogram statistics counter recording register
CSC_C00_REG	0x00C0	CSC coefficient 00 register
CSC_C01_REG	0x00C4	CSC coefficient 01 register
CSC_C02_REG	0x00C8	CSC coefficient 02 register
CSC_C03_REG	0x00CC	CSC constant 03 register
CSC_C10_REG	0x00D0	CSC coefficient 10 register
CSC_C11_REG	0x00D4	CSC coefficient 11 register
CSC_C12_REG	0x00D8	CSC coefficient 12 register
CSC_C13_REG	0x00DC	CSC constant 13 register
CSC_C20_REG	0x00E0	CSC coefficient 20 register

CSC_C21_REG	0x00E4	CSC coefficient 21 register
CSC_C22_REG	0x00E8	CSC coefficient 22 register
CSC_C23_REG	0x00EC	CSC constant 23 register
DRC_SPACOFF_REGN	0x0F0 + N*4 (N=0,1,2)	DRC spatial coefficient registers
DRC_INTCOFF_REGN	0x100 + N*4 (N=0 ~ 63)	DRC intensity coefficient registers
DRC_LGCOFF_REGN	0x200 + N*4 (N=0 ~ 127)	DRC Luminance gain coefficient registers

5.4.3 Register Description

5.4.3.1 General control register (Default Value: 0x0000_0000)

Offset: 0x0000			Register Name: GNECTL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	BIST_EN BIST enable 0x0: disable 0x1: enable
30:10	/	/	/
09:08	R/W	0x0	MOD Work mode selection. If bit 0 of the register is set ZERO, the following setting will be ignored. 0x0: reserved 0x1: reserved 0x2: DRC mode 0x3: reserved
07:05	/	/	/
04	R/W	0x0	COEF_SWITCH_EN 0x0: DONOT switch 0x1: Switch RAM use external RDY_EN Note: When LCD SYNC negative edge comes and COEF_SWITCH_EN is 1, DRC_LGCOFF_REGN will switch to the latest updated RAM if external RDY_EN equal to 1, and then the bit will also be self-cleared if switch action successes.
03:02	/	/	/
01	R/W	0x0	Reserved Note: Must set to 0.
00	R/W	0x0	DRC_EN DRC module enable 0x0: disabled 0x1: enable

			Note: When module disable, the input data will be bypassed to output, and the clock of calculation circuit will be gated automatically.
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5.4.3.2 DRC size setting register (Default Value: 0x0000_0000)

Offset: 0x0004			Register Name: DRC_SIZE_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	DRC_HEIGHT Display height The real display height = The value of these bits + 1. Note: Must set to 7~2047 when module enable.
15:13	/	/	/
12:00	R/W	0x0	DRC_WIDTH Display width The real display width = The value of these bits + 1.

5.4.3.3 DRC control register (Default Value: 0x0000_0000)

Offset: 0x0010			Register Name: DRC_CTL_REG
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/
09	R/W	0x0	HSV_MODE_EN Enable Using component V calculated from three source components to process Luma Histogram and DRC. 0x0: disable 0x1: enable Note: Must set to 1.
08	R/W	0x0	DRC_WIN_EN Output window function enable 0x0: disable 0x1: enable
07:01	/	/	/
00	R/W	0x0	DRC_DB_EN DRC double buffer function enable control 0x0: disable 0x1: enable

5.4.3.4 DRC setting register (Default Value: 0x0000_0000)

Offset: 0x0018			Register Name: DRC_SET_REG
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:08	R/W	0x0	DRC_LGC_ABSLUMPERVAL Abs luminance percent value

07:02	/	/	/
01	R/W	0x0	DRC_ADJUST_EN DRC adjust enable 0x0: disable 0x1: enable
00	R/W	0x0	DRC_LGC_ABSLUMSHF Abs luminance shift bits 0x0: shift 8bits 0x1: shift 9bits

5.4.3.5 DRC window position 0 register (Default Value: 0x0000_0000)

Offset: 0x001C			Register Name: DRC_WP0_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0x0	DRC_WIN_TOP Window Top position Top position is the left-top y coordinate of display window in pixels
15:12	/	/	/
11:00	R/W	0x0	DRC_WIN_LEFT Window Left position Left position is left-top x coordinate of display window in pixels

5.4.3.6 DRC window position 1 register (Default Value: 0x0000_0000)

Offset: 0x0020			Register Name: DRC_WP1_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0x0	DRC_WIN_BOT Window Bottom position Bottom position is the right-bottom y coordinate of display window in pixels
15:12	/	/	/
11:0	R/W	0x0	DRC_WIN_RIGHT Window Right position Right position is the right-bottom x coordinate of display window in pixels

5.4.3.7 Luminance histogram control register (Default Value: 0x0000_0000)

Offset: 0x0030			Register Name: LH_CTL_REG
Bit	Read/Write	Default/Hex	Description
31:02	/	/	/
1	R/W	0x0	LH_MOD 0: Current frame case 1: Average case
0	R/W	0x0	LH_REC_CLR

			If the bit is set, the all of the luminance statistics recording registers will be clear, and the bit will self-clear when the recording registers is clear done.
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5.4.3.8 Luminance histogram threshold setting 0 register (Default Value: 0x8060_4020)

Offset: 0x0034			Register Name: LH_THR0_REG
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x80	LH_THRES_VAL4 Step4 threshold value
23:16	R/W	0x60	LH_THRES_VAL3 Step3 threshold value
15:8	R/W	0x40	LH_THRES_VAL2 Step2 threshold value
7:0	R/W	0x20	LH_THRES_VAL1 Step1 threshold value

5.4.3.9 Luminance histogram threshold setting 1 register (Default Value: 0x00E0_C0A0)

Offset: 0x0038			Register Name: LH_THR1_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0xE0	LH_THRES_VAL7 Step7 threshold value
15:08	R/W	0xC0	LH_THRES_VAL6 Step6 threshold value
07:00	R/W	0xA0	LH_THRES_VAL5 Step5 threshold value

Note: When setting LHT_REG0 and LHT_REG1, make sure that THRES_VAL1<THRES_VAL2<...<THRES_VAL7.

5.4.3.10 Luminance histogram statistics lum recording register N (N = 0~7) (Default Value: 0x0000_0000)

Offset: 0x0040 + N*4			Register Name: LH_SLUM_REGN
Bit	Read/Write	Default/Hex	Description
31:00	R/W	0x0	LH_LUM_DATA Luminance statistics data

5.4.3.11 Luminance histogram statistics counter recording register N (N = 0~7) (Default Value: 0x0000_0000)

Offset: 0x0060 + N*4			Register Name: LH_SCNT_REGN
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:00	R/W	0x0	LH_CNT_DATA Luminance statistics data

5.4.3.12 CSC coefficient 00 register (Default Value: 0x0000_04A7)

Offset: 0x00C0			Register Name: CSC_C00_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x04A7	C00 Note: must set to 0x0400.

5.4.3.13 CSC coefficient 01 register (Default Value: 0x0000_1E6F)

Offset: 0x00C4			Register Name: CSC_C01_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x1E6F	C01 Note: must set to 0x0000.

5.4.3.14 CSC coefficient 02 register (Default Value: 0x0000_1CBF)

Offset: 0x00C8			Register Name: CSC_C02_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x1CBF	C02 Note: must set to 0x0000.

5.4.3.15 CSC constant 03 register (Default Value: 0x0000_0877)

Offset: 0x00CC			Register Name: CSC_C03_REG
Bit	Read/Write	Default/Hex	Description
31:14	/	/	/
13:0	R/W	0x0877	C03 Note: must set to 0x0000.

5.4.3.16 CSC coefficient 10 register (Default Value: 0x0000_04A7)

Offset: 0x00D0			Register Name: CSC_C10_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x04A7	C10 Note: must set to 0x0000.

5.4.3.17 CSC coefficient 11 register (Default Value: 0x0000_0000)

Offset: 0x00D4			Register Name: CSC_C11_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0000	C11

			Note: must set to 0x0400.
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5.4.3.18 CSC coefficient 12 register (Default Value: 0x0000_0662)

Offset: 0x00D8			Register Name: CSC_C12_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0662	C12 Note: must set to 0x0000.

5.4.3.19 CSC constant 13 register (Default Value: 0x0000_3211)

Offset: 0x00DC			Register Name: CSC_C13_REG
Bit	Read/Write	Default/Hex	Description
31:14	/	/	/
13:0	R/W	0x3211	C13 Note: must set to 0x0000.

5.4.3.20 CSC coefficient 20 register (Default Value: 0x0000_04A7)

Offset: 0x00E0			Register Name: CSC_C20_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x04A7	C20 Note: must set to 0x0000.

5.4.3.21 CSC coefficient 21 register (Default Value: 0x0000_0812)

Offset: 0x00E4			Register Name: CSC_C21_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0812	C21 Note: must set to 0x0000.

5.4.3.22 CSC coefficient 22 register (Default Value: 0x0000_0000)

Offset: 0x00E8			Register Name: CSC_C22_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0000	C22 Note: must set to 0x0400.

5.4.3.23 CSC constant 23 register (Default Value: 0x0000_2EB1)

Offset: 0x00EC			Register Name: CSC_C23_REG
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Bit	Read/Write	Default/Hex	Description
31:14	/	/	/
13:0	R/W	0x2EB1	C23 Note: must set to 0x0000.

5.4.3.24 DRC spatial coefficient register N (N=0~2) (Default Value: 0x0000_0000)

Offset: 0x00F0 + N*4			Register Name: DRC_SPACOFF_REGN
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	SPA_COEF2 8 bits unsigned spatial coefficient data
15:08	R/W	0x0	SPA_COEF1 8 bits unsigned spatial coefficient data
07:00	R/W	0x0	SPA_COEF0 8 bits unsigned spatial coefficient data

5.4.3.25 DRC intensity coefficient register N (N=0~63) (Default Value: 0x0000_0000)

Offset: 0x0100 + N*4			Register Name: DRC_INTCOFF_REGN
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	INT_COEF3 8 bits unsigned intensity coefficient data
23:16	R/W	0x0	INT_COEF2 8 bits unsigned intensity coefficient data
15:08	R/W	0x0	INT_COEF1 8 bits unsigned intensity coefficient data
07:00	R/W	0x0	INT_COEF0 8 bits unsigned intensity coefficient data

5.4.3.26 DRC luminance gain coefficient register N (N=0~127)

Offset: 0x0200 + N*4			Register Name: DRC_LGCOFF_REGN
Bit	Read/Write	Default/Hex	Description
31:16	R/W	UDF	LGC_COEF1 16bits luminance gain coefficient, unsigned data The high 5 bits is the integer part The low 11 bits is the decimal part
15:00	R/W	UDF	LGC_COEF0 16bits luminance gain coefficient, unsigned data The high 5 bits is the integer part The low 11 bits is the decimal part

Note: Double-buffered registers. When SYNC negative edge comes and COEF_SWITCH_EN is 1, coefficient RAM will switch to the latest updated RAM if external RDY_EN equal to 1.

5.5 DE FCC Specification

5.5.1 Overview

Fancy color curvature change (FCC) is to adjust fancy colors so that a better vivid vision effect can be achieved. The FCC has the following features:

- Support RGB888 input and output format
- Support window clipping up to 8192x8192 pixels
- Support local adjustment for hue/ saturation in HSV space
- Support red/green/blue/cyan/magenta/yellow areas modifying in local adjustment mode

5.5.2 Register List

Module name	Memory Range	Offset Address
FCC	8K	0xAA000

Register name	Offset	Description
FCC_CTL_REG	0x000	FCC control register
FCC_INPUT_SIZE_REG	0x004	FCC input size register
FCC_OUTPUT_WIN0_REG	0x008	FCC output window setting register0
FCC_OUTPUT_WIN1_REG	0x00c	FCC output window setting register1
FCC_HUE_RANGE_REG	0x010 - 0x024	FCC hue range register
FCC_LOCAL_GAIN_REG	0x030 - 0x044	FCC local gain register
INCSC_BYPASS_REG	0x050	
INCSC_COEFF0_REG	0x060 - 0x068	
INCSC_CONST0_REG	0x06c	
INCSC_COEFF1_REG	0x070 - 0x078	
INCSC_CONST1_REG	0x07c	
INCSC_COEFF2_REG	0x080 - 0x088	
INCSC_CONST2_REG	0x08c	
GLB_ALPHA_REG	0x90	

5.5.3 Register Description

5.5.3.1 FCC_CTRL_REG(Default Value: 0x0000_0000)

Offset: 0x000			Register Name: FCC_CTL_REG
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8	R/W	0x0	WIN_EN Output window function enable 0: disable

			1: enable
7:1	/	/	/
0	R/W	0x0	Enable Enable control 0:disable 1:enable If the bit is disabled, the input data will by-pass to next module.

5.5.3.2 FCC_INPUT_SIZE_REG(Default Value: 0x0000_0000)

Offset: 0x004			Register Name: FCC_INPUT_SIZE_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0	HEIGHT Processing height The real display height = The value of these bits + 1.
15:13	/	/	/
12:00	R/W	0	WIDTH Processing width The real display width = The value of these bits + 1.

5.5.3.3 FCC_OUTPUT_WIN0_REG(Default Value: 0x0000_0000)

Offset: 0x008			Register Name: FCC_OUTPUT_WIN0_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0	WIN_TOP Window Top position Top position is the left-top y coordinate of display window in pixels
15:13	/	/	/
12:00	R/W	0	WIN_LEFT Window Left position Left position is left-top x coordinate of display window in pixels

5.5.3.4 FCC_OUTPUT_WIN1_REG(Default Value: 0x0000_0000)

Offset: 0x00C			Register Name: FCC_OUTPUT_WIN1_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0	WIN_BOT Window Bottom position Bottom position is right-bottom y coordinate of display window in pixels
15:12	/	/	/
12:00	R/W	0	WIN_RIGHT Window Right position Right position is right-bottom x coordinate of display window in pixels

5.5.3.5 FCC_HUE_RANGE_REG(Default Value: 0x0000_0000)

Offset: H/R component: 0x010 H/G component: 0x014 H/B component: 0x018 H/C component: 0x01c H/M component: 0x020 H/Y component: 0x024			Register Name: FCC_HUE_RANGE_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0	HMAX the max value of hue
15:12	/	/	/
11:0	R/W	0	HMIN the min value of hue

5.5.3.6 FCC_LOCAL_GAIN_REG

Offset: H/R component: 0x030 H/G component: 0x034 H/B component: 0x038 H/C component: 0x03c H/M component: 0x040 H/Y component: 0x044			Register Name: FCC_LOCAL_HGAIN_REG
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24:16	R/W	0	HGAIN the local gain of hue(Hgain),this is a sign number.
15:9	/	/	/
8:0	R/W	0	SGAIN the local gain of Saturation (Sgain),this is a sign number.

5.5.3.7 INCSC_BYPASS_REG

Offset: 0x050			Register Name: INCSC_BYPASS_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	CSC_BYPASS CSC bypass 0: Bypass 1: Not bypass

5.5.3.8 INCSC_COEFF0_REG(Default Value: 0x0000_0000)

Offset: C00 component: 0x060 C01 component: 0x064 C02 component: 0x068			Register Name: INCSC_COEFF0_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0	COEFF The value equals to coefficient*2 ¹⁰

5.5.3.9 INCSC_CONST0_REG(Default Value: 0x0000_0000)

Offset: C03 component: 0x06C			Register Name: INCSC_CONST0_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	CONST The value equals to coefficient*2 ¹⁰

5.5.3.10 INCSC_COEFF1_REG(Default Value: 0x0000_0000)

Offset: C10 component: 0x070 C11 component: 0x074 C12 component: 0x078			Register Name: INCSC_COEFF1_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0	COEFF The value equals to coefficient*2 ¹⁰

5.5.3.11 INCSC_CONST1_REG(Default Value: 0x0000_0000)

Offset: C13 component: 0x07C			Register Name: INCSC_CONST1_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	CONST The value equals to coefficient*2 ¹⁰

5.5.3.12 INCSC_COEFF2_REG(Default Value: 0x0000_0000)

Offset: C20 component: 0x080 C21 component: 0x084 C22 component: 0x088			Register Name: INCSC_COEFF2_REG
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Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0	COEFF The value equals to coefficient*2 ¹⁰

5.5.3.13 INCSC_CONST2_REG(Default Value: 0x0000_0000)

Offset: C23 component: 0x08C			Register Name: INCSC_CONST2_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	CONST The value equals to coefficient*2 ¹⁰

5.5.3.14 GLB_ALPHA_REG

Offset: 0x090			Register Name: GLB_ALPHA_REG
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0xFF	GLOBAL ALPHA The global alpha value for video channel
23:0	/	/	/

5.6 DE FCE Specification

5.6.1 Overview

The FCE(Fresh and Contrast Enhancement) features:

- Support data format with 8-bit pre channel.
- Support Maximum input frame size 4096×2160
- Interface: Stream-to-stream pixel interface

5.6.2 Register List

Module name	Memory Range	Offset Address
FCE	8K	0xA0000

Register name	Offset	Description
GCTRL_REG	0x000	Control register
FCE_SIZE_REG	0x004	Size setting register
FCE_WIN0_REG	0x008	Window setting register0
FCE_WIN1_REG	0x00C	Window setting register1
LCE_GAIN_REG	0x010	LCE gain setting register
HIST_SUM_REG	0x020	Histogram sum register
HIST_STATUS_REG	0x024	Histogram status register
CE_STATUS_REG	0x028	CE LUT status register
FTC_GAIN_REG	0x030	FTC gain setting register
FCE_INCSC_BYPASS_REG	0x040	Input CSC bypass setting register
CE_LUT_REGN	0x100+N*4	CE LUT register N (N=0:63)
HIST_CNT_REGN	0x200+N*4	Histogram count register N (N=0:255)

5.6.3 Register Description

5.6.3.1 GCTRL_REG

Offset: 0x000			Register Name: GCTRL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	WINDOW_EN LCE/CE/FTD/FTC window function enable 0:Disable 1:Enable Note: When window function enable, only the area inside the window will be processed. Histogram function will be not effected by window function. Note: double-buffered register.

30:21	/	/	/
20	R/W	0x0	FTC_EN FTC function enable 0: Disable 1: Enable Note: double-buffered register.
19	R/W	0x0	FTD_EN FTD function enable 0: Disable 1: Enable Note: double-buffered register.
18	R/W	0x0	LCE_EN LCE function enable 0: Disable 1: Enable Note: double-buffered register.
17	R/W	0x0	CE_EN CE function enable 0: Disable 1: Enable Note: double-buffered register.
16	R/W	0x0	HIST_EN Histogram function enable 0: Disable 1: Enable Note: double-buffered register.
15:1	/	/	/
0	R/W	0x0	EN FCE module enable 0: Disable 1: Enable Note: When module disable, the clock of the calculation circuit will be gated automatically.

5.6.3.2 FCE_SIZE_REG

Offset: 0x004			Register Name: FCE_SIZE_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0x0	HEIGHT Input height The actual height is register value + 1
15:12	/	/	/
11:0	R/W	0x0	WIDTH Input width The actual width is register value + 1

5.6.3.3 FCE_WIN0_REG

Offset: 0x008			Register Name: FCE_WIN0_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0	WIN_TOP Window Top position Top position is the left-top y coordinate of display window in pixels
15:12	/	/	/
11:0	R/W	0	WIN_LEFT Window Left position Left position is left-top x coordinate of display window in pixels

5.6.3.4 FCE_WIN1_REG

Offset: 0x00C			Register Name: FCE_WIN1_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0	WIN_BOT Window Bottom position Bottom position is right-bottom y coordinate of display window in pixels
15:12	/	/	/
11:0	R/W	0	WIN_RIGHT Window Right position Right position is right-bottom x coordinate of display window in pixels

5.6.3.5 LCE_GAIN_REG

Offset: 0x010			Register Name: LCE_GAIN_REG
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:8	R/W	0x0	LCE_BLEND Blending ratio of C2 and SOURCE Note: U0.8 format. Range from 0 to (255/256).
7:6	/	/	/
5:0	R/W	0x0	LCE_GAIN LCE gain Note: U2.4 format. Range from 0 to (63/16).

5.6.3.6 HIST_SUM_REG

Offset: 0x020			Register Name: HIST_SUM_REG
Bit	Read/Write	Default/Hex	Description
31:0	R	0x0	SUM Pixel level sum

			Note: Register will clear when reset module. And double-buffered register refresh when pixel counter equal to WIDTH × HEIGHT.
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5.6.3.7 HIST_STATUS_REG

Offset: 0x024			Register Name: HIST_STATUS_REG
Bit	Read/Write	Default/Hex	Description
31:8	R	0x0	HIST_NUM Histogram pixel number counter
7:2	/	/	/
1	R/W	0x0	BIST_EN BIST enable 0: Disable 1: Enable
0	R	0x0	HIST_CNT_VALID 0: AHB access not valid 1: AHB access valid Note: This bit switch to 0 when HIST_CNT_REG are updated.

5.6.3.8 CE_STATUS_REG

Offset: 0x028			Register Name: CE_STATUS_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	CELUT_ACCESS_SWITCH 0: Module access 1: AHB access Note: When module access, CELUT registers can't access through AHB bus. When AHB access, CELUT will return the input address for data output.

5.6.3.9 FTC_GAIN_REG

Offset: 0x030			Register Name: FTC_GAIN_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0	FTC_H_GAIN_2 FTC gain value2
15:8	/	/	/
7:0	R/W	0	FTC_H_GAIN_1 FTC gain value1

5.6.3.10 FCE_INCSC_BYPASS_REG

Offset: 0x040			Register Name: FCE_INCSC_BYPASS_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/

0	R/W	0x0	CSC_BYPASS CSC bypass 0: Bypass 1: Not bypass
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5.6.3.11 CE_LUT_REGN (N = 0:63)

Offset: 0x100 + N*4			Register Name: CE_LUT_REGN
Bit	Read/Write	Default/Hex	Description
31:24	R/W	UDF	CE_LUT3 (N*4+3) level LUT
23:16	R/W	UDF	CE_LUT2 (N*4+2) level LUT
15:8	R/W	UDF	CE_LUT1 (N*4+1) level LUT
7:0	R/W	UDF	CE_LUT0 (N*4) level LUT

Note: Registers can access in burst mode.

5.6.3.12 HIST_CNT_REGN (N = 0:255)

Offset: 0x200 + N*4			Register Name: HIST_CNT_REGN
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/
21:0	R/W	UDF	HIST Level N counter. Note: 1.Registers switch to AHB bus access mode when HIST_CNT_VALID = 1. When HIST_CNT_VALID = 0, accessing registers will return 0x00000000. 2.Registers will be clear to zero when LCD SYNC positive edge comes. 3.Registers can access in burst mode.

5.7 DE UIS Specification

5.7.1 Overview

UI Scaler(UIS) provides RGB format image resizing function for display engine. It receives data from overlay module, performs the image resizing function, and outputs to routing modules.

The UIS can receive ARGB8888 data format, and then converts to a required size ARGB8888 image for display. Horizontal and vertical direction scaling are implemented independently.

The US features:

- Support ARGB8888 data format with 8-bit pre channel.
- Support input and output size from 4×2 to 4096×4096
- Support 1/16× to 32× resize ratio.
- Support 16-phase 4-tap horizontal anti-alias filter, 16-phase linear filter in vertical.
- Point-to-point display size up to **W** pixels/line. (**W** is RTL programmable, default value 2048)

5.7.2 Register List

Register name	Offset	Description
UIS_CTRL_REG	0x000	Control register
UIS_STATUS_REG	0x008	Status register
UIS_FIELD_CTRL_REG	0x00C	Field control register
UIS_BIST_REG	0x010	BIST control register
UIS_OUTSIZE_REG	0x040	Output size register
UIS_INSIZE_REG	0x080	Input size register
UIS_HSTEP_REG	0x088	Horizontal step register
UIS_VSTEP_REG	0x08C	Vertical step register
UIS_HPHASE_REG	0x090	Horizontal initial phase register
UIS_VPHASE0_REG	0x098	Vertical initial phase 0 register
UIS_VPHASE1_REG	0x09C	Vertical initial phase 1 register
UIS_HCOEF_REGN	0x200+N*4	Horizontal filter coefficient register N (N=0:15)

Note: All registers except some bits in **UIS_CTRL_REG**, **UIS_FIELD_CTRL_REG**, **UIS_STATUS_REG** are double-buffered refreshed by **REG_RDY**.

5.7.3 Register Description

5.7.3.1 UIS_CTRL_REG

Offset: 0x000			Register Name: UIS_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	BIST_EN

			BIST enable 0: Disable 1: Enable
30	R/W	0x0	CORE_RST Core circuit reset 0: Do noting 1: Reset core circuit
29:5	/	/	/
4	R/W	0x0	COEF_SWITCH_RDY Coefficients RAM switch 0: DONOT switch 1: Switch RAM use REG_RDY Note: When LCD_SYNC go low and COEF_SWITCH_RDY is 1, coefficient RAM will switch to the latest updated RAM if REG_RDY is 1, and then the bit will also be self-cleared if switch action successes.
3:1	/	/	/
0	R/W	0x0	EN UI Scaler enable 0: Disable 1: Enable Note: When module disabled, the core clock to the core circuit will be gated, and the input data will be bypassed to down-stream module.

Note: Only bit **EN** is double-buffered.

5.7.3.2 UIS_STATUS_REG

Offset: 0x008			Register Name: UIS_STATUS_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R	0x0	LINE_CNT Output line number
15:5	/	/	/
4	R	0x0	BUSY Core circuit status 0: idle (finish, module disable, waiting for LCD_SYNC negative edge) 1: busy (core circuit calculating)
3:0	/	/	/

Note: Whole WORD is non-double-buffered.

5.7.3.3 UIS_FIELD_CTRL_REG

Offset: 0x00C			Register Name: UIS_FIELD_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5	R/W	0x0	LCD_SYNC_REVERSE Reverse LCD_SYNC 0: DONOT reverse 1: Reverse

4	R/W	0x0	LCD_FILED_REVERSE Reverse LCD FILED 0: DONOT reverse 1: Reverse
3:1	/	/	/
0	R/W	0x0	FIELD_SEL_VPHASE_EN Vertical initial phase switch control 0: Vertical initial phase fix to phase0 1: Switch Vertical initial phase by LCD FIELD (Switch to phase0 when LCD FILED is 1, and switch to phase1 when LCD FIELD is 0)

Note: Only bit **FIELD_SEL_VPHASE_EN** is double-buffered.

5.7.3.4 UIS_BIST_REG

Offset: 0x010			Register Name: UIS_BIST_REG
Bit	Read/Write	Default/Hex	Description
31:21	/	/	/
20:16	W	0x0	BIST_RAM_SEL Select the index of RAM for BIST
15:1	/	/	/
0	W	0x0	BIST_EN BIST enable 0: Disable 1: Enable

Note: Whole WORD is non-double-buffered.

5.7.3.5 UIS_OUTSIZE_REG

Offset: 0x040			Register Name: UIS_OUTSIZE_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	OUT_HEIGHT Output height The actual height is register value + 1
15:13	/	/	/
12:0	R/W	0x0	OUT_WIDTH Output width The actual width is register value + 1

5.7.3.6 UIS_INSIZE_REG

Offset: 0x080			Register Name: UIS_INSIZE_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	IN_HEIGHT Input height The actual height is register value + 1
15:13	/	/	/

12:0	R/W	0x0	IN_WIDTH Input width The actual width is register value + 1
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5.7.3.7 UIS_HSTEP_REG

Offset: 0x088			Register Name: UIS_HSTEP_REG
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24:20	R/W	0x0	HSTEP_INT The integer part of horizontal scale ratio
19:2	R/W	0x0	HSTEP_FRAC The fraction part of horizontal scale ratio
1:0	/	/	/

5.7.3.8 UIS_VSTEP_REG

Offset: 0x08C			Register Name: UIS_VSTEP_REG
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24:20	R/W	0x0	VSTEP_INT The integer part of vertical scale ratio
19:2	R/W	0x0	VSTEP_FRAC The fraction part of vertical scale ratio
1:0	/	/	/

5.7.3.9 UIS_HPHASE_REG

Offset: 0x090			Register Name: UIS_HPHASE_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:20	R/W	0x0	HPHASE_INT The integer part of horizontal initial phase
19:2	R/W	0x0	HPHASE_FRAC The fraction part of horizontal initial phase
1:0	/	/	/

Note: HPHASE is a SIGNED parameter.

5.7.3.10 UIS_VPHASE0_REG

Offset: 0x098			Register Name: UIS_VPHASE0_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:20	R/W	0x0	VPHASE0_INT The integer part of vertical initial phase0
19:2	R/W	0x0	VPHASE0_FRAC The fraction part of vertical initial phase0

1:0	/	/	/
-----	---	---	---

Note: *VPHASE0* is a SIGNED parameter.

5.7.3.11 UIS_VPHASE1_REG

Offset: 0x09C			Register Name: UIS_VPHASE1_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:20	R/W	0x0	VPHASE1_INT The integer part of vertical initial phase1
19:2	R/W	0x0	VPHASE1_FRAC The fraction part of vertical initial phase1
1:0	/	/	/

Note: *VPHASE1* is a SIGNED parameter.

5.7.3.12 UIS_HCOEF_REGN (N = 0:15)

Offset: 0x200 + N*4			Register Name: UIS_HCOEF_REGN
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	COEF3 The most right hand-side pixel coefficient
23:16	R/W	0x0	COEF2 The right hand-side pixel coefficient
15:8	R/W	0x0	COEF1 The left hand-side pixel coefficient
7:0	R/W	0x0	COEF0 The most left hand-side pixel coefficient

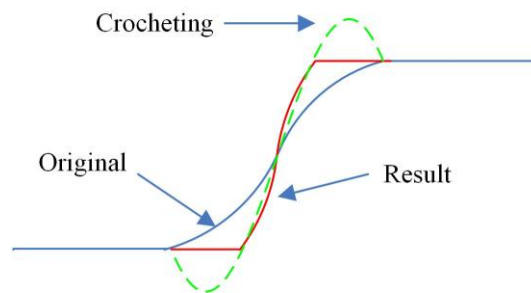
Note: HCOEFF is a two's complement. The register value equals to coefficient*2⁶. **N** represents the phase.

5.8 DE LTI Specification

5.8.1 Overview

Luminance transient improvement (DLTI) is a method to enhance the image quality . The method of the luminance signal transition bands were detected, when the luminance signal transition, so that the signal transition zone is narrow, steep edge, so as to improve the clarity of the image, the image is more clear and bright.

This algorithm uses the one-dimensional stacking hook signal method commonly used, through a hook signal superimposed on the input signal, the signal edge becomes steeper, and then calculate the signal of a window within the scope of the maximum and minimum values, will limit the signal after superposition in the minimum and maximum range, to get the final results. Sketch of the algorithm is as follows:



5.8.2 Register List

Module name	Memory Range	Offset Address
LTI	8K	0xA4000

Register name	Offset	Description
LTI_EN	0x000	Global control register
LTI_SIZE	0x00C	LTI size register
LTI_FIR_COFF0	0x010	LTI FIR filter coefficient register0
LTI_FIR_COFF1	0x014	LTI FIR filter coefficient register1
LTI_FIR_COFF2	0x018	LTI FIR filter coefficient register2
LTI_FIR_GAIN	0x01C	LTI FIR filter gain register
LTI_COR_TH	0x020	LTI coring threshold register
LTI_DIFF_CTL	0x024	LTI scaling coefficient/offset of frist derivation
LTI_EDGE_GAIN	0x028	LTI edge gain
LTI_OS_CON	0x02C	LTI coring/clipping threshold of Y shoot value
LTI_WIN_EXPANSION	0x030	LTI window range control register
LTI_EDGE_LEVEL_TH	0x034	LTI edge level threshold in edge-adaptive filtering
LTI_WIN0_REG	0x038	LTI window setting register0

LTI_WIN1_REG	0x03c	LTI window setting register1
LTI_INCSC_BYPASS_REG	0x050	Input CSC bypass setting register
LTI_INCSC_COEFF0_REG	0x060, 0x064, 0x068	Input CSC CH0 coefficients registers
LTI_INCSC_CONST0_REG	0x06C	Input CSC CH0 constant register
LTI_INCSC_COEFF1_REG	0x070, 0x074, 0x078	Input CSC CH1 coefficients registers
LTI_INCSC_CONST1_REG	0x07C	Input CSC CH1 constant register
LTI_INCSC_COEFF2_REG	0x080, 0x084, 0x088	Input CSC CH2 coefficients registers
LTI_INCSC_CONST2_REG	0x08C	Input CSC CH2 constant register
LTI_OUTCSC_BYPASS_REG	0x090	Output CSC bypass setting register
LTI_OUTCSC_COEFF0_REG	0x0A0, 0x0A4, 0x0A8	Output CSC CH0 coefficients registers
LTI_OUTCSC_CONST0_REG	0x0AC	Output CSC CH0 constant register
LTI_OUTCSC_COEFF1_REG	0x0B0, 0x0B4, 0x0B8	Output CSC CH1 coefficients registers
LTI_OUTCSC_CONST1_REG	0x0BC	Output CSC CH1 constant register
LTI_OUTCSC_COEFF2_REG	0x0C0, 0x0C4, 0x0C8	Output CSC CH2 coefficients registers
LTI_OUTCSC_CONST2_REG	0x0CC	Output CSC CH2 constant register

5.8.3 Register Description

5.8.3.1 Global control register

Offset: 0x000			Register Name: LTI_CTL
Bit	Read/Write	Default/Hex	Description
31:25	/	/	Reserved
24	R/W	0x0	WIN_EN 0: disable 1: enable
23:17	/	/	/
16	R/W	0x0	NONLINEAR_LIMIT_EN 0: disable 1: enable
15:9	/	/	/
8	R/W	0x0	SEL 0: select window range 1: select first differ
7:1	/	/	/
0	R/W	0x0	LTI_EN 0: LTI close 1: LTI open

5.8.3.2 LTI size register

Offset: 0x00C			Register Name: LTI_SIZE
Bit	Read/Write	Default/Hex	Description
31:28	/	/	Reserved
27:16	R/W	0x0	HEIGHT

			The real height = The value of these bits add 1
15:12	/	/	/
11:0	R/W	0x0	WIDTH The real width = The value of these bits add 1

5.8.3.3 LTI FIR filter coefficient register0

Offset: 0x010			Register Name: LTI_FIR_COFF0
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	C1 FIR filter coefficient C1 Note: in two's complement.
15:8	/	/	/
7:0	R/W	0x0	C0 FIR filter coefficient C0 Note: in two's complement.

5.8.3.4 LTI FIR filter coefficient register1

Offset: 0x014			Register Name: LTI_FIR_COFF1
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	C3 FIR filter coefficient C3 Note: in two's complement.
15:8	/	/	/
7:0	R/W	0x0	C2 FIR filter coefficient C2 Note: in two's complement.

5.8.3.5 LTI FIR filter coefficient register2

Offset: 0x018			Register Name: LTI_FIR_COFF2
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	C4 FIR filter coefficient C4 Note: in two's complement.

5.8.3.6 LTI FIR filter gain register

Offset: 0x01C			Register Name: LTI_FIR_GAIN
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3:0	R/W	0x0	FIR FILTER GAIN

5.8.3.7 LTI coring threshold register

Offset: 0x020			Register Name: LTI_COR_TH
Bit	Read/Write	Default/Hex	Description
31:10	/	/	/
9:0	R/W	0x0	LTI_COR_TH Coring threshold

5.8.3.8 LTI differ filter control register

Offset: 0x024			Register Name: LTI_DIFF_CTL
Bit	Read/Write	Default/Hex	Description
31:21	/	/	/
20:16	R/W	0x0	DIFF_SLOPE Differ gain slope control
15:8	/	/	/
7:0	R/W	0x0	DIFF_OFFSET Differ gain offset control

5.8.3.9 LTI adjustable gain parameter register

Offset: 0x028			Register Name: LTI_EDGE_GAIN
Bit	Read/Write	Default/Hex	Description
31:5	/	/	/
4:0	R/W	0x0	LTI_EDGE_GAIN Edge adjustable gain parameter

5.8.3.10 LTI The overshoot of control register

Offset: 0x02C			Register Name: LTI_OS_CON
Bit	Read/Write	Default/Hex	Description
31	/	/	/
30:28	R/W	0x0	PEAK_LIMIT Shifting bit for nonlinear limit
27:24	/	/	/
23:16	R/W	0x0	CLIP_Y Clipping threshold of Y shoot value
15:8	/	/	/
7:0	R/W	0x0	CORE_X Coring threshold of Y shoot value

5.8.3.11 LTI window range expansion register

Offset: 0x030			Register Name: LTI_WIN_EXPANSION
Bit	Read/Write	Default/Hex	Description

31:8	/	/	/
7:0	R/W	0x0	LTI_WIN_EXPANSION Window expansion size

5.8.3.12 LTI edge strength threshold register

Offset: 0x034			Register Name: LTI_EDGE_ELVEL_TH
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	LTI_EDGE_ELVEL_TH Edge level threshold in edge-adaptive filtering

5.8.3.13 LTI_WIN0_REG

Offset: 0x038			Register Name: LTI_EDGE_ELVEL_TH
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
27:16	R/W	0x0	WIN_TOP Window Top position
15:12	/	/	/
11:0	R/W	0x0	LTI_LEFT Window Left position

5.8.3.14 LTI_WIN1_REG

Offset: 0x03c			Register Name: LTI_EDGE_ELVEL_TH
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
27:16	R/W	0x0	WIN_BOT Window bottom position
15:12	/	/	/
11:0	R/W	0x0	LTI_WIGHT Window wright position

5.8.3.15 LTI_INCSC_BYPASS_REG

Offset: 0x50			Register Name: LTI_INCSC_BYPASS_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	CSC_BYPASS CSC bypass 0: Bypass 1: Not bypass

5.8.3.16 LTI_INCSC_COEFF0_REG

Offset: C00 component: 0x60 C01 component: 0x64 C02 component: 0x68			Register Name: LTI_INCSC_COEFF0_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0	COEFF The value equals to coefficient*2 ¹⁰

5.8.3.17 LTI_INCSC_CONST0_REG

Offset: 0x6C			Register Name: LTI_INCSC_CONST0_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	CONST The value equals to coefficient*2 ¹⁰

5.8.3.18 LTI_INCSC_COEFF1_REG

Offset: C10 component: 0x70 C11 component: 0x74 C12 component: 0x78			Register Name: LTI_INCSC_COEFF1_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0	COEFF The value equals to coefficient*2 ¹⁰

5.8.3.19 LTI_INCSC_CONST1_REG

Offset: 0x7C			Register Name: LTI_INCSC_CONST1_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	CONST The value equals to coefficient*2 ¹⁰

5.8.3.20 LTI_INCSC_COEFF2_REG

Offset: C20 component: 0x80 C21 component: 0x84 C22 component: 0x88			Register Name: LTI_INCSC_COEFF2_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/

12:0	R/W	0x0	COEFF The value equals to coefficient*2 ¹⁰
------	-----	-----	---

5.8.3.21 LTI_INCSC_CONST2_REG

Offset: 0x8C			Register Name: LTI_INCSC_CONST2_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	CONST The value equals to coefficient*2 ¹⁰

5.8.3.22 LTI_OUTCSC_BYPASS_REG

Offset: 0x90			Register Name: LTI_OUTCSC_BYPASS_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	CSC_BYPASS CSC bypass 0: Bypass 1: Not bypass

5.8.3.23 LTI_OUTCSC_COEFF0_REG

Offset: C00 component: 0xA0 C01 component: 0xA4 C02 component: 0xA8			Register Name: LTI_OUTCSC_COEFF0_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0	COEFF The value equals to coefficient*2 ¹⁰

5.8.3.24 LTI_OUTCSC_CONST0_REG

Offset: 0xAC			Register Name: LTI_OUTCSC_CONST0_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	CONST The value equals to coefficient*2 ¹⁰

5.8.3.25 LTI_OUTCSC_COEFF1_REG

Offset: C10 component: 0xB0 C11 component: 0xB4 C12 component: 0xB8			Register Name: LTI_OUTCSC_COEFF1_REG
Bit	Read/Write	Default/Hex	Description

31:13	/	/	/
12:0	R/W	0x0	COEFF The value equals to coefficient*2 ¹⁰

5.8.3.26 LTI_OUTCSC_CONST1_REG

Offset: 0xBC			Register Name: LTI_OUTCSC_CONST1_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	CONST The value equals to coefficient*2 ¹⁰

5.8.3.27 LTI_OUTCSC_COEFF2_REG

Offset: C20 component: 0xC0 C21 component: 0xC4 C22 component: 0xC8			Register Name: LTI_OUTCSC_COEFF2_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0	COEFF The value equals to coefficient*2 ¹⁰

5.8.3.28 LTI_OUTCSC_CONST2_REG

Offset: 0xCC			Register Name: LTI_OUTCSC_CONST2_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	CONST The value equals to coefficient*2 ¹⁰

5.9 DE Peaking Specification

5.9.1 Overview

The Luma Peaking module enhances mid-high frequency of luma channel in images. The chroma channels will bypass.

The Luma Peaking module including following feature:

- Maximum input frame size: 4096×2160.
- Data width: 8bit /10bit(hardware programmable) per channel.
- Interface: Stream-to-stream pixel interface.

5.9.2 Block Diagram

Figure 5-1 shows the block diagram of luma peaking module. It is a stream-to-stream module with uniform input and output interface. It contains some parts and their function list followed:

- Control logic: Status machine control, registers operation.
- Peaking calculation unit: luma peaking, chroma bypassing.
- MUX: select processed data or input data (input data pass-through works without clock)

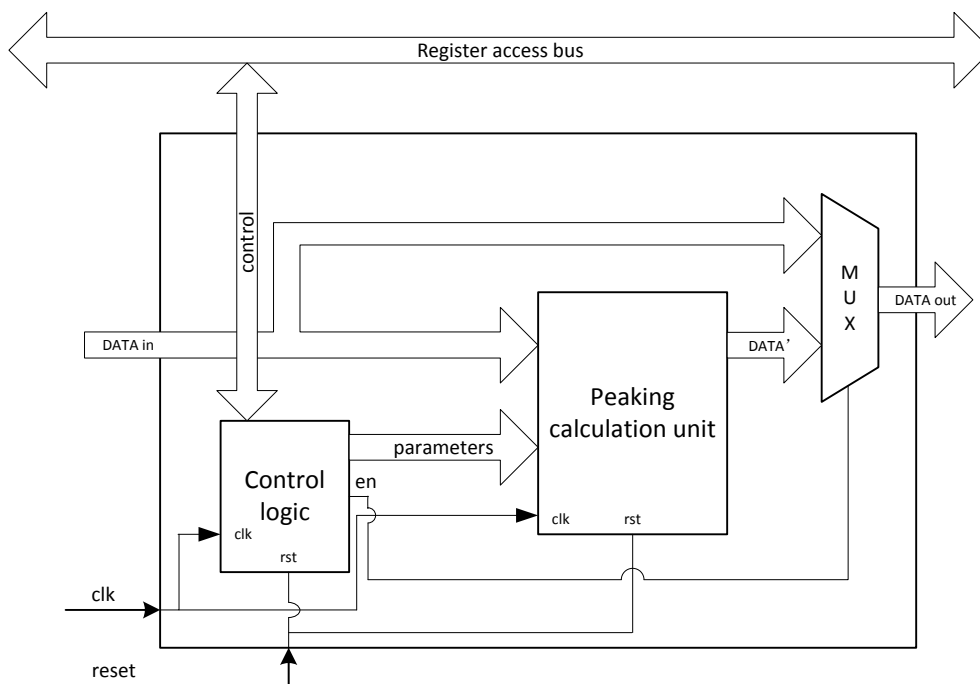


Figure 5-1. Luma peaking module block diagram

5.9.3 Register list

Module name	Memory Range	Offset Address
PEAKING	8K	0xA6000

Register Name	Offset	Description
LP_CTRL_REG	0x000	LP module control register
LP_SIZE_REG	0x004	LP size setting register
LP_WIN0_REG	0x008	LP window setting register0
LP_WIN1_REG	0x00C	LP window setting register1
LP_FILTER_REG	0x010	LP filter setting register
LP_CSTM_FILTER0_REG	0x014	LP custom filter setting register0
LP_CSTM_FILTER1_REG	0x018	LP custom filter setting register1
LP_CSTM_FILTER2_REG	0x01C	LP custom filter setting register2
LP_GAIN_REG	0x020	LP gain setting register
LP_GAINCTRL_REG	0x024	LP gain control setting register
LP_SHOOTCTRL_REG	0x028	LP shoot control setting register
LP_CORING_REG	0x02C	LP coring setting register
LP_INCSC_BYPASS_REG	0x050	Input CSC bypass setting register
LP_INCSC_COEFF0_REG	0x060, 0x064, 0x068	Input CSC CH0 coefficients registers
LP_INCSC_CONST0_REG	0x06C	Input CSC CH0 constant register
LP_INCSC_COEFF1_REG	0x070, 0x074, 0x078	Input CSC CH1 coefficients registers
LP_INCSC_CONST1_REG	0x07C	Input CSC CH1 constant register
LP_INCSC_COEFF2_REG	0x080, 0x084, 0x088	Input CSC CH2 coefficients registers
LP_INCSC_CONST2_REG	0x08C	Input CSC CH2 constant register
LP_OUTCSC_BYPASS_REG	0x090	Output CSC bypass setting register
LP_OUTCSC_COEFF0_REG	0x0A0, 0x0A4, 0x0A8	Output CSC CH0 coefficients registers
LP_OUTCSC_CONST0_REG	0x0AC	Output CSC CH0 constant register
LP_OUTCSC_COEFF1_REG	0x0B0, 0x0B4, 0x0B8	Output CSC CH1 coefficients registers
LP_OUTCSC_CONST1_REG	0x0BC	Output CSC CH1 constant register
LP_OUTCSC_COEFF2_REG	0x0C0, 0x0C4, 0x0C8	Output CSC CH2 coefficients registers
LP_OUTCSC_CONST2_REG	0x0CC	Output CSC CH2 constant register

5.9.4 Register Description

5.9.4.1 LP_CTRL_REG

Offset: 0x00			Register name: LP_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8	R/W	0x0	WIN_EN Output window function enable 0: disable

			1: enable
7:1	/	/	/
0	R/W	0x0	EN LP Module enable 0: Disable 1: Enable

5.9.4.2 LP_SIZE_REG

Offset: 0x04			Register name: LP_SIZE_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0	HEIGHT Processing height The real display height = The value of these bits + 1.
15:12	/	/	/
11:00	R/W	0	WIDTH Processing width The real display width = The value of these bits + 1.

5.9.4.3 LP_WIN0_REG

Offset: 0x08			Register name: LP_WIN0_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0	WIN_TOP Window Top position Top position is the left-top y coordinate of display window in pixels
15:12	/	/	/
11:00	R/W	0	WIN_LEFT Window Left position Left position is left-top x coordinate of display window in pixels

5.9.4.4 LP_WIN1_REG

Offset: 0x0C			Register name: LP_WIN1_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R/W	0	WIN_BOT Window Bottom position Bottom position is right-bottom y coordinate of display window in pixels
15:12	/	/	/
11:00	R/W	0	WIN_RIGHT Window Right position Right position is right-bottom x coordinate of display window in pixels

5.9.4.5 LP_FILTER_REG

Offset: 0x10			Register Name: LP_FILTER_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	FILTER_SEL Filter selection: 0: Default filter. 1: Custom filter.
30:22	/	/	/
21:16	R/W	0x0	HP_RATIO Default high-pass filter ratio Note: in two's complement.
15:14	/	/	/
13:8	R/W	0x0	BP0_RATIO Default band-pass filter0 ratio Note: in two's complement.
7:6	/	/	/
5:0	R/W	0x0	BP1_RATIO Default band-pass filter1 ratio Note: in two's complement.

5.9.4.6 LP_CSTM_FILTER0_REG

Offset: 0x14			Register Name: LP_CSTM_FILTER0_REG
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24:16	R/W	0x0	C1 Custom filter coefficient C1. Note: in two's complement.
15:9	/	/	/
8:0	R/W	0x0	C0 Custom filter coefficient C0. Note: in two's complement.

5.9.4.7 LP_CSTM_FILTER1_REG

Offset: 0x18			Register Name: LP_CSTM_FILTER1_REG
Bit	Read/Write	Default/Hex	Description
31:25	/	/	/
24:16	R/W	0x0	C3 Custom filter coefficient C3. Note: in two's complement.
15:9	/	/	/
8:0	R/W	0x0	C2

			Custom filter coefficient C2. Note: in two's complement.
--	--	--	---

5.9.4.8 LP_CSTM_FILTER2_REG

Offset: 0x1C			Register Name: LP_CSTM_FILTER2_REG
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8:0	R/W	0x0	C4 Custom filter coefficient C4. Note: in two's complement.

5.9.4.9 LP_GAIN_REG

Offset: 0x20			Register Name: LP_CSTM_FILTER0_REG
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	GAIN Peaking gain setting.

5.9.4.10 LP_GAINCTRL_REG

Offset: 0x24			Register Name: LP_GAINCTRL_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	DIF_UP Gain control: limitation threshold. Note: use low 8bits for 8bit data width.
15:8	/	/	/
4:0	R/W	0x0	BETA Gain control: large gain limitation. Note: in two's complement.

5.9.4.11 LP_SHOOTCTRL_REG

Offset: 0x28			Register Name: LP_SHOOTCTRL_REG
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5:0	R/W	0x0	NEG_GAIN Undershoot gain control.

5.9.4.12 LP_CORING_REG

Offset: 0x2C			Register Name: LP_CORING_REG
Bit	Read/Write	Default/Hex	Description

31:8	/	/	/
7:0	R/W	0x0	CORTHR Coring threshold. Note: use low 8bits for 8bit data width.

5.9.4.13 LP_INCSC_BYPASS_REG

Offset: 0x50			Register Name: LP_INCSC_BYPASS_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	CSC_BYPASS CSC bypass 0: Bypass 1: Not bypass

5.9.4.14 LP_INCSC_COEFF0_REG

Offset: C00 component: 0x60 C01 component: 0x64 C02 component: 0x68			Register Name: LP_INCSC_COEFF0_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0	COEFF The value equals to coefficient*2 ¹⁰

5.9.4.15 LP_INCSC_CONST0_REG

Offset: 0x6C			Register Name: LP_INCSC_CONST0_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	CONST The value equals to coefficient*2 ¹⁰

5.9.4.16 LP_INCSC_COEFF1_REG

Offset: C10 component: 0x70 C11 component: 0x74 C12 component: 0x78			Register Name: LP_INCSC_COEFF1_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0	COEFF The value equals to coefficient*2 ¹⁰

5.9.4.17 LP_INCSC_CONST1_REG

Offset: 0x7C			Register Name: LP_INCSC_CONST1_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	CONST The value equals to coefficient*2 ¹⁰

5.9.4.18 LP_INCSC_COEFF2_REG

Offset: C20 component: 0x80 C21 component: 0x84 C22 component: 0x88			Register Name: LP_INCSC_COEFF2_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0	COEFF The value equals to coefficient*2 ¹⁰

5.9.4.19 LP_INCSC_CONST2_REG

Offset: 0x8C			Register Name: LP_INCSC_CONST2_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	CONST The value equals to coefficient*2 ¹⁰

5.9.4.20 LP_OUTCSC_BYPASS_REG

Offset: 0x90			Register Name: LP_OUTCSC_BYPASS_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	CSC_BYPASS CSC bypass 0: Bypass 1: Not bypass

5.9.4.21 LP_OUTCSC_COEFF0_REG

Offset: C00 component: 0xA0 C01 component: 0xA4 C02 component: 0xA8			Register Name: LP_OUTCSC_COEFF0_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0	COEFF The value equals to coefficient*2 ¹⁰

5.9.4.22 LP_OUTCSC_CONST0_REG

Offset: 0xAC			Register Name: LP_OUTCSC_CONST0_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	CONST The value equals to coefficient*2 ¹⁰

5.9.4.23 LP_OUTCSC_COEFF1_REG

Offset: C10 component: 0xB0 C11 component: 0xB4 C12 component: 0xB8			Register Name: LP_OUTCSC_COEFF1_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0	COEFF The value equals to coefficient*2 ¹⁰

5.9.4.24 LP_OUTCSC_CONST1_REG

Offset: 0xBC			Register Name: LP_OUTCSC_CONST1_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	CONST The value equals to coefficient*2 ¹⁰

5.9.4.25 LP_OUTCSC_COEFF2_REG

Offset: C20 component: 0xC0 C21 component: 0xC4 C22 component: 0xC8			Register Name: LP_OUTCSC_COEFF2_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R/W	0x0	COEFF The value equals to coefficient*2 ¹⁰

5.9.4.26 LP_OUTCSC_CONST2_REG

Offset: 0xCC			Register Name: LP_OUTCSC_CONST2_REG
Bit	Read/Write	Default/Hex	Description
31:20	/	/	/
19:0	R/W	0x0	CONST The value equals to coefficient*2 ¹⁰

5.10 DE RT-MIXER Specification

5.10.1 Overview

The RT-mixer Core consist of dma, overlay, scaler and blender block. It supports 4 layers overlay in one pipe, and its result can scaler up or down to blender in the next processing. There's the feature description as follows:

- Support layer size up to 2048x204 pixels
- Support pre-multiply alpha image data
- Support four layers overlay in every pipe and four pipes alpha blending
- Support color key
- Support *Porter-Duff* alpha blending
- Support input format *YUV422/YUV420/YUV411/ARGB8888/XRGB8888/RGB888/ARGB4444/ ARGB1555 and RGB565*
- Support 3D format image data

5.10.2 Block Diagram

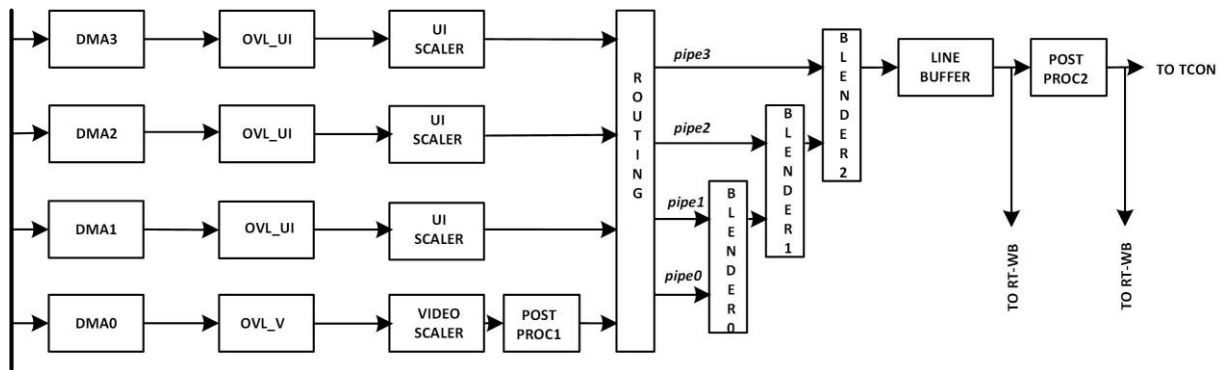


Figure 5-2. RT-Mixer General Diagram

5.10.3 DE RT-Mixer Description

5.10.3.1 Input Data Memory Layout

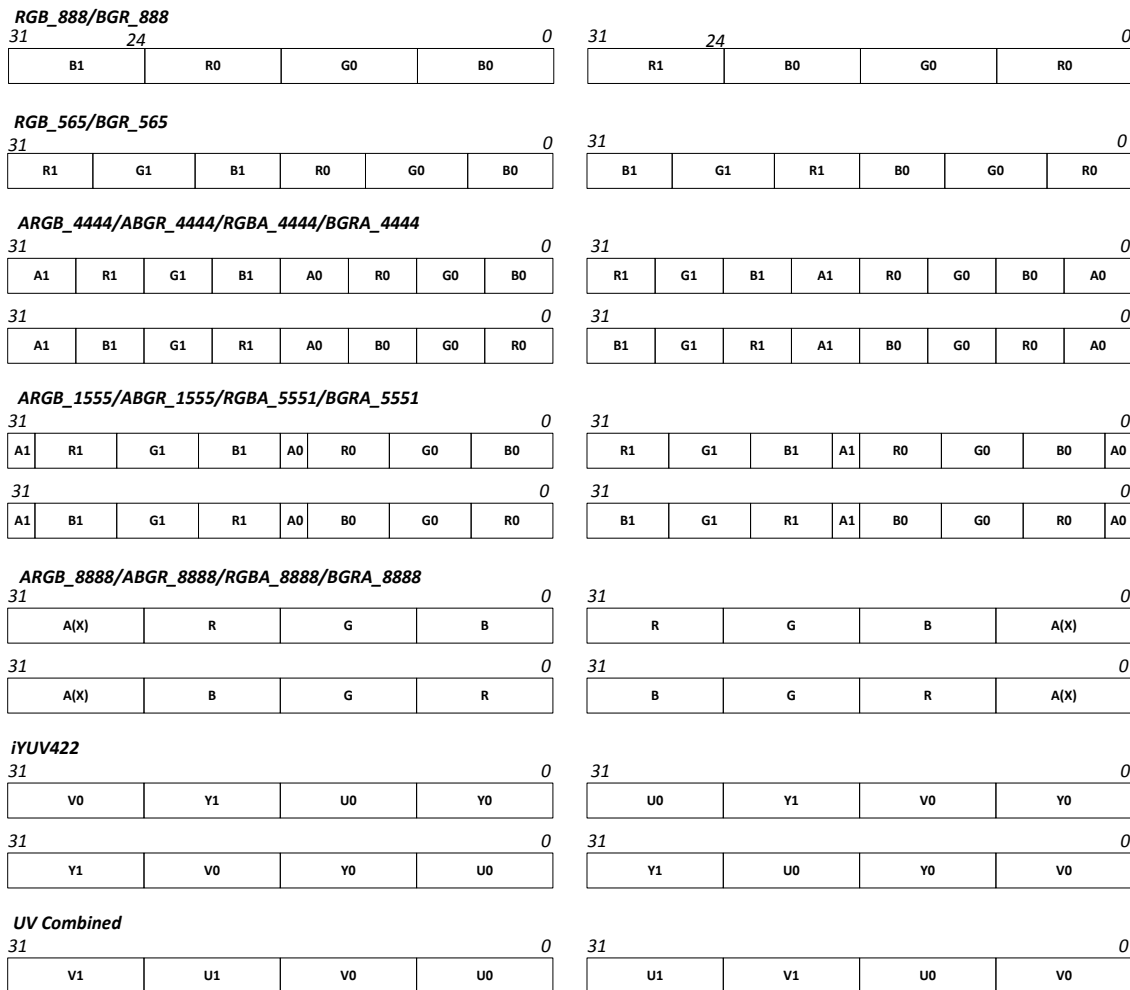


Figure 5-3. Input data pixel sequence

5.10.3.2 Overlay

Figure 5-4 is the overlay processing include layer memory data access and overlay relationship, the detail as following.

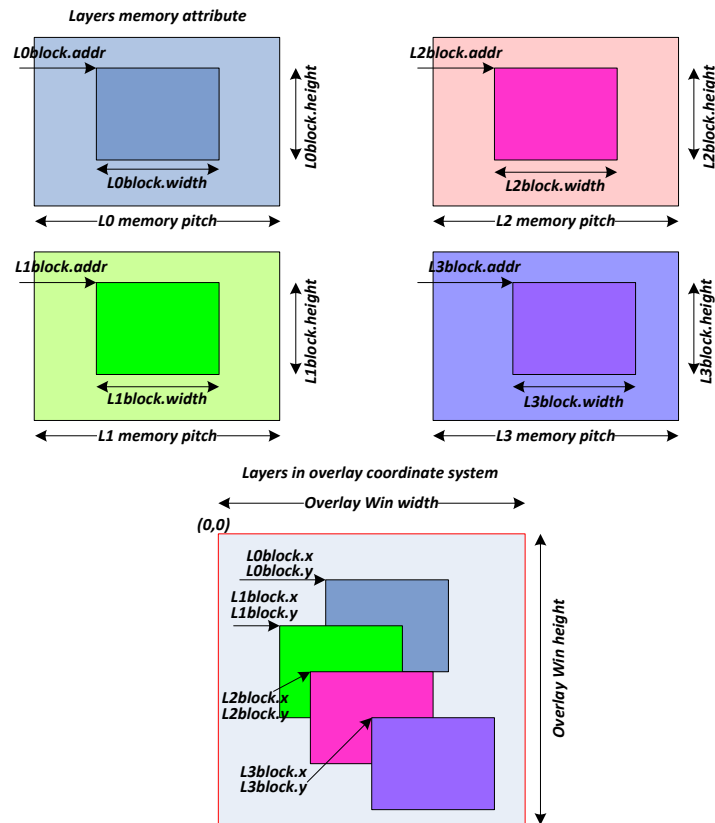


Figure 5-4. Layers memory access and overlay processing

Note: the layer priority is layer3>layer2>layer1>layer0

5.10.3.3 Scaling

UI scaler : Please refer to << Display_Engine2.0_UI_Scaler_SPEC>>

Video scaler: Please refer to << Display_Engine2.0_Video_Scaler_SPEC>>

Routing: The routing connects N channels and N input of blending together. Any channel can connect to any pipe. Programmer should make sure every pipe should connect one different channel only.

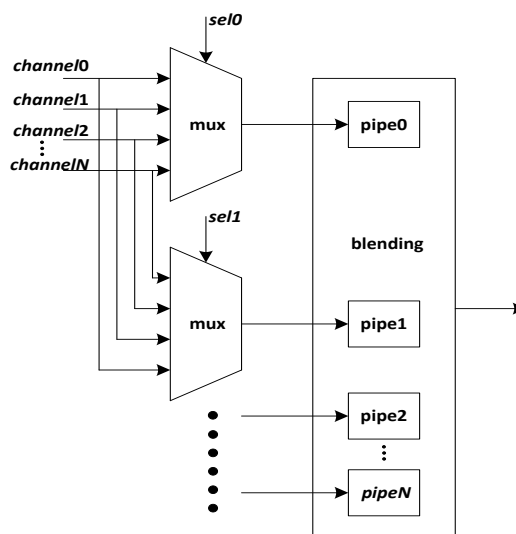


Figure 5-5. N Channel select pipe route processing

5.10.3.4 Blender

Blender input data storing:

The following diagram is about overlay data remapping in the blender pipe.

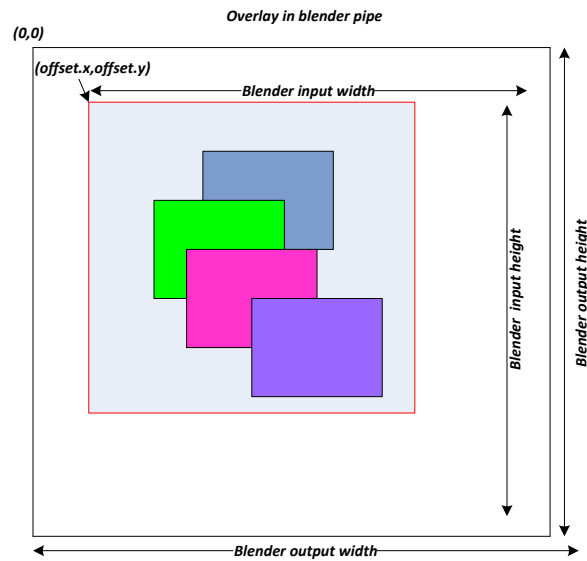


Figure 5-6. Pipe memory access diagram

5.10.4 Memory Mapping List

Module name	Memory Range	Offset Address
GLB	4K	0x00000
BLD	4K	0x01000
OVL_V(CH0)	4K	0x02000
OVL_UI(CH1)	4K	0x03000
OVL_UI(CH2)	4K	0x04000
OVL_UI(CH3)	4K	0x05000
VIDEO_SCALER(CH0)	128K	0x20000
UI_SCALER1(CH1)	64K	0x40000
UI_SCALER2(CH2)	64K	0x50000
UI_SCALER3(CH3)	64K	0x60000
POST_PROC1	64K	0xA0000
POST_PROC2	64K	0xB0000
DMA	128K	0xC0000

5.10.5 Register List

Register name	Offset	Description
GLB_CTL	0x000	Global control register
GLB_STS	0x004	Global status register
GLB_DBUFFER	0x008	Global double buffer control register
GLB_SIZE	0x00C	Global size register

OVL_V_ATTCTL	0x000 + N*0x30	OVL_V attribute control register(N=0,1,2,3)
OVL_V_MBSIZE	0x004 + N*0x30	OVL_V memory block size register(N=0,1,2,3)
OVL_V_COOR	0x008 + N*0x30	OVL_V memory block coordinate register(N=0,1,2,3)
OVL_V_PITCH0	0x00C + N*0x30	OVL_V memory pitch register0(N=0,1,2,3)
OVL_V_PITCH1	0x010 + N*0x30	OVL_V memory pitch register1(N=0,1,2,3)
OVL_V_PITCH2	0x014 + N*0x30	OVL_V memory pitch register2(N=0,1,2,3)
OVL_V_TOP_LADD0	0x018 + N*0x30	OVL_V top field memory block low address register0(N=0,1,2,3)
OVL_V_TOP_LADD1	0x01C + N*0x30	OVL_V top field memory block low address register1(N=0,1,2,3)
OVL_V_TOP_LADD2	0x020 + N*0x30	OVL_V top field memory block low address register2(N=0,1,2,3)
OVL_V_BOT_LADD0	0x024 + N*0x30	OVL_V bottom field memory block low address register0(N=0,1,2,3)
OVL_V_BOT_LADD1	0x028 + N*0x30	OVL_V bottom field memory block low address register1(N=0,1,2,3)
OVL_V_BOT_LADD2	0x02C + N*0x30	OVL_V bottom field memory block low address register2(N=0,1,2,3)
OVL_V_FILL_COLOR	0x0C0 + N*0x4	OVL_V fill color register(N=0,1,2,3)
OVL_V_TOP_HADD0	0x0D0	OVL_V top field memory block high address register0
OVL_V_TOP_HADD1	0x0D4	OVL_V top field memory block high address register1
OVL_V_TOP_HADD2	0x0D8	OVL_V top field memory block high address register2
OVL_V_BOT_HADD0	0x0DC	OVL_V bottom field memory block high address register0
OVL_V_BOT_HADD1	0x0E0	OVL_V bottom field memory block high address register1
OVL_V_BOT_HADD2	0x0E4	OVL_V bottom field memory block high address register2
OVL_V_SIZE	0x0E8	OVL_V overlay window size register
OVL_V_HDS_CTL0	0x0F0	OVL_V horizontal down sample control register0
OVL_V_HDS_CTL1	0x0F4	OVL_V horizontal down sample control register1
OVL_V_VDS_CTL0	0x0F8	OVL_V vertical down sample control register0
OVL_V_VDS_CTL1	0x0FC	OVL_V vertical down sample control register1
OVL_UI_ATTCTL	0x000 + N*0x20	OVL_UI attribute control register(N=0,1,2,3)
OVL_UI_MBSIZE	0x004 + N*0x20	OVL_UI memory block size register(N=0,1,2,3)
OVL_UI_COOR	0x008 + N*0x20	OVL_UI memory block coordinate register(N=0,1,2,3)
OVL_UI_PITCH	0x00C + N*0x20	OVL_UI memory pitch register(N=0,1,2,3)
OVL_UI_TOP_LADD	0x010 + N*0x20	OVL_UI top field memory block low address register(N=0,1,2,3)
OVL_UI_BOT_LADD	0x014 + N*0x20	OVL_UI bottom field memory block low address register(N=0,1,2,3)
OVL_UI_FILL_COLOR	0x018 + N*0x20	OVL_UI fill color register(N=0,1,2,3)
OVL_UI_TOP_HADD	0x080	OVL_UI top field memory block high address register
OVL_UI_BOT_HADD	0x084	OVL_UI bottom field memory block high address register
OVL_UI_SIZE	0x088	OVL_UI overlay window size register
BLD_FILLCOLOR_CTL	0x000	BLD fill color control register
BLD_FILL_COLOR	0x004 + N*0x14	BLD fill color register(N=0,1,2,3,4)
BLD_CH_ISIZE	0x008 + N*0x14	BLD input memory size register(N=0,1,2,3,4)
BLD_CH_OFFSET	0x00C + N*0x14	BLD input memory offset register(N=0,1,2,3,4)
BLD_CH_RTCTL	0x080	BLD routing control register
BLD_PREMUL_CTL	0x084	BLD pre-multiply control register
BLD_BK_COLOR	0x088	BLD background color register
BLD_SIZE	0x08C	BLD output size setting register
BLD_CTL	0x090 – 0x09C	BLD control register

BLD_KEY_CTL	0x0B0	BLD color key control register
BLD_KEY_CON	0x0B4	BLD color key configuration register
BLD_KEY_MAX	0x0C0 – 0x0CC	BLD color key max register
BLD_KEY_MIN	0x0E0 – 0x0EC	BLD color key min register
BLD_OUT_COLOR	0x0FC	BLD output color control register

5.10.6 GLB Register Description

Note: all registers in GLB are not double buffer.

Note: all registers in GLB are not double buffer.

5.10.6.1 Global control register

Offset: 0x000			Register Name: GLB_CTL
Bit	Read/Write	Default/Hex	Description
31:14	/	/	/
13:12	R/W	0x0	OUT_DATA_WB Output data for RT-WB 0:RT-WB fetch data after DEP port 1:RT-WB fetch data before DEP port Other: Reserved
11:10	/	/	/
9	R/W	0x0	lcd flied in reverse
8	R/W	0x0	lcd sync in reverse
7:6	/	/	/
5	R/W	0x0	ERROR_IRQ_EN Hardware error IRQ enable 0:disable 1:enable
4	R/W	0x0	FINISH_IRQ_EN Mission finish IRQ enable 0:disable 1:enable
3:1	/	/	/
0	R/W	0x0	EN RT enable/disable 0: disable 1: enable

5.10.6.2 Global status register

Offset: 0x004			Register Name: GLB_STS
Bit	Read/Write	Default/Hex	Description
31:9	/	/	/
8	R	0x0	EVEN_ODD 0: even field

			1: odd field This bit is the flag for output data in interlace mode
7:6	/	/	/
5	R	0x0	ERROR Hardware error status
4	R	0x0	BUSY Module working status 0:idle 1:running busy
3:2	/	/	/
1	R/W	0x0	ERROR_IRQ Hardware error IRQ It will be set when hardware error occur, and cleared by writing 1.
0	R/W	0x0	FINISH_IRQ Mission finish IRQ It will be set when 1 frame operation accomplished, and cleared by writing 1.

5.10.6.3 Global double buffer control register

Offset: 0x008			Register Name: GLB_DBUFFER
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	DOUBLE_BUFFER_RDY 0: no change 1: register value be ready for update Note: This bit is self-cleared by writing 1 after update.

5.10.6.4 Global size register

Offset: 0x00C			Register Name: GLB_SIZE
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	HEIGHT Height The Height = The value of these bits add 1
15:13	/	/	/
12:0	R/W	0x0	WIDTH Width The Width = The value of these bits add 1

5.10.7 OVL_V Register Description

Note: all registers in OVL_V are double buffer.

5.10.7.1 OVL_V attribute control register

Offset: 0x000+N*0x30 (N=0,1,2,3)			Register Name: OVL_V_ATTCTL
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23	R/W	0x0	LAY0_TOP_BOTTOM_ADDR_EN 0: disable 1: enable When this bit is disable the layer address use top field address in both top and bottom field, otherwise use the top and bottom field address separate in top and bottom field.
22:16	/	/	/
15	R/W	0x0	Video_UI_SEL Video Overlay or UI Overlay Select 0: Video Overlay(using Video Overlay Layer Input data format) 1: UI Overlay(using UI Overlay Layer Input data format)
14:13	/	/	/
12:8	R/W	0x0	LAY_FBFMT Video Overlay Layer Input data format 0x00: Interleaved YUV422(V0Y1U0Y0) 0x01: Interleaved YUV422(Y1V0Y0U0) 0x02: Interleaved YUV422(U0Y1V0Y0) 0x03: Interleaved YUV422(Y1U0Y0V0) 0x04: Planar YUV422 UV combined(V1U1V0U0) 0x05: Planar YUV422 UV combined(U1V1U0V0) 0x06: Planar YUV422 0x07: Reserved 0x08: Planar YUV420 UV combined(V1U1V0U0) 0x09: Planar YUV420 UV combined(U1V1U0V0) 0x0A: Planar YUV420 0x0B: Reserved 0x0C: Planar YUV411 UV combined(V1U1V0U0) 0x0D: Planar YUV411 UV combined(U1V1U0V0) 0x0E: Planar YUV411 Other: Reserved All video layers must be the same format, programmer should confirm it. UI Overlay Layer Input data format 0x00: ARGB_8888 0x01: ABGR_8888 0x02: RGBA_8888 0x03: BGRA_8888 0x04: XRGB_8888 0x05: XBGR_8888 0x06: RGBX_8888

			0x07: BGRX_8888 0x08: RGB_888 0x09: BGR_888 0x0A: RGB_565 0x0B: BGR_565 0x0C: ARGB_4444 0x0D: ABGR_4444 0x0E: RGBA_4444 0x0F: BGRA_4444 0x10: ARGB_1555 0x11: ABGR_1555 0x12: RGBA_5551 0x13: BGRA_5551 Other: Reserved All ui layers' alpha is useless.
7:5	/	/	/
4	R/W	0x0	LAY_FILLCOLOR_EN 0: disable 1:enable When the layer fill-color is enabled, the layer data will use the fill-color.
3:1	/	/	/
0	R/W	0x0	LAYO_EN Layer0 enable/disable 0: disabled 1: enabled

Note: the layer priority is layer3>layer2>layer1>layer0

5.10.7.2 OVL_V memory block size register

Offset: 0x004+N*0x30 (N=0,1,2,3)			Register Name: OVL_V_MBSIZE
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	LAY_HEIGHT Layer Height The Layer Height = The value of these bits add 1
15:13	/	/	/
12:0	R/W	0x0	LAY_WIDTH Layer Width The Layer Width = The value of these bits add 1

5.10.7.3 OVL_V memory block coordinate register

Offset: 0x008+N*0x30 (N=0,1,2,3)	Register Name: OVL_V_COOR
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Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	LAY_YCOOR Y coordinate Y is the left-top y coordinate of layer on overlay window in pixels
15:0	R/W	0x0	LAY_XCOOR X coordinate X is left-top x coordinate of the layer on overlay window in pixels

Setting the layer0-layer3 the coordinate (left-top) on overlay window control information

5.10.7.4 OVL_V memory pitch register0

Offset: 0x00C+N*0x30 (N=0,1,2,3)			Register Name: OVL_V_PITCH0
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LAY_PITCH Layer memory pitch in bytes

Note: The setting of this register is Y channel.

5.10.7.5 OVL_V memory pitch register1

Offset: 0x010+N*0x30 (N=0,1,2,3)			Register Name: OVL_V_PITCH1
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LAY_PITCH Layer memory pitch in bytes

Note: The setting of this register is U/UV channel.

5.10.7.6 OVL_V memory pitch register2

Offset: 0x014+N*0x30 (N=0,1,2,3)			Register Name: OVL_V_PITCH2
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LAY_PITCH Layer memory pitch in bytes

Note: The setting of this register is V channel.

5.10.7.7 OVL_V top field memory block low address register0

Offset: 0x018+N*0x30 (N=0,1,2,3)			Register Name: OVL_V_TOP_LADD0
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LAYMB_LADD Memory Block Start Address Layer Memory Block Address in bytes

Note: The setting of this register is Y channel address.

5.10.7.8 OVL_V top field memory block low address register1

Offset: 0x01C+N*0x30 (N=0,1,2,3)			Register Name: OVL_V_TOP_LADD1
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LAYMB_LADD Memory Block Start Address Layer Memory Block Address in bytes

Note: The setting of this register is U/UV channel address.

5.10.7.9 OVL_V top field memory block low address register2

Offset: 0x020+N*0x30 (N=0,1,2,3)			Register Name: OVL_V_TOP_LADD2
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LAYMB_LADD Memory Block Start Address Layer Memory Block Address in bytes

Note: The setting of this register is V channel address.

5.10.7.10 OVL_V bottom field memory block low address register0

Offset: 0x024+N*0x30 (N=0,1,2,3)			Register Name: OVL_V_BOT_LADD0
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LAYMB_LADD Memory Block Start Address Layer Memory Block Address in bytes

Note: The setting of this register is Y channel address.

5.10.7.11 OVL_V bottom field memory block low address register1

Offset: 0x028+N*0x30 (N=0,1,2,3)			Register Name: OVL_V_BOT_LADD1
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LAYMB_LADD Memory Block Start Address Layer Memory Block Address in bytes

Note: The setting of this register is U/UV channel address.

5.10.7.12 OVL_V bottom field memory block low address register2

Offset: 0x02C+N*0x30 (N=0,1,2,3)			Register Name: OVL_V_BOT_LADD2
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Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LAYMB_LADD Memory Block Start Address Layer Memory Block Address in bytes

Note: The setting of this register is V channel address.

5.10.7.13 OVL_V fill color register

Offset: 0x0C0+N*0x4 (N=0,1,2,3)			Register Name: OVL_V_FILL_COLOR
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	Y/R Y/Red fill color value
15:8	R/W	0x0	U/G U/Green fill color value
7:0	R/W	0x0	V/B V/Blue fill color value

5.10.7.14 OVL_V top field memory block high address register0

Offset: 0x0D0			Register Name: OVL_V_TOP_HADD0
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	LAY3MB_HADD Layer3 Layer Memory Block Address in bytes
23:16	R/W	0x0	LAY2MB_HADD Layer2 Layer Memory Block Address in bytes
15:8	R/W	0x0	LAY1MB_HADD Layer1 Layer Memory Block Address in bytes
7:0	R/W	0x0	LAY0MB_HADD Layer0 Layer Memory Block Address in bytes

Note: The setting of this register is Y channel address.

5.10.7.15 OVL_V top field memory block high address register1

Offset: 0x0D4			Register Name: OVL_V_TOP_HADD1
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	LAY3MB_HADD Layer3 Layer Memory Block Address in bytes
23:16	R/W	0x0	LAY2MB_HADD

			Layer2 Layer Memory Block Address in bytes
15:8	R/W	0x0	LAY1MB_HADD Layer1 Layer Memory Block Address in bytes
7:0	R/W	0x0	LAY0MB_HADD Layer0 Layer Memory Block Address in bytes

Note: The setting of this register is U/UV channel address.

5.10.7.16 OVL_V top field memory block high address register2

Offset: 0x0D8			Register Name: OVL_V_TOP_HADD2
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	LAY3MB_HADD Layer3 Layer Memory Block Address in bytes
23:16	R/W	0x0	LAY2MB_HADD Layer2 Layer Memory Block Address in bytes
15:8	R/W	0x0	LAY1MB_HADD Layer1 Layer Memory Block Address in bytes
7:0	R/W	0x0	LAY0MB_HADD Layer0 Layer Memory Block Address in bytes

Note: The setting of this register is V channel address.

5.10.7.17 OVL_V bottom field memory block high address register0

Offset: 0x0DC			Register Name: OVL_V_BOT_HADD0
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	LAY3MB_HADD Layer3 Layer Memory Block Address in bytes
23:16	R/W	0x0	LAY2MB_HADD Layer2 Layer Memory Block Address in bytes
15:8	R/W	0x0	LAY1MB_HADD Layer1 Layer Memory Block Address in bytes
7:0	R/W	0x0	LAY0MB_HADD Layer0 Layer Memory Block Address in bytes

Note: The setting of this register is Y channel address.

5.10.7.18 OVL_V bottom field memory block high address register1

Offset: 0x0E0			Register Name: OVL_V_BOT_HADD1
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	LAY3MB_HADD Layer3 Layer Memory Block Address in bytes
23:16	R/W	0x0	LAY2MB_HADD Layer2 Layer Memory Block Address in bytes
15:8	R/W	0x0	LAY1MB_HADD Layer1 Layer Memory Block Address in bytes
7:0	R/W	0x0	LAY0MB_HADD Layer0 Layer Memory Block Address in bytes

Note: The setting of this register is U/UV channel address.

5.10.7.19 OVL_V bottom field memory block high address register2

Offset: 0x0E4			Register Name: OVL_V_BOT_HADD2
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	LAY3MB_HADD Layer3 Layer Memory Block Address in bytes
23:16	R/W	0x0	LAY2MB_HADD Layer2 Layer Memory Block Address in bytes
15:8	R/W	0x0	LAY1MB_HADD Layer1 Layer Memory Block Address in bytes
7:0	R/W	0x0	LAY0MB_HADD Layer0 Layer Memory Block Address in bytes

Note: The setting of this register is V channel address.

5.10.7.20 OVL_V overlay window size register

Offset: 0x0E8			Register Name: OVL_V_SIZE
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	HEIGHT Overlay Height The Overlay Height = The value of these bits add 1

15:13	/	/	/
12:0	R/W	0x0	WIDTH Overlay Width The Overlay Width = The value of these bits add 1

when all the layers are disable the overlay has no output data, and by pass.

5.10.7.21 OVL_V horizontal down sampling control register0

Offset: 0x0F0			Register Name: OVL_V_HDS_CTL0
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R/W	0x0	N The step size for overlay data fetch. Note: Refer to "Figure 6 Video overlay data fetch rule"
15:14	/	/	/
13:0	R/W	0x0	M The counter threshold for fetch data Note: Refer to "Figure 6 Video overlay data fetch rule"

The setting of this register is Y channel.

5.10.7.22 OVL_V horizontal down sampling control register1

Offset: 0x0F4			Register Name: OVL_V_HDS_CTL1
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R/W	0x0	N The step size for overlay data fetch. Note: Refer to "Figure 6 Video overlay data fetch rule"
15:14	/	/	/
13:0	R/W	0x0	M The counter threshold for fetch data Note: Refer to "Figure 6 Video overlay data fetch rule"

The setting of this register is UV channel.

5.10.7.23 OVL_V vertical down sampling control register0

Offset: 0x0F8			Register Name: OVL_V_VDS_CTL0
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R/W	0x0	N The step size for overlay data fetch. Note: Refer to "Figure 6 Video overlay data fetch rule"
15:14	/	/	/
13:0	R/W	0x0	M The counter threshold for fetch data

Note: Refer to "Figure 6 Video overlay data fetch rule"

The setting of this register is Y channel.

5.10.7.24 OVL_V vertical down sampling control register1

Offset: 0x0FC			Register Name: OVL_V_VDS_CTL1
Bit	Read/Write	Default/Hex	Description
31:30	/	/	/
29:16	R/W	0x0	N The step size for overlay data fetch. Note: Refer to "Figure 6 Video overlay data fetch rule"
15:14	/	/	/
13:0	R/W	0x0	M The counter threshold for fetch data Note: Refer to "Figure 6 Video overlay data fetch rule"

The setting of this register is UV channel.

5.10.8 OVL_UI Register Description

Note: all registers in OVL_UI are double buffer.

5.10.8.1 OVL_UI attribute control register

Offset: 0x000+N*0x20 (N=0,1,2,3)			Register Name: OVL_UI_ATTR_CTL
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	LAY_GLBALPHA Globe alpha value Alpha value is used for this layer
23	R/W	0x0	TOP_BOTTOM_ADDR_EN 0: disable 1: enable When this bit is disable the layer address use top field address in both top and bottom field, otherwise use the top and bottom field address separate in top and bottom field.
22:18	/	/	/
17:16	R/W	0x0	LAY_PREMUL_CTL Layer input pre-multiply alpha control 0x0: Input layer data is non-pre-multiply data and it needn't to unify pre-multiply data. 0x1: Input layer data is non-pre-multiply data and it need to unify pre-multiply data. 0x2: pre-multiply input layer Other: Reserved
15:13	/	/	/

12:8	R/W	0x0	LAY_FBFMT Input data format 0x00: ARGB_8888 0x01: ABGR_8888 0x02: RGBA_8888 0x03: BGRA_8888 0x04: XRGB_8888 0x05: XBGR_8888 0x06: RGBX_8888 0x07: BGRX_8888 0x08: RGB_888 0x09: BGR_888 0x0A: RGB_565 0x0B: BGR_565 0x0C: ARGB_4444 0x0D: ABGR_4444 0x0E: RGBA_4444 0x0F: BGRA_4444 0x10: ARGB_1555 0x11: ABGR_1555 0x12: RGBA_5551 0x13: BGRA_5551 Other: Reserved
7:5	/	/	/
4	R/W	0x0	LAY_FILLCOLOR_EN 0: disable 1:enable When the layer fill-color is enabled, the layer data will use the fill-color.
3	/	/	/
2:1	R/W	0x0	LAY_ALPHA_MODE Layer input alpha mode 0:Ignore Input alpha value = pixels alpha, if no pixel alpha, the alpha value equal 0xff 1:Globe alpha enable Ignore pixel alpha value Input alpha value = globe alpha value 2: Globe alpha mix pixel alpha Input alpha value = globe alpha value * pixels alpha value 3:Reserved
0	R/W	0x0	LAY_EN Layer enable/disable 0: disabled 1: enabled

Note: the layer priority is layer3>layer2>layer1>layer0

5.10.8.2 OVL_UI memory block size register

Offset: 0x004+N*0x20 (N=0,1,2,3)			Register Name: OVL_UI_MBSIZE
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	LAY_HEIGHT Layer Height The Layer Height = The value of these bits add 1
15:13	/	/	/
12:0	R/W	0x0	LAY_WIDTH Layer Width The Layer Width = The value of these bits add 1

5.10.8.3 OVL_UI memory block coordinate register

Offset: 0x008+N*0x20 (N=0,1,2,3)			Register Name: OVL_UI_COOR
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	LAY_YCOOR Y coordinate Y is the left-top y coordinate of layer on overlay window in pixels
15:0	R/W	0x0	LAY_XCOOR X coordinate X is left-top x coordinate of the layer on overlay window in pixels

Setting the layer0-layer3 the coordinate (left-top) on overlay window control information

5.10.8.4 OVL_UI memory pitch register

Offset: 0x00C+N*0x20 (N=0,1,2,3)			Register Name: OVL_UI_PITCH
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LAY_PITCH Layer memory pitch in bytes

5.10.8.5 OVL_UI top field memory block low address register

Offset: 0x010+N*0x20 (N=0,1,2,3)			Register Name: OVL_UI_TOP_LADD
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LAYMB_LADD Memory Block Start Address Layer Memory Block Address in bytes

5.10.8.6 OVL_UI bottom field memory block low address register

Offset: 0x014+N*0x20 (N=0,1,2,3)			Register Name: OVL_UI_BOT_LADD
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LAYMB_LADD Memory Block Start Address Layer Memory Block Address in bytes

5.10.8.7 OVL_UI fill color register

Offset: 0x018+N*0x20 (N=0,1,2,3)			Register Name: OVL_UI_FILL_COLOR
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	Alpha Alpha fill color value
23:16	R/W	0x0	RED Red fill color value
15:8	R/W	0x0	GREEN Green fill color value
7:0	R/W	0x0	BLUE Blue fill color value

5.10.8.8 OVL_UI top field memory block high address register

Offset: 0x080			Register Name: OVL_UI_TOP_HADD
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	LAY3MB_HADD Layer3 Layer Memory Block Address in bytes
23:16	R/W	0x0	LAY2MB_HADD Layer2 Layer Memory Block Address in bytes
15:8	R/W	0x0	LAY1MB_HADD Layer1 Layer Memory Block Address in bytes
7:0	R/W	0x0	LAY0MB_HADD Layer0 Layer Memory Block Address in bytes

5.10.8.9 OVL_UI bottom field memory block high address register

Offset: 0x084			Register Name: OVL_UI_BOT_HADD
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	LAY3MB_HADD Layer3

			Layer Memory Block Address in bytes
23:16	R/W	0x0	LAY2MB_HADD Layer2 Layer Memory Block Address in bytes
15:8	R/W	0x0	LAY1MB_HADD Layer1 Layer Memory Block Address in bytes
7:0	R/W	0x0	LAY0MB_HADD Layer0 Layer Memory Block Address in bytes

5.10.8.10 OVL_UI overlay window size register

Offset: 0x088			Register Name: OVL_UI_SIZE
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	HEIGHT Overlay Height The Overlay Height = The value of these bits add 1
15:13	/	/	/
12:0	R/W	0x0	WIDTH Overlay Width The Overlay Width = The value of these bits add 1

Note: When all the layers are disable the overlay has no output data, and by pass.

Channel2~4 register definition is same as OVL_UI's register definition.

5.10.9 BLD Register Description

Note: all registers in BLD are double buffer.

5.10.9.1 BLD fill color control register

Offset: 0x000			Register Name: BLD_FILL_COLOR_CTL
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12	R/W	0x0	P4_EN Pipe4 enable/disable 0: disabled 1: enabled
11	R/W	0x0	P3_EN Pipe3 enable/disable 0: disabled 1: enabled
10	R/W	0x0	P2_EN Pipe2 enable/disable

			0: disabled 1: enabled
9	R/W	0x0	P1_EN Pipe1 enable/disable 0: disabled 1: enabled
8	R/W	0x0	P0_EN Pipe0 enable/disable 0: disabled 1: enabled
7:5	/	/	/
4	R/W	0x0	P4_FCEN Pipe4 fill color enable/disable 0: disabled 1: enabled
3	R/W	0x0	P3_FCEN Pipe3 fill color enable/disable 0: disabled 1: enabled
2	R/W	0x0	P2_FCEN Pipe2 fill color enable/disable 0: disabled 1: enabled
1	R/W	0x0	P1_FCEN Pipe1 fill color enable/disable 0: disabled 1: enabled
0	R/W	0x0	P0_FCEN Pipe0 fill color enable/disable 0: disabled 1: enabled

5.10.9.2 BLD fill color register

Offset: 0x004+N*0x10 (N=0,1,2,3,4)			Register Name: BLD_FILL_COLOR
Bit	Read/Write	Default/Hex	Description
31:24	R/W	0x0	Alpha Alpha fill color value
23:16	R/W	0x0	RED Red fill color value
15:8	R/W	0x0	GREEN Green fill color value
7:0	R/W	0x0	BLUE Blue fill color value

5.10.9.3 BLD input memory size register

Offset: 0x008+N*0x10 (N=0,1,2,3,4)			Register Name: BLD_CH_ISIZE
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	HEIGHT Pipe input memory height The input height = The value of these bits add 1
15:13	/	/	/
12:0	R/W	0x0	WIDTH Pipe input memory width The input width = The value of these bits add 1

5.10.9.4 BLD input memory offset register

Offset: 0x00C+N*0x10 (N=0,1,2,3,4)			Register Name: BLD_CH_OFFSET
Bit	Read/Write	Default/Hex	Description
31:16	R/W	0x0	YCOOR Y coordinate Y is the left-top y coordinate of the pipe on blender memory window in pixels
15:0	R/W	0x0	XCOOR X coordinate X is the left-top x coordinate of the pipe on blender memory window in pixels

5.10.9.5 BLD routing control register

Offset: 0x080			Register Name: BLD_CH_RTCTL
Bit	Read/Write	Default/Hex	Description
31:16	/	/	/
15:12	R/W	0x3	P3_RTCTL Pipe3 select channel control 0:from channel0 1:from channel1 2:from channel2 3:from channel3 Other: reserved
11:8	R/W	0x2	P2_RTCTL Pipe2 select channel control 0:from channel0 1:from channel1 2:from channel2 3:from channel3 Other: reserved
7:4	R/W	0x1	P1_RTCTL

			Pipe1 select channel control 0:from channel0 1:from channel1 2:from channel2 3:from channel3 Other: reserved
3:0	R/W	0x0	PO_RTCTL Pipe0 select channel control 0:from channel0 1:from channel1 2:from channel2 3:from channel3 Other: reserved

Note: Setting 2 or more channels in the same pipe is illegal, programmer should confirm it.

5.10.9.6 BLD pre-multiply control register

Offset: 0x084			Register Name: BLD_PREMUL_CTL
Bit	Read/Write	Default/Hex	Description
31:4	/	/	/
3	R/W	0x0	P3_ALPHA_MODE Pipe3 input alpha mode 0:all alpha data is no-pre-multiply alpha 1:all alpha data is pre-multiply alpha
2	R/W	0x0	P2_ALPHA_MODE Pipe2 input alpha mode 0:all alpha data is no-pre-multiply alpha 1:all alpha data is pre-multiply alpha
1	R/W	0x0	P1_ALPHA_MODE Pipe1 input alpha mode 0:all alpha data is no-pre-multiply alpha 1:all alpha data is pre-multiply alpha
0	R/W	0x0	P0_ALPHA_MODE Pipe0 input alpha mode 0:all alpha data is no-pre-multiply alpha 1:all alpha data is pre-multiply alpha

5.10.9.7 BLD background color register

Offset: 0x088			Register Name: BLD_BK_COLOR
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	RED Red background color value
15:8	R/W	0x0	GREEN

			Green background color value
7:0	R/W	0x0	BLUE Blue background color value

5.10.9.8 BLD output size setting register

Offset: 0x08C			Register Name: BLD_SIZE
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	BLD_HEIGHT Blender height The real blender height = The value of these bits add 1
15:13	/	/	/
12:0	R/W	0x0	BLD_WIDTH Blender width The real blender width = The value of these bits add 1

5.10.9.9 BLD control register

Offset: Blender0: 0x090 Blender1: 0x094 Blender2: 0x098 Blender3: 0x09C			Register Name: BLD_CTL
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:24	R/W	0x3	BLEND_AFD Specifies the coefficient that used in destination alpha data Q_d . 0x0: 0 0x1: 1 0x2: A_s 0x3: $1-A_s$ Other: Reserved
23:20	/	/	/
19:16	R/W	0x1	BLEND_AFS Specifies the coefficient that used in source alpha data Q_s . 0x0: 0 0x1: 1 0x2: A_d 0x3: $1-A_d$ Other: Reserved
15:12	/	/	/
11:8	R/W	0x3	BLEND_PFD Specifies the coefficient that used in destination pixel data F_d . 0x0: 0

			0x1: 1 0x2: A_s 0x3: $1-A_s$ Other: Reserved
7:4	/	/	/
3:0	R/W	0x1	BLEND_PFS Specifies the coefficient that used in source pixel data F_s . 0x0: 0 0x1: 1 0x2: A_d 0x3: $1-A_d$ Other: Reserved

5.10.9.10 BLD color key control register

Offset: 0x0B0			Register Name: BLD_KEY_CTL
Bit	Read/Write	Default/Hex	Description
31:15	/	/	
14:13	R/W	0x0	KEY3_MATCH_DIR In Alpha Blender3 0: when the pixel value matches destination image, it displays the pixel form source image. 1: when the pixel value matches source image, it displays the pixel form destination image. 1x: Reserved
12	R/W	0x0	KEY3_EN Enable color key 0: disabled color key 1: enable color key in Alpha Blender3.
11	/	/	/
10:9	R/W	0x0	KEY2_MATCH_DIR In Alpha Blender2 0: when the pixel value matches destination image, it displays the pixel form source image. 1: when the pixel value matches source image, it displays the pixel form destination image. 1x: Reserved
8	R/W	0x0	KEY2_EN Enable color key 0: disabled color key 1: enable color key in Alpha Blender2.
7	/	/	
6:5	R/W	0x0	KEY1_MATCH_DIR In Alpha Blender1 0: when the pixel value matches destination image, it displays the pixel form source image.

			1: when the pixel value matches source image, it displays the pixel form destination image. 1x: Reserved
4	R/W	0x0	KEY1_EN Enable color key 0: disabled color key 1: enable color key in Alpha Blender1.
3	/	/	
2:1	R/W	0x0	KEY0_MATCH_DIR In Alpha Blender0 0: when the pixel value matches destination image, it displays the pixel form source image. 1: when the pixel value matches source image, it displays the pixel form destination image. 1x: Reserved
0	R/W	0x0	KEY0_EN Enable color key 0: disabled color key 1: enable color key in Alpha Blender0.

5.10.9.11 BLD color key configuration register

Offset: 0x0B4			Register Name: BLD_KEY_CON
Bit	Read/Write	Default/Hex	Description
31:27	/	/	/
26	R/W	0x0	KEY3R_MATCH Red Match Rule 0: match if (Color Min=<Color<=Color Max) the red condition is true, else the condition is false. 1: match if (Color>Color Max or Color<Color Min) the red condition is true, else the condition is false.
25	R/W	0x0	KEY3G_MATCH Green Match Rule 0: match if (Color Min=<Color<=Color Max) the green condition is true, else the condition is false. 1: match if (Color>Color Max or Color<Color Min) the green condition is true, else the condition is false.
24	R/W	0x0	KEY3B_MATCH Blue Match Rule 0: match if (Color Min=<Color<=Color Max) the blue condition is true, else the condition is false. 1: match if (Color>Color Max or Color<Color Min) the blue condition is true, else the condition is false.
23:19	/	/	/
18	R/W	0x0	KEY2R_MATCH Red Match Rule

			0: match if (Color Min=<Color<=Color Max) the red condition is true, else the condition is false. 1: match if (Color>Color Max or Color<Color Min) the red condition is true, else the condition is false.
17	R/W	0x0	KEY2G_MATCH Green Match Rule 0: match if (Color Min=<Color<=Color Max) the green condition is true, else the condition is false. 1: match if (Color>Color Max or Color<Color Min) the green condition is true, else the condition is false.
16	R/W	0x0	KEY2B_MATCH Blue Match Rule 0: match if (Color Min=<Color<=Color Max) the blue condition is true, else the condition is false. 1: match if (Color>Color Max or Color<Color Min) the blue condition is true, else the condition is false.
15:11	/	/	/
10	R/W	0x0	KEY1R_MATCH Red Match Rule 0: match if (Color Min=<Color<=Color Max) the red condition is true, else the condition is false. 1: match if (Color>Color Max or Color<Color Min) the red condition is true, else the condition is false.
9	R/W	0x0	KEY1G_MATCH Green Match Rule 0: match if (Color Min=<Color<=Color Max) the green condition is true, else the condition is false. 1: match if (Color>Color Max or Color<Color Min) the green condition is true, else the condition is false.
8	R/W	0x0	KEY1B_MATCH Blue Match Rule 0: match if (Color Min=<Color<=Color Max) the blue condition is true, else the condition is false. 1: match if (Color>Color Max or Color<Color Min) the blue condition is true, else the condition is false.
7:3	/	/	/
2	R/W	0x0	KEY0R_MATCH Red Match Rule 0: match if (Color Min=<Color<=Color Max) the red condition is true, else the condition is false. 1: match if (Color>Color Max or Color<Color Min) the red condition is true, else the condition is false.
1	R/W	0x0	KEY0G_MATCH Green Match Rule 0: match if (Color Min=<Color<=Color Max) the green condition is true, else the condition is false.

			1: match if (Color>Color Max or Color<Color Min) the green condition is true, else the condition is false.
0	R/W	0x0	KEY0B_MATCH Blue Match Rule 0: match if (Color Min=<Color<=Color Max) the blue condition is true, else the condition is false. 1: match if (Color>Color Max or Color<Color Min) the blue condition is true, else the condition is false.

Note: when R/G/B channel condition is true the color pass through otherwise is false.

5.10.9.12 BLD color key max register

Offset: CK0: 0x0C0 CK1: 0x0C4 CK2: 0x0C8 CK3: 0x0CC			Register Name: BLD_KEY_MAX
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	MAX_R Red Red color key max
15:8	R/W	0x0	MAX_G Green Green color key max
7:0	R/W	0x0	MAX_B Blue Blue color key max

5.10.9.13 BLD color key min register

Offset: CK0: 0x0E0 CK1: 0x0E4 CK2: 0x0E8 CK3: 0x0EC			Register Name: BLD_KEY_MIN
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	MIN_R Red Red color key min
15:8	R/W	0x0	MIN_G Green Green color key min
7:0	R/W	0x0	MIN_B

			Blue Blue color key min
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5.10.9.14 BLD output color control register

Offset: 0x0FC			Register Name: BLD_OUT_COLOR
Bit	Read/Write	Default/Hex	Description
31:2	/	/	/
1	R/W	0x0	ITLMOD_EN Output interlace mode enable 0:disable 1:enable When output interlace mode software programmer should confirm the blender output height is even.
0	R/W	0x0	PREMUL_EN Output color control 0: output normal data(A ' will be A _{abcde} ') 1: output pre-multiply data(A ' = 1)

5.11 DE RT-WB Controller Specification

5.11.1 Overview

The Real-time write-back controller (RT-WB) provides data capture function for display engine. It captures data from RT-mixer module, performs the image resizing function, and then write-back to SDRAM.

The RT-WB can receive RGB888 or YUV444 data format, and then converts to YV12/NV12/NV21 or interleaved RGB888/pRGB888/pBGR888 for write-back. Horizontal and vertical direction scaling-down are implemented independently.

The RT-WB features:

- Support RGB888 and YUV444 input data format
- Support input size from 8×4 to 4096×4096
- Support output size from 8×4 to **W**×4096 (**W** is RTL programmable)
- Support fine down scaling ratio from 1× to 1/2×, and the anti-aliasing filter is 16-phase 4-tap in horizontal , 16-phase 2-tap filter in vertical.
- Support coarse down scaling.

Table 5-1. RTL programmable parameters table

Parameter name	Default	Possible value	Description
W	2048	1280/2048/4096	3 channels line buffer length
FINE_SCALE_EXIST	1	1/0	Define fine scaler exist or not.
PORT_NUM	4	2/4/8	Input port number
M	16	16	Phase number
FRAC	18	18	Phase adder fraction part bit width

5.11.2 Block Diagram

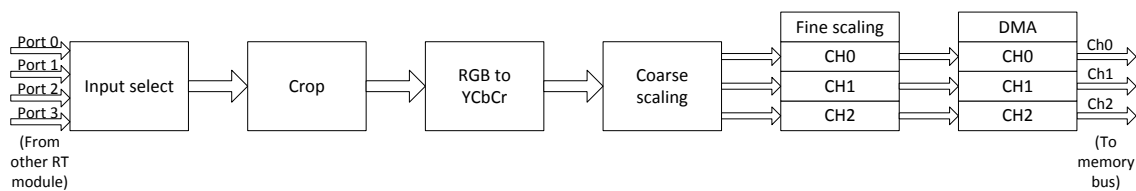


Figure 5-7. RT-WB Block Diagram

5.11.3 Operations and Function Descriptions

5.11.3.1 Write-back flow

The main flow when new frame starts description in Figure 5-8.

The new frame starts with a positive edge of *START*, which is a input from RT-TOP. RT-WB module will check signals like *BUSY*, *OVERFLOW*, *FINISH* to determine the last frame is success to write-back or not. If *BUSY* is still 1, *TIME_OUT* will be set, and users will know that the last frame did not write-back successfully. The *OVERFLOW* is also a error status signal.

It represents the module is too slow to write-back all the data to memory bus in last frame. These two errors will cause a local reset action to clear the whole scaler and some parts in DMA circuit.

A write-back address switch circuit can select the next write-back address automatically. The detail switch function description in Figure 5-9.

After local reset action or address switch, the circuit will check the WB_START is set or not. When WB_START is 1, the write-back function will be activated and this frame will be processed.

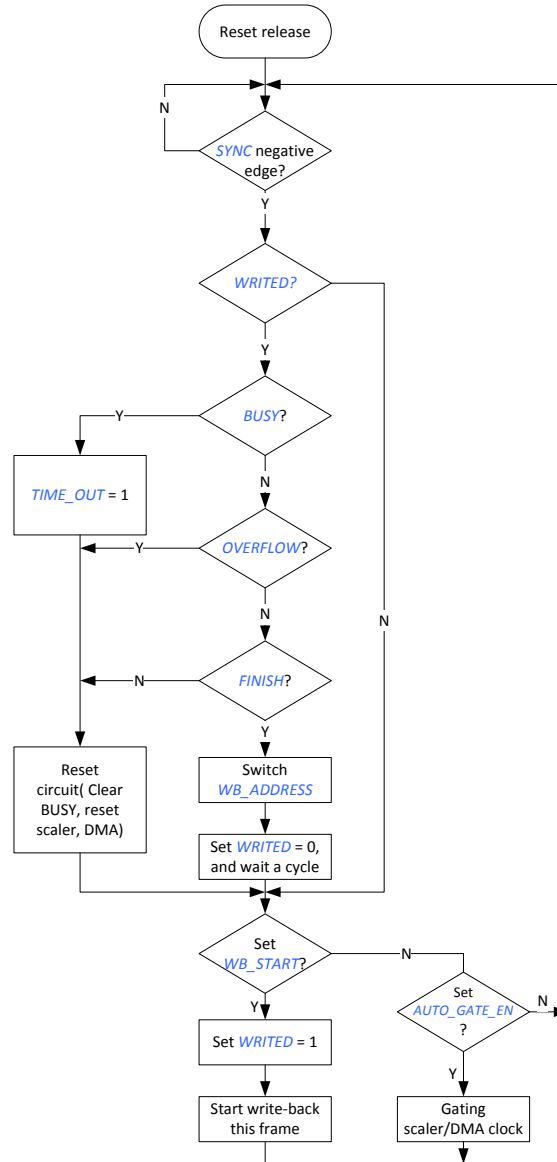


Figure 5-8. The main control flow of RT-WB

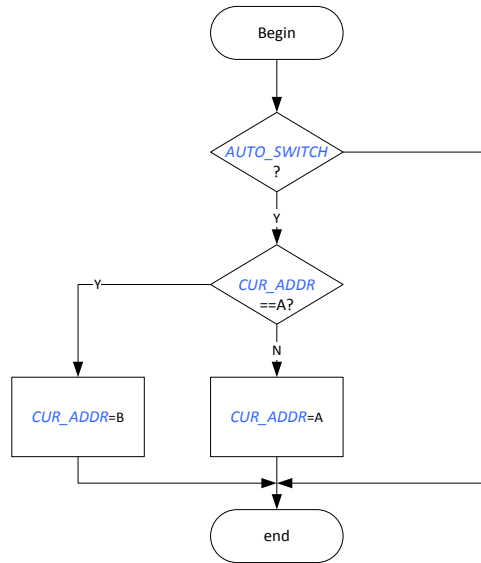


Figure 5-9. Write-back address switch function

5.11.3.2 Key signal and key register

Signal	Type	Description	Note
BUSY	Read only register	Scaler or DMA busy status	Becomes 1 when WB_START takes effect, becomes 0 when FIFO overflowed, write-back time-out or write-back finished, or local reset, or global reset.
WB_START	R/W register	Write-back enable register	Set 1 to start a new frame write-back, becomes 0 when write-back started, or global reset.
SYNC	Input signal	New frame start signal from RT TOP module	Negative edge of SYNC represents a new display frame.
OVERFLOW	R/W register	FIFO overflowed flag	Becomes 1 when FIFO overflowed (write-back too slow), becomes 0 when write 1 to it, or global reset.
FINISH	R/W register	Write-back successful finish flag	Becomes 1 when write-back finished successfully, becomes 0 when write 1 to it, or global reset.
TIME_OUT	R/W register	Write-back time out flag	When SYNC negative edge comes, write-back has not finished yet , this bit will set 1. Becomes 0 when write 1 to it, or global reset.
WRITED	Internal signal	One frame has been written-back	Becomes 1 when WB_START takes effect, becomes 0 when next SYNC negative edge comes, or global reset.
IRQ	R/W register	Write-back end flag	When WRITED negative edge comes, IRQ becomes 1, becomes 0 when write 1 to it, or global reset.
INTR	Output signal	Interrupt signal to GIC module	INTR = IRQ & INT_EN
GLOBAL RESET	Input signal	Global reset from RT TOP module	Clear the whole scaler circuit, some parts of DMA circuit, and status register/WB_START.

AUTO SWITCH	R/W register	Write-back address automatic switch enable	Set 1 to enable function, set 0 to disable.
CUR_ADDRx (x=0,1,2)	Internal register	Channel x current write back start address	
AUTO_GATE_EN	R/W register	Enable automatic gating clock.	Set 1 to enable function, set 0 to disable. If AUTO_GATE_EN==1, when module idle, the clock to scaler and DMA will be gated. And when a write-back frame starts, the clock gate will be released.
CLK_GATE	R/W register	Clock gate of scaler and DMA module	If AUTO_GATE_EN==0, set 1 to release clock, set 0 to gating clock. No use when AUTO_GATE_EN==1.

5.11.4 RT-WB Register List

Module name	Memory Range	Offset Address
RT-WB	64K	0x01010000

Register name	Offset	Description
WB_GCTRL_REG	0x000	Module general control register
WB_SIZE_REG	0x004	Input size register
WB_CROP_COORD_REG	0x008	Cropping coordinate register
WB_CROP_SIZE_REG	0x00c	Cropping size register
WB_A_CH0_ADDR_REG	0x010	Write-back Group A channel 0 address register
WB_A_CH1_ADDR_REG	0x014	Write-back Group A channel 1 address register
WB_A_CH2_ADDR_REG	0x018	Write-back Group A channel 2 address register
WB_A_HIGH_ADDR_REG	0x01c	Write-back Group A address high bit register
WB_B_CH0_ADDR_REG	0x020	Write-back Group B channel 0 address register
WB_B_CH1_ADDR_REG	0x024	Write-back Group B channel 1 address register
WB_B_CH2_ADDR_REG	0x028	Write-back Group B channel 2 address register
WB_B_HIGH_ADDR_REG	0x02c	Write-back Group B address high bit register
WB_CH0_PITCH_REG	0x030	Write-back channel 0 pitch register
WB_CH12_PITCH_REG	0x034	Write-back channel 1/2 pitch register
WB_ADDR_SWITCH_REG	0x040	Write-back address switch setting register
WB_FORMAT_REG	0x044	Output format register
WB_INT_REG	0x048	Interrupt control register
WB_STATUS_REG	0x04c	Module status register
WB_BYPASS_REG	0x054	Bypass control register
WB_CS_HORZ_REG	0x070	Coarse scaling horizontal setting register
WB_CS_VERT_REG	0x074	Coarse scaling vertical setting register
WB_FS_INSIZE_REG	0x080	Fine scaling input size register
WB_FS_OUTSIZE_REG	0x084	Fine scaling output size register
WB_FS_HSTEP_REG	0x088	Fine scaling horizontal step register

WB_FS_VSTEP_REG	0x08C	Fine scaling vertical step register
WB_DEBUG_REG	0x0FC	Debug register
WB_CH0_HCOEF_REGN	0x200 + N*4	Channel 0 horizontal coefficient register N (N = 0,1,2,...,15)
WB_CH1_HCOEF_REGN	0x280 + N*4	Channel 1/2 horizontal coefficient register N (N = 0,1,2,...,15)

Note: None of these registers is double-buffer.

5.11.5 RT-WB Register Description

5.11.5.1 WB_GCTRL_REG

Offset: 0X000			Register name: WB_GCTRL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	BIST_EN Enable BIST mode 0: Disable 1: Enable
30:	/	/	/
29	R/W	0x0	CLK_GATE Clock gate of scaler and DMA module 0: Gate 1: Release
28	R/W	0x0	AUTO_GATE_EN Enable automatic clock gating 0: Disable 1: Enable
27:18	/	/	/
17:16	R/W	0x0	IN_PORT_SEL Input port selection 0: port 0 1: port 1 2: port 2 3: port 3
15:5	/	/	/
4	R/W	0x0	SOFT_RESET Reset the whole scaler and DMA. 0:Reset release. 1:Reset hold.
3:1	/	/	/
0	R/W	0x0	WB_START Start write-back process. 0: Do nothing. 1: Start.

5.11.5.2 WB_SIZE_REG

Offset: 0X004			Register name: WB_SIZE_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	HEIGHT Input height The real input height = The value of these bits + 1.
15:13	/	/	/
12:0	R/W	0x0	WIDTH Input width The real input width = The value of these bits + 1.

5.11.5.3 WB_CROP_COORD_REG

Offset: 0X008			Register name: WB_CROP_COORD_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	CROP_TOP Cropping top position Top position is the left-top y coordinate of input window in pixels
15:13	/	/	/
12:0	R/W	0x0	CROP_LEFT Cropping left position Left position is left-top x coordinate of input window in pixels

5.11.5.4 WB_CROP_SIZE_REG

Offset: 0X00C			Register name: WB_CROP_SIZE_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	CROP_HEIGHT Cropping region height The real cropping region height = the value of these bits + 1.
15:13	/	/	/
12:0	R/W	0x0	CROP_WIDTH Cropping region width The real cropping region width = the value of these bits + 1.

5.11.5.5 WB_A_CH0_ADDR_REG

Offset: 0X010			Register name: WB_A_CH0_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	ADDR

			Note: In BYTE. When output format is RGB, ADDR must 4 bytes aligning.
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5.11.5.6 WB_A_CH1_ADDR_REG

Offset: 0X014			Register name: WB_A_CH1_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	ADDR Note: In BYTE.

5.11.5.7 WB_A_CH2_ADDR_REG

Offset: 0X018			Register name: WB_A_CH2_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0	ADDR Note: In BYTE.

5.11.5.8 WB_A_HIGH_ADDR_REG

Offset: 0X01C			Register name: WB_A_HIGH_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	CH2_H_ADDR
15:8	R/W	0x0	CH1_H_ADDR
7:0	R/W	0x0	CH0_H_ADDR

5.11.5.9 WB_B_CH0_ADDR_REG

Offset: 0X020			Register name: WB_B_CH0_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:00	R/W	0x0	ADDR Note: In BYTE. When output format is RGB, ADDR must 4 bytes aligning.

5.11.5.10 WB_B_CH1_ADDR_REG

Offset: 0X024			Register name: WB_B_CH1_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	ADDR Note: In BYTE.

5.11.5.11 WB_B_CH2_ADDR_REG

Offset: 0X028			Register name: WB_B_CH2_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	ADDR Note: In BYTE.

5.11.5.12 WB_B_HIGH_ADDR_REG

Offset: 0X02C			Register name: WB_B_HIGH_ADDR_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:16	R/W	0x0	CH2_H_ADDR
15:8	R/W	0x0	CH1_H_ADDR
7:0	R/W	0x0	CH0_H_ADDR

5.11.5.13 WB_CH0_PITCH_REG

Offset: 0X030			Register name: WB_CH0_PITCH_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	PITCH Write-back channel 0 pitch in BYTE.

5.11.5.14 WB_CH12_PITCH_REG

Offset: 0X034			Register name: WBC_CH12_PITCH_REG
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	PITCH Write-back channel 1/2 pitch in BYTE.

5.11.5.15 WB_ADDR_SWITCH_REG

Offset: 0X040			Register name: WB_ADDR_SWITCH_REG
Bit	Read/Write	Default/Hex	Description
31:21	/	/	/
20	R/W	0x0	MANUAL_GROUP When AUTO_SWITCH is 0, set this bit will switch the group. 0: Group A 1: Group B
19:17	/	/	/
16	R/W	0x0	AUTO_SWITCH Write-back address automatic switch enable 0: Disable 1: Enable
15:01	/	/	/
0	R	0x0	CUR_GROUP When AUTO_SWITCH is 1, this bit will show the index of group using. 0: Group A 1: Group B

5.11.5.16 WB_FORMAT_REG

Offset: 0X044			Register name: WB_FORMAT_REG
Bit	Read/Write	Default/Hex	Description

31:04	/	/	/
03:0	R/W	0x0	FORMAT Output format selection 0000: RGB888 (R in high address) 0001: BGR888 (B in high address) 0010: Reserved 0011: Reserved 0100: pRGB888(pad equal to 0xff)(p in high address) 0101: pBGR888(pad equal to 0xff)(p in high address) 0110: BGRp888(pad equal to 0xff)(B in high address) 0111: RGBp888(pad equal to 0xff)(R in high address) 1000: Planar YUV420 1001: Reserved 1010: Reserved 1011: Reserved 1100: Planar YUV420 UV combined (V1U1V0U0, V1 in high address) 1101: Planar YUV420 UV combined (U1V1U0V0, U1 in high address) 1110: Reserved 1111: Reserved

5.11.5.17 WB_INT_REG

Offset: 0X048			Register name: WB_INT_REG
Bit	Read/Write	Default/Hex	Description
31:1	/	/	/
0	R/W	0x0	INT_EN Write-back interrupt enable 0: Disable 1: Enable

5.11.5.18 WB_STATUS_REG

Offset: 0X04C			Register name: WB_STATUS_REG
Bit	Read/Write	Default/Hex	Description
31:09	/	/	/
8	R	0x0	BUSY Write-back process status 0: write-back end or write-back disable 1: write-back in process
7	/	/	/
6	R/W	0x0	TIME_OUT Write-back TIME_OUT error flag 0: No error 1: Error
5	R/W	0x0	OVERFLOW Write-back FIFO overflow error flag

			0: No error 1: Error
4	R/W	0x0	FINISH Write-back process finish flag 0: write-back not finish or fail 1: write-back finished successfully
3:1	/	/	/
0	R/W	0x0	IRQ Write-back process end flag 0: write-back not end 1: write-back end

5.11.5.19 WB_BYPASS_REG

Offset: 0x054			Register Name: WB_BYPASS_REG
Bit	Read/Write	Default/Hex	Description
31:3	/	/	/
2	R/W	0x0	FS_EN Enable Fine Scaling function 0: Bypass Fine scaling 1: Enable Fine scaling
1	R/W	0x0	CS_EN Enable Coarse Scaling function 0: Bypass Coarse scaling 1: Enable Coarse scaling
0	R/W	0x0	CSC_EN Enable RGB to YPbPr color space conversion 0: Bypass CSC 1: Enable CSC

5.11.5.20 WB_CS_HORZ_REG

Offset: 0x070			Register Name: WB_CS_HORZ_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	N
15:13	/	/	/
12:0	R/W	0x0	M

5.11.5.21 WB_CS_VERT_REG

Offset: 0x074			Register Name: WB_CS_VERT_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	N

15:13	/	/	/
12:0	R/W	0x0	M

5.11.5.22 WB_FS_INSIZE_REG

Offset: 0x080			Register Name: WB_FS_INSIZE_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	IN_HEIGHT Channel 0 input height The actual height is register value + 1
15:13	/	/	/
12:0	R/W	0x0	IN_WIDTH Channel 0 input width The actual width is register value + 1

5.11.5.23 WB_FS_OUTSIZE_REG

Offset: 0x084			Register Name: WB_FS_OUTSIZE_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	OUT_HEIGHT Channel 0 output height The actual height is register value + 1
15:13	/	/	/
12:0	R/W	0x0	OUT_WIDTH Channel 0 output width The actual width is register value + 1

5.11.5.24 WB_FS_HSTEP_REG

Offset: 0x088			Register Name: WB_FS_HSTEP_REG
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/
21:20	R/W	0x0	HSTEP_INT The integer part of channel 0 horizontal scale ratio
19:2	R/W	0x0	HSTEP_FRAC The fraction part of channel 0 horizontal scale ratio
1:0	/	/	/

5.11.5.25 WB_FS_VSTEP_REG

Offset: 0x08C			Register Name: WB_FS_VSTEP_REG
Bit	Read/Write	Default/Hex	Description
31:22	/	/	/

21:20	R/W	0x0	VSTEP_INT The integer part of channel 0 vertical scale ratio
19:2	R/W	0x0	VSTEP_FRAC The fraction part of channel 0 vertical scale ratio
1:0	/	/	/

5.11.5.26 WB_DEBUG_REG

Offset: 0x0FC			Register Name: WB_DEBUG_REG
Bit	Read/Write	Default/Hex	Description
31:13	/	/	/
12:0	R	0x0	INPUT_LINE_CNT Input line counter

5.11.5.27 WB_CH0_HCOEF_REGN (N = 0:15)

Offset: 0x200 + N*4			Register Name: WB_CH0_HCOEF_REGN
Bit	Read/Write	Default/Hex	Description
31:24	W	UDF	COEF3 The most right hand-side pixel coefficient
23:16	W	UDF	COEF2 The right hand-side pixel coefficient
15:8	W	UDF	COEF1 The left hand-side pixel coefficient
7:0	W	UDF	COEF0 The most left hand-side pixel coefficient

Note: This coefficients are signed. The coefficient value equals to coefficient*2⁶. N represents the phase.

5.11.5.28 WB_CH1_HCOEF_REGN (N = 0:15)

Offset: 0x280 + N*4			Register Name: WB_CH1_HCOEF_REGN
Bit	Read/Write	Default/Hex	Description
31:24	W	UDF	COEF3 The most right hand-side pixel coefficient
23:16	W	UDF	COEF2 The right hand-side pixel coefficient
15:8	W	UDF	COEF1 The left hand-side pixel coefficient
7:0	W	UDF	COEF0 The most left hand-side pixel coefficient

Note: This coefficients are signed. The coefficient value equals to coefficient*2⁶. N represents the phase.

5.12 DE VSU Specification

5.12.1 Overview

The Video Scaler (VS) provides YUV format image resizing function for display engine. It receives data from overlay module, performs the image resizing function, and outputs to video post-processing modules.

The VS can receive YUV420/YUV422/YUV411 data format, and then converts to YUV444 for display. Horizontal and vertical direction scaling are implemented independently.

The VS features:

- Support YUV420/YUV422/YUV411 data format with 8-bit pre channel
- Support input and output size from 8×4 to 4096×4096
- Support 1/16× to 32× resize ratio
- Support **M**-phase 8-tap horizontal anti-alias filter, **M**-phase 4-tap vertical anti-alias filter (**M** = 16/32 etc)
- Point-to-point display size up to **W** pixels/line. (**W** is RTL programmable)

Table 5-2. RTL programmable parameters table

Parameter name	Default	Possible value	Description
M	32	16/32	Phase number
W	2048	1280/2048/4096	Y channel line buffer length, U/V channel should divide by 2
FRAC	19	18/19	Phase adder fraction part bit width
VCLIP	6	0~6 integer	Vertical filtering right shift bit width

5.12.2 Block Diagram

Figure 5-10 shows the block diagram of Video Scaler. It is a stream-to-stream module. The input interface which contains independent F/Y/U/V channel, receives data separately from up-stream module. The four channels also have independent resizing path for U/V format conversion. Together with the Y channel, the F channel is a 1-bit data channel that scales using the nearest neighborhood method which data represents the valid data flag generated by overlay module. After resizing to FYUV1444, data outputs in pixel mode to down-stream module. It contains some parts and their function list followed:

- Control logic: status machine control, registers operation, VPHASE selection.
- Resizing: three independent channels line buffer control, horizontal resize, vertical resize.

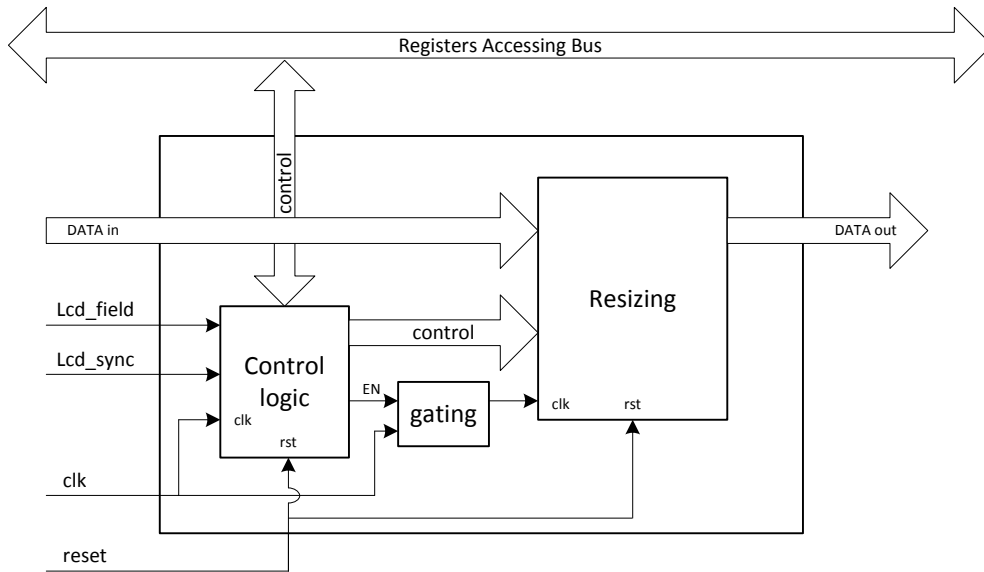


Figure 5-10. VS block diagram

5.12.3 Video Scaler Register List

Register name	Offset	Description
VS_CTRL_REG	0x000	Control register
VS_STATUS_REG	0x008	Status register
VS_FIELD_CTRL_REG	0x00C	Field control register
VS_OUT_SIZE_REG	0x040	Output size register
VS_Y_SIZE_REG	0x080	Y channel size register
VS_Y_HSTEP_REG	0x088	Y channel horizontal step register
VS_Y_VSTEP_REG	0x08C	Y channel vertical step register
VS_Y_HPHASE_REG	0x090	Y channel horizontal initial phase register
VS_Y_VPHASE0_REG	0x098	Y channel vertical initial phase 0 register
VS_Y_VPHASE1_REG	0x09C	Y channel vertical initial phase 1 register
VS_C_SIZE_REG	0x0C0	C channel size register
VS_C_HSTEP_REG	0x0C8	C channel horizontal step register
VS_C_VSTEP_REG	0x0CC	C channel vertical step register
VS_C_HPHASE_REG	0x0D0	C channel horizontal initial phase register
VS_C_VPHASE0_REG	0x0D8	C channel vertical initial phase 0 register
VS_C_VPHASE1_REG	0x0DC	C channel vertical initial phase 1 register
VS_Y_HCOEF0_REGN	0x200+N*4	Y channel horizontal filter coefficient0 register N (N=0:(M-1))
VS_Y_HCOEF1_REGN	0x300+N*4	Y channel horizontal filter coefficient1 register N (N=0:(M-1))
VS_Y_VCOEF_REGN	0x400+N*4	Y channel vertical filter coefficient register N (N=0:(M-1))
VS_C_HCOEF0_REGN	0x600+N*4	C channel horizontal filter coefficient0 register N (N=0:(M-1))
VS_C_HCOEF1_REGN	0x700+N*4	C channel horizontal filter coefficient1 register N (N=0:(M-1))
VS_C_VCOEF_REGN	0x800+N*4	C channel vertical filter coefficient register N (N=0:(M-1))

Note: All registers except some bits in **VS_CTRL_REG**, **VS_FIELD_CTRL_REG**, **VS_STATUS_REG** are double-buffered refreshed by **REG_RDY**.

5.12.4 Video Scaler Register Description

5.12.4.1 VS_CTRL_REG

Note: Only bit EN is double-buffered.

Offset: 0x000			Register Name: VS_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	BIST_EN BIST enable 0: Disable 1: Enable
30	R/W	0x0	CORE_RST Core circuit reset 0: Do noting 1: reset core circuit
29:5	/	/	/
4	R/W	0x0	COEF_SWITCH_EN Coefficients RAM switch 0: DONOT switch 1: Switch RAM use REG_RDY Note: When LCD SYNC go low and COEF_SWITCH_EN is 1, coefficient RAM will switch to the latest updated RAM if REG_RDY is 1, and then the bit will also be self-cleared if switch action successes.
3:1	/	/	/
0	R/W	0x0	EN Video Scaler enable 0: Disable 1: Enable Note: When module disabled, the core clock to the core circuit will be gated.

5.12.4.2 VS_STATUS_REG

Note: Whole WORD is non-double-buffered.

Offset: 0x008			Register Name: VS_STATUS_REG
Bit	Read/Write	Default/Hex	Description
31:28	/	/	/
27:16	R	0x0	LINE_CNT Output line number
15:5	/	/	/
4	R	0x0	BUSY Core circuit status 0: idle (finish, module disable, waiting for LCD SYNC negative edge) 1: busy (core circuit calculating)
3:0	/	/	/

5.12.4.3 VS_FIELD_CTRL_REG

Note: Only bit FIELD_SEL_VPHASE_EN is double-buffered.

Offset: 0x00C			Register Name: VS_FIELD_CTRL_REG
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5	R/W	0x0	LCD_SYNC_REVERSE Reverse LCD SYNC 0: DONOT reverse 1: Reverse
4	R/W	0x0	LCD_FILED_REVERSE Reverse LCD FILED 0: DONOT reverse 1: Reverse
3:1	/	/	/
0	R/W	0x0	FIELD_SEL_VPHASE_EN Vertical initial phase switch control 0: Vertical initial phase fix to phase0 1: Switch Vertical initial phase by LCD FIELD (Switch to phase0 when LCD FILED is 1, and switch to phase1 when LCD FIELD is 0)

5.12.4.4 VS_OUT_SIZE_REG

Offset: 0x040			Register Name: VS_OUT_SIZE_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	OUT_HEIGHT Output height The actual height is register value + 1
15:13	/	/	/
12:0	R/W	0x0	OUT_WIDTH Output width The actual width is register value + 1

5.12.4.5 VS_Y_SIZE_REG

Offset: 0x080			Register Name: VS_Y_SIZE_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	Y_HEIGHT Y channel input height The actual height is register value + 1
15:13	/	/	/
12:0	R/W	0x0	Y_WIDTH Y channel input width The actual width is register value + 1

5.12.4.6 VS_Y_HSTEP_REG

Offset: 0x088			Register Name: VS_Y_HSTEP_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:20	R/W	0x0	Y_HSTEP_INT The integer part of Y channel horizontal scale ratio
19:1	R/W	0x0	Y_HSTEP_FRAC The fraction part of Y channel horizontal scale ratio
0	/	/	/

5.12.4.7 VS_Y_VSTEP_REG

Offset: 0x08C			Register Name: VS_Y_VSTEP_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:20	R/W	0x0	Y_VSTEP_INT The integer part of Y channel vertical scale ratio
19:1	R/W	0x0	Y_VSTEP_FRAC The fraction part of Y channel vertical scale ratio
0	/	/	/

5.12.4.8 VS_Y_HPHASE_REG

Offset: 0x090			Register Name: VS_Y_HPHASE_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:20	R/W	0x0	Y_HPHASE_INT The integer part of Y channel horizontal initial phase
19:1	R/W	0x0	Y_HPHASE_FRAC The fraction part of Y channel horizontal initial phase
0	/	/	/

5.12.4.9 VS_Y_VPHASE0_REG

Offset: 0x098			Register Name: VS_Y_VPHASE0_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:20	R/W	0x0	Y_VPHASE0_INT The integer part of Y channel vertical initial phase0
19:1	R/W	0x0	Y_VPHASE0_FRAC The fraction part of Y channel vertical initial phase0
0	/	/	/

5.12.4.10 VS_Y_VPHASE1_REG

Offset: 0x09C			Register Name: VS_Y_VPHASE1_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:20	R/W	0x0	Y_VPHASE1_INT The integer part of Y channel vertical initial phase1
19:1	R/W	0x0	Y_VPHASE1_FRAC The fraction part of Y channel vertical initial phase1
0	/	/	/

5.12.4.11 VS_C_SIZE_REG

Offset: 0x0C0			Register Name: VS_C_SIZE_REG
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	C_HEIGHT C channel input height The actual height is register value + 1
15:13	/	/	/
12:0	R/W	0x0	C_WIDTH C channel input width The actual width is register value + 1

5.12.4.12 VS_C_HSTEP_REG

Offset: 0x0C8			Register Name: VS_C_HSTEP_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:20	R/W	0x0	C_HSTEP_INT The integer part of C channel horizontal scale ratio
19:1	R/W	0x0	C_HSTEP_FRAC The fraction part of C channel horizontal scale ratio
0	/	/	/

5.12.4.13 VS_C_VSTEP_REG

Offset: 0x0CC			Register Name: VS_C_VSTEP_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:20	R/W	0x0	C_VSTEP_INT The integer part of C channel vertical scale ratio
19:1	R/W	0x0	C_VSTEP_FRAC The fraction part of C channel vertical scale ratio
0	/	/	/

5.12.4.14 VS_C_HPHASE_REG

Offset: 0x0D0			Register Name: VS_C_HPHASE_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:20	R/W	0x0	C_HPHASE_INT The integer part of C channel horizontal initial phase
19:1	R/W	0x0	C_HPHASE_FRAC The fraction part of C channel horizontal initial phase
0	/	/	/

5.12.4.15 VS_Y_VPHASE0_REG

Offset: 0x0D8			Register Name: VS_Y_VPHASE0_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:20	R/W	0x0	C_VPHASE0_INT The integer part of C channel vertical initial phase0
19:1	R/W	0x0	C_VPHASE0_FRAC The fraction part of C channel vertical initial phase0
0	/	/	/

5.12.4.16 VS_C_VPHASE1_REG

Offset: 0x0DC			Register Name: VS_C_VPHASE1_REG
Bit	Read/Write	Default/Hex	Description
31:24	/	/	/
23:20	R/W	0x0	C_VPHASE1_INT The integer part of C channel vertical initial phase1
19:1	R/W	0x0	C_VPHASE1_FRAC The fraction part of C channel vertical initial phase1
0	/	/	/

5.12.4.17 VS_Y_HCOEF0_REGN (N = 0:(M-1))

Offset: 0x200 + N*4			Register Name: VS_Y_HCOEF0_REGN
Bit	Read/Write	Default/Hex	Description
31:24	R/W	UDF	COEF3
23:16	R/W	UDF	COEF2
15:8	R/W	UDF	COEF1
7:0	R/W	UDF	COEF0

Note: HCOEFF0 is a two's complement. The register value equals to coefficient*2⁶. N represents the phase.

5.12.4.18 VS_Y_HCOEF1_REGN (N = 0:(M-1))

Offset: 0x300 + N*4			Register Name: VS_Y_HCOEF1_REGN
---------------------	--	--	---------------------------------

Bit	Read/Write	Default/Hex	Description
31:24	R/W	UDF	COEF7
23:16	R/W	UDF	COEF6
15:8	R/W	UDF	COEF5
7:0	R/W	UDF	COEF4

Note: HCOEFF1 is a two's complement. The register value equals to coefficient*2⁶. N represents the phase.

5.12.4.19 VS_Y_VCOEF_REGN (N = 0:(M-1))

Offset: 0x400 + N*4			Register Name: VS_Y_VCOEF_REGN
Bit	Read/Write	Default/Hex	Description
31:24	R/W	UDF	COEF3
23:16	R/W	UDF	COEF2
15:8	R/W	UDF	COEF1
7:0	R/W	UDF	COEF0

Note: VCOEFF is a two's complement. The register value equals to coefficient*2⁶. N represents the phase.

5.12.4.20 VS_C_HCOEF0_REGN (N = 0:(M-1))

Offset: 0x600 + N*4			Register Name: VS_C_HCOEF0_REGN
Bit	Read/Write	Default/Hex	Description
31:24	R/W	UDF	COEF3
23:16	R/W	UDF	COEF2
15:8	R/W	UDF	COEF1
7:0	R/W	UDF	COEF0

Note: HCOEFF0 is a two's complement. The register value equals to coefficient*2⁶. N represents the phase.

5.12.4.21 VS_C_HCOEF1_REGN (N = 0:(M-1))

Offset: 0x700 + N*4			Register Name: VS_C_HCOEF1_REGN
Bit	Read/Write	Default/Hex	Description
31:24	R/W	UDF	COEF7
23:16	R/W	UDF	COEF6
15:8	R/W	UDF	COEF5
7:0	R/W	UDF	COEF4

Note: HCOEFF1 is a two's complement. The register value equals to coefficient*2⁶. N represents the phase.

5.12.4.22 VS_C_VCOEF_REGN (N = 0:(M-1))

Offset: 0x800 + N*4			Register Name: VS_C_VCOEF_REGN
Bit	Read/Write	Default/Hex	Description
31:24	R/W	UDF	COEF3
23:16	R/W	UDF	COEF2
15:8	R/W	UDF	COEF1
7:0	R/W	UDF	COEF0

Note: VCOEFF is a two's complement. The register value equals to coefficient*2⁶. N represents the phase.

5.13 DE Rotation Specification

5.13.1 Overview

- Support 1/2/3 address data copy
- Support input and output size from 8x8 to 2048x2048
- Support horizontal and vertical flip, clockwise 0/90/180/270 degree rotate
- Support input/output format YUV422/YUV420/ARGB8888/XRGB8888

5.13.2 Block Diagram

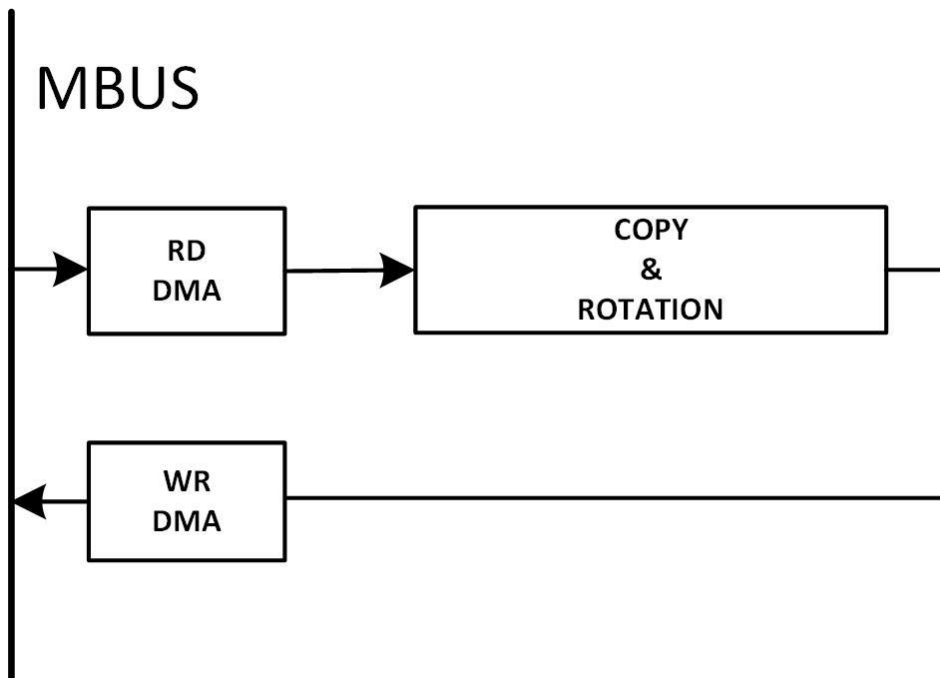


Figure 5-11. CSR Block Diagram

5.13.3 Description

5.13.4 Copy & Rotation

There are several types of rotation: clockwise 0/90/180/270 degree Rotation and H-Flip/V-Flip. Operation of Copy is the same as a 0 degree rotation.

Copy & Rotation will not change RGB color format. But for YUV422/YUV420 input, all output will be in YUV 420 planar.

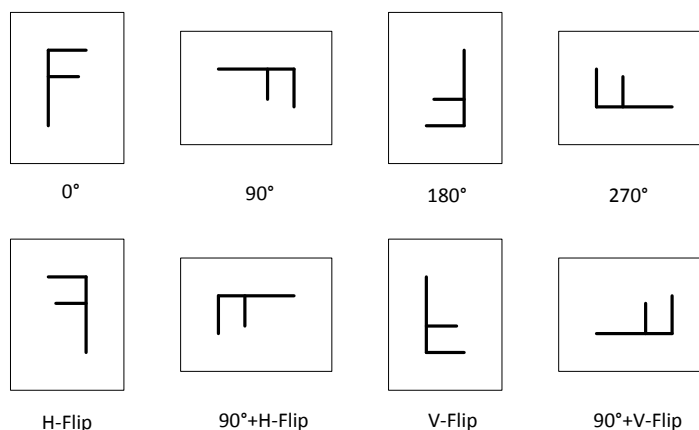


Figure 5-12. Rotation Diagram

Speed requirement:

For ARGB : 1pixel/1cycle

For RGB565 : 2pixel/1cycle

For YUV420 : 4pixel/1cycle

Test Format	Rotation Angle	Test Resolution	Boundary Condition		
ARGB8888	0,90,180,270 0+HFlip; 90+HFlip; 0+VFlip; 90+VFlip;	4x4, 4096x4,4x4096 2048x1536 Typical Resolution	In/Out Pitch:	16byte Align	
			In/Out Width/Height:	1pixel Align	
			Input Address:	16byte Align	
			Output Address:	4byte Align	
YUV420 /YUV422 Semi-Planar		8x8, 4096x8, 8x4096 2048x1536 Typical Resolution	In/Out Pitch: U Pitch = V Pitch	16byte Align	
			In/Out Width/Height:	2pixel Align	
Input Address:			16byte Align		
Y Output Address:			2byte Align		
UV Output Address:			1byte Align When the format is Planar, the remainder of U and V base address divided by 32byte must be equal.		
YUV420 /YUV422 Planar			4x4, 4096x4,4x4096 2048x1536 Typical Resolution	In/Out Pitch:	16byte Align
				In/Out Width/Height:	1pixel Align
		Input Address:		16byte Align	
	Output Address:	(4*N + 3)byte (4*N + 6)byte (4*N + 9)byte (4*N + 12)byte			

5.13.5 Register List

Register name	Offset	Description
GLB_CTL	0x000	Global control register

INT	0x004	Interrupt register
IFMT	0x020	Input data attribute register
IDATA_SIZE	0x024	Input data size register
IDATA_MEN_PITCH0	0x030	Input Y/RGB/ARGB memory pitch register
IDATA_MEN_PITCH1	0x034	Input U/UV memory pitch register
IDATA_MEN_PITCH2	0x038	Input V memory pitch register
IMEN_LADD0	0x040	Input Y/RGB/ARGB memory address register0
IMEN_HADD0	0x044	Input Y/RGB/ARGB memory address register1
IMEN_LADD1	0x048	Input U/UV memory address register0
IMEN_HADD1	0x04C	Input U/UV memory address register1
IMEN_LADD2	0x050	Input V memory address register0
IMEN_HADD2	0x054	Input V memory address register1
ODATA_SIZE	0x084	Output data size register
ODATA_MEN_PITCH0	0x090	Output Y/RGB/ARGB memory pitch register
ODATA_MEN_PITCH1	0x094	Output U/UV memory pitch register
ODATA_MEN_PITCH2	0x098	Output V memory pitch register
OMEN_LADD0	0x0A0	Output Y/RGB/ARGB memory address register0
OMEN_HADD0	0x0A4	Output Y/RGB/ARGB memory address register1
OMEN_LADD1	0x0A8	Output U/UV memory address register0
OMEN_HADD1	0x0AC	Output U/UV memory address register1
OMEN_LADD2	0x0B0	Output V memory address register0
OMEN_HADD2	0x0B4	Output V memory address register1

5.13.6 Register Description

5.13.6.1 Global control register

Offset: 0x000			Register Name: CSR_CTL
Bit	Read/Write	Default/Hex	Description
31	R/W	0x0	START 0: idle 1: start Write 1 to this register bit will start the CSR, when finish, it's self-cleaned by hardware.
30	R/W	0x0	RESET 0: normal 1: assert reset
29:22	/	/	reserved
21:16	R/W	0x3f	Burst length 0x07: burst8 0x0f: burst16 0x1f: burst32 0x3f: burst64
15:8	/	/	/
7	R/W	0x0	H-FLIP

			0: disable 1: enable
6	R/W	0x0	V-FLIP 0: disable 1: enable
5:4	R/W	0x0	Rotation function 00: clockwise 0 degree 01: clockwise 90 degree 10: clockwise 180 degree 11: clockwise 270 degree
3:2	/	/	reserved
1:0	R/W	0x0	MODE_SEL CSR mode select 00: disable 01: copy& rotation other: reserved

5.13.6.2 Interrupt register

Offset: 0x004			Register Name: INT
Bit	Read/Write	Default/Hex	Description
31:17	/	/	/
16	R/W	0x0	FINISH_IRQ_EN 0: disable FINISH IRQ 1: enable FINISH IRQ
15:2	/	/	/
0	R/W	0x0	FINISH Mission finish flag It will be set when job is finished, and cleared by writing 1.

5.13.6.3 Input data attribute register

Offset: 0x020			Register Name: IFMT
Bit	Read/Write	Default/Hex	Description
31:6	/	/	/
5:0	R/W	0x0	IFMT Input data format 0x00: ARGB_8888 0x01: ABGR_8888 0x02: RGBA_8888 0x03: BGRA_8888 0x04: XRGB_8888 0x05: XBGR_8888 0x06: RGBX_8888 0x07: BGRX_8888

			0x08~0x23: Reserved 0x24: Planar YUV422 UV combined(V1U1V0U0) 0x25: Planar YUV422 UV combined(U1V1U0V0) 0x26: Planar YUV422 0x27: Reserved 0x28: Planar YUV420 UV combined(V1U1V0U0) 0x29: Planar YUV420 UV combined(U1V1U0V0) 0x2A: Planar YUV420 Other: Reserved
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5.13.6.4 Input data size register

Offset: 0x024			Register Name: IDATA_SIZE
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	HEIGHT The Height = The value of these bits add 1
15:13	/	/	/
12:0	R/W	0x0	WIDTH The Width = The value of these bits add 1

5.13.6.5 Input Y/RGB/ARGB memory pitch register

Offset: 0x030			Register Name: IDATA_MEN_PITCH0
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	PITCH0 Input Y/RGB data memory pitch in bytes Should be 128bit aligned.

Note: The setting of this register is Y/RGB channel address.

5.13.6.6 Input U/UV memory pitch register

Offset: 0x034			Register Name: IDATA_MEN_PITCH 1
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	PITCH1 Input U/UV data memory pitch in bytes Should be 128bit aligned.

Note: The setting of this register is U/UV channel address.

5.13.6.7 Input V memory pitch register

Offset: 0x038			Register Name: IDATA_MEN_PITCH2
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	PITCH2 Input V data memory pitch in bytes.

			Should be 128bit aligned. Must be equal to input U/UV pitch.
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Note: The setting of this register is V channel address.

5.13.6.8 Input Y/RGB/ARGB memory address register0

Offset: 0x040			Register Name: IMEN_LADD0
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LADD Memory Block Start Address bit[31:0] Should be 128bit aligned.

Note: The setting of this register is Y/RGB channel address.

5.13.6.9 Input Y/RGB/ARGB memory address register1

Offset: 0x044			Register Name: IMEN_HADD0
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	HADD Memory Block Start Address bit[39:32] Layer Memory Block Address in bytes

Note: The setting of this register is Y/RGB channel address.

5.13.6.10 Input U/UV memory address register0

Offset: 0x048			Register Name: IMEN_LADD1
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LADD Memory Block Start Address bit[31:0] Should be 128bit aligned.

Note: The setting of this register is U/UV channel address.

5.13.6.11 Input U/UV memory address register1

Offset: 0x04c			Register Name: IMEN_HADD1
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	HADD Memory Block Start Address bit[39:32] Layer Memory Block Address in bytes

Note: The setting of this register is U/UV channel address.

5.13.6.12 Input V memory address register0

Offset: 0x050			Register Name: IMEN_LADD2
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Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LADD Memory Block Start Address bit[31:0] Should be 128bit aligned.

Note: The setting of this register is V channel address.

5.13.6.13 Input V memory address register1

Offset: 0x054			Register Name: IMEN_HADD2
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	HADD Memory Block Start Address bit[39:32] Layer Memory Block Address in bytes

Note: The setting of this register is V channel address.

5.13.6.14 Output data size register

Offset: 0x084			Register Name: ODATA_SIZE
Bit	Read/Write	Default/Hex	Description
31:29	/	/	/
28:16	R/W	0x0	HEIGHT The Height = The value of these bits add 1
15:13	/	/	/
12:0	R/W	0x0	WIDTH The Width = The value of these bits add 1

5.13.6.15 Output Y/RGB/ARGB memory pitch register

Offset: 0x090			Register Name: ODATA_MEN_PITCH0
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	PITCH0 output Y/RGB data memory pitch in bytes Should be 128bit aligned.

Note: The setting of this register is Y/RGB channel address.

5.13.6.16 Output U/UV memory pitch register

Offset: 0x094			Register Name: ODATA_MEN_PITCH 1
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	PITCH1 U/UV data memory pitch in bytes Should be 128bit aligned.

Note: The setting of this register is U/UV channel address.

5.13.6.17 Output V memory pitch register

Offset: 0x098			Register Name: ODATA_MEN_PITCH2
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	PITCH2 Output V data memory pitch in bytes Should be 128bit aligned. Must be equal to output U/UV pitch.

Note: The setting of this register is V channel address.

5.13.6.18 Output Y/RGB/ARGB memory address register0

Offset: 0x0a0			Register Name: OMEN_LADD0
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LADD Memory Block Start Address bit[31:0]

Note: The setting of this register is Y/RGB channel address.

5.13.6.19 Output Y/RGB/ARGB memory address register1

Offset: 0x0a4			Register Name: OMEN_HADD0
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	HADD Memory Block Start Address bit[39:32] Layer Memory Block Address in bytes

Note: The setting of this register is Y/RGB channel address.

5.13.6.20 Output U/UV memory address register0

Offset: 0x0a8			Register Name: OMEN_LADD1
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LADD Memory Block Start Address bit[31:0]

Note: The setting of this register is U/UV channel address.

5.13.6.21 Output U/UV memory address register1

Offset: 0x0ac			Register Name: OMEN_HADD1
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	HADD Memory Block Start Address bit[39:32] Layer Memory Block Address in bytes

Note: The setting of this register is U/UV channel address.

5.13.6.22 Output V memory address register0

Offset: 0x0b0			Register Name: OMEN_LADD2
Bit	Read/Write	Default/Hex	Description
31:0	R/W	0x0	LADD Memory Block Start Address bit[31:0]

Note: The setting of this register is V channel address.

5.13.6.23 Output V memory address register1

Offset: 0x0b4			Register Name: OMEN_HADD2
Bit	Read/Write	Default/Hex	Description
31:8	/	/	/
7:0	R/W	0x0	HADD Memory Block Start Address bit[39:32] Layer Memory Block Address in bytes

Note: The setting of this register is V channel address.